A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING

A Thesis Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In partial fulfillment of the requirement for the degree
Bachelor of Science in Computer Science
Specialize in Intelligent Systems

JUSTINE MILES E. DEL ROSARIO
LOUISE ANDREI SAPIN
JOHN DAVE A. SILVA

ROWAN N. ELOMINA, PhD

JUNE 2026

APPROVAL SHEET

The thesis entitled " **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING**" prepared and submitted by **Del Rosario, Justine Miles E., Sapin, Louise Andrei** and **Silva, John Dave A.** in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

ROWAN N. ELOMINA, PhD
Adviser

Approved and accepted by the Committee on Oral Examination with a grade of _____.

MIKE PHILIP T. RAMOS, MIT
Member

SHERWIN B. SAPIN, EdD
Member

JOCELYN O. PADALLAN, MIT
Member

JEFFERSON L. LERIOS, DIT
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science specialized in Intelligent Systems.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGEMENT

All those whose authority, assistance, and cooperation contributed to the completion of the research deserve sincere gratitude.

First and foremost, the researchers express a deepest gratitude to God Almighty for granting strength, wisdom, and energy throughout the research journey. In moments of doubt and struggle, His divine guidance provided stability and direction. All glory and honor belong to Him, as the completion of this research would not have been possible without His grace.

The researchers extend sincere gratitude to the University President, Dr. Mario R. Briones, and the OIC Campus Director, Engr. Manuel Luis R. Alvarez, for exemplary leadership and continued commitment to academic excellence at Laguna State Polytechnic University.

Heartfelt appreciation is given by the researchers to the Associate Dean of the College of Computer Studies, Dr. Loyd S. Echalar, and to the Head of the Research Implementing Unit, Dr. Jefferson L. Lerios, for support and encouragement throughout the development of the research.

Sincere appreciation is likewise expressed by the researchers to Mr. Jonardo R. Asor, whose suggestions and professional advice were instrumental in identifying the key ideas and focus of the research. His guidance significantly contributed to shaping the conceptual direction and relevance of the research.

The researchers also express gratitude to the Thesis Adviser, Dr. Rowan N. Elomina, for guidance, expertise, and unwavering support that were instrumental in shaping the direction and quality of the research. Patience,

encouragement, and thoughtful mentorship greatly contributed to the successful completion of the research.

To distinguished Members of Approval Committee, Mr. Mike Philip T. Ramos, Dr. Sherwin B. Sapin, and Ms. Jocelyn O. Padallan, the researchers are profoundly grateful for the constructive feedback, valuable insights, and time in evaluating the work.

The researchers also acknowledge Sir Harold Rebong, Head of Agriculture in Victoria, Laguna, and Ms. Esperanza Daing of Barangay San Benito, for kind assistance and recommendations that were essential in locating the leads and data sources necessary for the research.

Sincere thanks are extended by the researchers to Mr. Eddie Mesina, farm owner, for warm accommodation and for providing crucial information and data for interviews and background research. Equal appreciation is given to Mr. Ruel Mesina and Mr. Raymond Mesina, whose generosity, assistance, and firsthand knowledge as farmers greatly contributed to the data-gathering process.

Finally, the researchers extend appreciation to all individuals who contributed to the completion of this research, whether directly or indirectly. All forms of support and assistance are deeply valued and sincerely appreciated.

The Researchers

DEDICATION

The thesis is fully dedicated to all who have been the source of strength, inspiration, and unconditional support throughout academic journey of the researchers.

Mr. Del Rosario, Michael O. and Mrs. Del Rosario, Lea E.: This journey was built upon unconditional love, unwavering support, and lasting motivation. This achievement would not have been possible without selfless sacrifices and continuous efforts, particularly the financial and emotional support provided throughout the years. Every opportunity to succeed was given, even during the most challenging times. Belief in the researchers' potential, patience through every struggle, and constant encouragement served as the greatest sources of strength. Commitment, diligence, and devotion stand as the true reasons behind this accomplishment, which goes beyond individual success.

Mr. Sapin, Ryan Q. and Ms. Sapin Aizalyn Q.: Unwavering support served as a constant source of motivation, encouraging perseverance and the courage to aim higher. The challenges and sacrifices faced with strength and resilience inspired determination, risk-taking, and the pursuit of a better future. Faith and trust became powerful reminders to continue moving forward despite difficulties, motivating the researchers to strive for goals once considered distant dreams.

Mr. Silva, Catalino L. and Mrs. Silva, Rosalema A.: Forever gratitude and appreciation are extended for unyielding guidance and enduring support. From the beginning of the academic journey to its completion, encouragement,

belief, and reassurance played a vital role in reaching this milestone. The love, memory, and aspirations of a late father remain a guiding force, providing strength and motivation in every step toward success.

To our **siblings, friends, and classmates**: Patience, understanding, and companionship served as constant sources of strength throughout the journey. Support during challenging moments and shared joy in times of success made the academic experience meaningful and fulfilling.

And to our **mentors and professors**: Wisdom, guidance, and belief in potential played a vital role in shaping the direction and success of this work. Dedication to teaching and the knowledge shared contributed significantly to the growth and development of the researchers.

Lastly, to the **quiet supporters**: Those whose unseen efforts, silent sacrifices, and unwavering presence provided strength through every stage of the journey. This work is dedicated to those who stood by the researchers during uncertain times, offering support not always through words, but through constant care and steadfast presence.

J.M.E.D.R

L.A.S.

J.D.A.S.

ABSTRACT

Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') is a high-value yellow-fleshed cultivar whose market price is strongly influenced by sweetness. However, post-harvest sweetness evaluation remains subjective and inconsistent, particularly under variable environmental conditions. This study investigates the correlation between external aesthetic traits and internal sweetness (°Brix) using image processing and hybrid machine learning techniques. A total of 150 harvested watermelons were imaged under controlled conditions, producing 900 high-resolution images that were expanded to 3,150 samples through data augmentation. Sweetness levels were measured using a handheld Brix refractometer, while physical attributes were recorded as supplementary metadata. Deep visual features were extracted using EfficientNet-B0, and multiple regression models were evaluated, including CNN-only and hybrid CNN–machine learning approaches. Model performance was assessed using R^2, MAE, and RMSE. Results indicate that visual features alone are insufficient for accurate sweetness prediction, though they enable reliable estimation within a general range. Among all configurations, the CNN + Random Forest model demonstrated the most stable performance (R^2 score = -1.4112, MAE = 0.2243, and RMSE = 0.0908), reflecting weak but non-zero correlation. Feature analysis revealed fruit color as the most influential visual trait. The study highlights both the potential and limitations of non-destructive, image-based sweetness estimation for agricultural quality assessment.

Keywords: *Sweet Gold watermelon, °Brix prediction, Image processing, Convolutional neural network, Random Forest, Non-destructive fruit quality assessment*

TABLE OF CONTENTS

Preliminaries		***Page***
Title page		
Approval Sheet		i
Acknowledgement		ii
Dedication		iv
Abstract		vi
Table of Contents		vii
List of Tables		ix
List of Figures		x
Definition of Terms		xii
CHAPTER		
I	**INTRODUCTION**	
	Background of the Study	1
	Research Objectives	4
	Significance of the Study	5
	Scope and Limitations of the Study	6
II	**THEORETICAL FRAMEWORK OF THE STUDY**	
	Fruit Ripeness and Sweetness	7
	Aesthetic and Physical Features of Fruits	10
	Image Processing Techniques in Agriculture	12
	Machine Learning in Fruit Quality Prediction	15
	Synthesis and Relevance of the Study	17
	Conceptual Framework of the Study	20
III	**METHODOLOGY**	
	Experimental Design	22
	Study Area	23
	Materials	24
	Hardware Specifications	25
	Software Specifications	25
	Data Acquisition Implementation	27
	Data Splitting	29
	Data Processing	29
	Data Augmentation	30
	Model Development	31
	Machine Learning Algorithms	33
	Convolutional Neural Network	33
	Random Forest	34
	Support Vector Regression	34
	Extreme Gradient Boosting	35
	Theorem Foundation	36
	Model Evaluation	37
	Ethical Considerations	38

IV RESULT AND DISCUSSION
High-resolution Images of Sweet Gold Watermelon 39
Aesthetic Traits and Sweetness Level of Sweet Gold Watermelon 40
Hybrid Machine Learning Algorithms for Best Performing Model 44
Evaluation for Hybrid Machine Learning Model Performance 47
Hyperparameter Configurations Across Machine Learning Models 49
Principal Component Analysis with CNN Features 55
Feature Importance for CNN + Random Forest 57
Correlation Heatmap 58

V SUMMARY, CONCLUSION, AND RECOMMENDATION
Summary 67
Conclusions 68
Recommendations 71

REFERENCES 73
APENDICIES
Appendix A Letter of Approval
Appendix B Certificate of Acceptance
Appendix C Survey Questionnaire
Appendix D Code Screenshot
Appendix E Revision Matrix
Appendix F Grammarly & Turnitin Report
Appendix G Publishable Paper (IEEE Format)
Appendix H Curriculum Vitae

LIST OF TABLES

No.	Title	Page
1	Literature Matrix	18
2	Hardware Specifications	25
3	Software Specifications	26
4	Sweetness/Brix Measurement Basis	28
5	Hybrid Machine Learning Models Performance Results	46
6	Hyperparameters used in EfficientNetB0 CNN-only	51
7	Hyperparameters used in EfficientNetB0 CNN + Metadata	52
8	Hyperparameters used in Hybrid Random Forest	53
9	Hyperparameters used in Hybrid Support Vector Regressor	54
10	Hyperparameters used in Hybrid Extreme Gradient Boosting	55
11	Actual Brix Measurement for Testing Images of Sweet Gold Watermelon with Day 3 and Day 5	65

LIST OF FIGURES

No.	Title	Page
1	Concept of Developing Model for Studying Sweet Gold Watermelon	21
2	Workflow for Sweetness Prediction of Sweet Gold Watermelon	23
3	Brgy. Wakat, Calauan – Nagcarlan Rd, Laguna, Maps: Google Maps	24
4	Data Collection Procedures of Sweet Gold Watermelon	27
5	Data Mapping Process	28
6	Data Splitting, Pie Chart	29
7	Image Processing, Grabcut	30
8	Data Augmentation Process, “Original vs. Augmented”	31
9	Convolutional Neural Network	33
10	Random Forest, Hybrid feature extraction	34
11	Structure of XGBoost Algorithm	36
12	Folder Structure and Multi-Angle Images of Sweet Gold Watermelons	40
13	Sweet Gold Watermelon Measurement Procedure	41
14	Sweet Gold Watermelon Metadata	42
15	Comparison of Hybrid Models for Sweetness Prediction	48
16	CNN Feature (Images only) Colored by Brix	56
17	Top 20 Important Feature (CNN + RF)	57
18	CNN Feature vs. Sweetness	58
19	Physical Traits vs. Sweetness	59
20	Predictive Tool for Sweet Gold Watermelon, Prototype	61

21	Sweet Gold Watermelon, Testing Images	61
22	Reference Data for Sweet Gold Watermelon	63
23	Sample Input/Output, Day 3 Image and Day 5 Image	64

DEFINITION OF TERMS

Brix Refractometer	A tool that calculates dissolved solids (mostly sugars) in liquids by measuring refractive index. Study uses a handheld version to test juice from the harvested Sweet Gold watermelons and assign a numerical sweetness value in degrees Brix.
Convolutional Neural Networks (CNN)	A deep learning method that extracts layered visual features from image via specialized convolutional layers. Serves as either a standalone predictor or feature extractor for hybrid models analysing Sweet Gold watermelon images to estimate Brix values.
Correlation Heatmap	A visual matrix-style plot that shows the strength and direction of relationships between variables using color intensity. In this study, it is used to display how different visual features (color values, texture values, etc.) correlate with the measured Brix values of Sweet Gold watermelons.
Data Augmentation	Machine learning tactic that expands dataset size by creating modified data copies to enhance model generalization. The study applies it to generate four extra image per original Sweet Gold watermelon photo, growing the dataset from 900 to 4500 samples for training.
EffientNet-B0	Pre-trained CNN architecture optimized for image classification efficiency by balancing depth, width, and resolution scaling. It acts as the core feature extractor for Sweet Gold watermelon image in hybrid models (CNN- Random Forest, CNN-Support Vector Regression, CNN-Extreme Gradient Boosting).
Extreme Gradient Boosting (XGBoost Regressor)	A supervised boosting algorithm that combines sequential decision trees with regularization to avoid overfitting. Paired with EfficientNet-B0 in the study to model links between Sweet Gold watermelon visual traits and continuous Brix values.

Feature Importance	A ranking measure that identifies which variables contribute the most to a model's prediction. In this research, feature importance indicates which extracted image features (from EfficientNet-B0) have the strongest influence on predicting Sweet Gold watermelon Brix values in hybrid models like RF, SVR, and XGBoost.
GrabCut Algorithm	An Image segmentation tool that separates foreground objects from backgrounds using iterative graph cut optimization. This study uses it to strip backgrounds from Sweet Gold watermelon images, focusing the model exclusively on the fruit`s visual characteristics.
High-Resolution Images	Photographs taken with small detail, clarity, and definition in order to extract surface characteristics from samples of watermelon and still retain some level of detail.
Hybrid Machine Learning Model	A predictive tool that merges two or more machine learning algorithms to leverage their unique strengths and boost accuracy. Refers to this study`s combinations of EfficientNet-B0 (feature extractor) with Random Forest, Support Vector Regression, or Extreme Gradient Boosting to predict Sweet Gold watermelon sweetness.
Image Processing	A computational process which examines and modifies digital images for useful information. Image processing has been applied in this research project to extract visual features like color, texture, shape, and size from the watermelon images.
Mean Absolute Error (MAE)	A Statistical metric that calculates the average absolute gap between predicted and actual values. It is used to measure the average error in predicted vs. Actual Brix values for Sweet Gold watermelons.
R^2 Score (Coefficient of Determination)	A statistical metric that shows the proportion of dependent variable variance explained by independent variables. Used to gauge how well Sweet Gold watermelon visual traits explain actual Brix values.
Principal Component Analysis	A dimensionality-reduction technique that transforms many correlated features

	into a smaller set of uncorrelated components called principal components. In this study, PCA is used to simplify and visualize extracted image features (e.g., from EfficientNet-B0) while still keeping most of the information related to Sweet Gold watermelon sweetness.
Random Forest (RF)	An ensemble learning method that combine multiple decision trees to reduce overfitting and improve prediction accuracy. Paired with EffiecientNet-B0 to model visual feature-sweetness correlation and estimate Sweet Gold watermelon Brix values.
Root Mean Squared Error (RMSE)	A statistical metric that computes square root of the average squared difference between predicted and actual values. Used to highlight larger errors in predicted vs. actual Brix values for Sweet Gold watermelon.
Statistical Learning Theory (SLT)	A mathematical framework that explains how machine learning models learn patterns from data to make accurate predictions underpinning supervised learning techniques. It is the Theoretical basis for all models use in the study to corelate Sweet Gold watermelon visual traits with internal sweetness levels.
Support Vector Machines (SVR)	A supervised machine learning algorithm from Support Vector Machine (SVM) that finds a hyperlane to minimize regression prediction error. In this research, it paired with EffiecientNet-B0 to predict continous Brix values for Sweet Gold watermelon.
***Sweet Gold Watermelon (*Citrullus lanatus* 'Sweet Gold')**	A yellow-Fleshed F1 hybrid watermelon cultivar in the Cucurbitaceae family, valued for high sugar content and tropical climate tolerance. It is the post-harvest fruit sample (grown in Brgy. Wakat, Laguna) whose visual traits (color, texture, Shape, size) are analyzed to predict internal sweetness in degrees Brix.
Visual Features	The features of a digital image such as color, texture, shape, and size, as visual input variables to predict watermelon sweetness.

CHAPTER 1

INTRODUCTION

Background of the Study

Watermelons originated in Africa over 5,000 years ago, where wild types were often hard-textured, pale, and bitter, unlike the sweet dessert types consumed today. The Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') is a yellow-fleshed cultivar belonging to the Cucurbitaceae family, known for producing large, juicy, and edible fruits (Paris, 2015). The sweet, domesticated cultivars, such as C. Lanatus, the delicate, vibrantly colored, and flavorful watermelons that are known worldwide, originated from a tiny ancestral population through a series of selection events. Sweet Gold, commonly mistaken as a honeydew because of its golden flesh, is still classified as a true watermelon, not a Cucumis melo. Its yellow color is attributed to the presence of β-carotene and xanthophylls, pigments responsible for its bright appearance and enhanced sweetness (Park et al., 2023). Watermelon variety studies reveal that C. lanatus var. Sweet Gold is a member of the lanatus group, which differs markedly from other subspecies in fruit quality traits such as size, soluble solids content, and sweetness (Cui et al., 2020). With over 1,000 cultivars available globally, Sweet Gold stands out not only for its distinctive color but also for its consumer appeal and market value as a high-sugar, non-GMO variety that matures rapidly and yields consistent quality fruit (NC Extension Gardener, n.d.). The yellow-fleshed F1 hybrid watermelon known as Sweet Gold (Citrullus lanatus 'Sweet Gold') is identified by its firm, crisp flesh, vibrant golden hue, and consistently high Brix value of 14-16° is considered very sweet (Santos,

2010). It is an excellent variety for producing in regions like Laguna because of its high tolerance for tropical conditions and resistance to common diseases, particularly in the dry season. Field trials in Victoria, Laguna, have proven that Sweet Gold can produce up to 7,000 kg per cropping season (January-April). However, during the rainy months (May-December), this yield drops to 3,000-4,000 kg due to excessive moisture, which degrades sweetness and causes fruit skin defects (Farmer Interview, 2025).

In terms of growing conditions, Sweet Gold grows best at temperatures between 25 and 35 °C and requires a sandy substrate that drains well and has a pH of 5.8 to 6.5 to grow and accumulate sweetness (Business Diary Philippines, 2023). Proper water management is essential: ample irrigation during early fruit development and reduced watering 1-2 weeks before harvest is key to enhancing sugar concentration and preventing over-ripeness or cracking (Permaterra Farm, 2024). Farmers do not use scientific tools like a Brix refractometer to determine maturity; instead, they rely on simple signs, such as a dry, curled tendril, a yellow spot where the fruit touches the ground, or a soft stem near the fruit. Fruits that are cracked, damaged, or overripe are rejected during traditional sorting and careful handling while packing preserves postharvest quality. Market-acceptable crops often sell for ₱35 per kilogram, but overripe or damaged fruits only earn ₱15 to ₱20 per kilogram or are rejected entirely (Farmer Interview, 2025).

The Philippines ranks 330th in melon exports and 51st in watermelon output by value, with 134.6 million kg in 2022 and USD 5.56 million in 2024, respectively. However, these figures include all varieties of watermelon, including yellow-fleshed cultivars like Sweet Gold (Palengke Boy, n.d.).

Southeast Asia's regional demand is increasing, despite Spain, Mexico, and Morocco leading worldwide in exports. Philippine agriculture is not yet among the top exporters in this subregion, but the growing interest in specialty cultivars like Sweet Gold worldwide offers the possibility of increased trade (Tridge, n.d.). Beyond its unique golden flesh and high Brix, Sweet Gold is grown for its disease resistance, small size (average weight of 5-7 kg), and adaptability to tropical climates for worldwide and domestic markets. However, production faces common cucurbit challenges such as aphids, whiteflies, root-knot nematodes, and fungal diseases like gummy stem blight, downy mildew, and Anthracnose, which threaten yield and fruit quality unless managed through integrated pest management and crop rotation (NC Extension Gardener, n.d.).

The study focuses on post-harvest Sweet Gold watermelons and aims to estimate its sweetness and ripeness level using only its visual appearance. Although each crop is technically harvest-ready, external appearance alone is not always a reliable indicator of internal sweetness, as it can differ greatly depending on factors such as rainfall, soil composition, and sunlight exposure. However, by using machine learning techniques, undetectable visual patterns that are not recognized by humans can be properly identified and associated with honest sweetness levels. This approach provides a non-destructive, data-driven solution that has the potential to reduce labor costs, improve yield quality, and support advancements in precision agriculture (Oliveira et al., 2025). This leads to the formulation of specific goals focused on developing a predictive model that supports consistency in quality assessment and informed decision-making in fruit marketing and handling.

Research Objectives

The primary objective of this study is to develop and evaluate a hybrid machine learning model that demonstrates the correlation between the sweetness level (Brix value regression) of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') and its external aesthetics such as color, texture, and shape using image processing techniques. Specifically, the study aims to:

1. Capture high-resolution images of Sweet Gold watermelons using a controlled imaging setup.
2. Analyze and extract visual features from the images to investigate their correlation with sweetness levels measured using a manual Brix refractometer.
3. Train and compare multiple hybrid machine learning algorithms to predict sweetness levels based on image processing techniques.
    a. Convolutional Neural Networks (CNNs)
    b. Random Forest (RF)
    c. Support Vector Regression (SVR)
    d. Extreme Gradient Boosting (XGBoost Regressor)
4. Assess model performance using evaluation metrics.
    a. Coefficient of determination (R^2 score)
    b. Mean Absolute Error (MAE)
    c. Root Mean Squared Error (RMSE)
5. Determine the extent to which external aesthetic features can reliably predict the sweetness level of Sweet Gold watermelons, and present the results through a visual demonstration of the trained model.

Significance of the Study

This section highlights the importance of developing a regression model that correlates the appearance of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') to its internal sweetness level. The goal of the study is to provide a way to assess the fruit sweetness quality of Sweet Gold watermelon despite similar external color across maturity, supporting better decision-making in fruit marketing and postharvest handling.

This study contributes to the UN Sustainable Development Goals through supporting innovative solutions in the agricultural sector. The study primarily supports SDG 9 “Industry, Innovation, and Infrastructure” by promoting the use of image processing and machine learning to modernize traditional fruit assessment practices. The study also supports SDG 2 “Zero Hunger” by improving postharvest quality assessment, helping reduce waste and increase the market value of produce. Lastly, this study supports the SDG 12 “Responsible Consumption and Production” by promoting accurate, non-destructive fruit evaluation that encourages better decision-making to producers and sellers, ethical handling, and improved efficiency across the supply chain (UN, 2020).

This study intends to have significant implications for various stakeholders:

Local Farmers. The main way that the results of this study will benefit these individuals is by providing a non-destructive, accurate method to assess fruit sweetness, which can support better harvesting and pricing decisions (Chawgien and Kiattisin, 2021).

Fruit Sellers and Distributors. The regression model can help these individuals apply a reliable method for grading fruit, reduce losses from misclassification, and improve customer satisfaction (FAO, 2021).

Agricultural Researchers, Future Researchers, and Technologists. The results of this study will provide many things for agricultural researchers and technologists. This outcome could provide a valuable reference for future research in this field, assisting in further innovations by offering a data-driven, non-invasive way to investigate fruit quality (Oliveira et al., 2025).

Scope and Limitations of the Study

This study aims to develop a regression model that estimates the sweetness level of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') based on external visual features using image processing and machine learning techniques. The dataset consists of 900 images captured from 150 harvested Sweet Gold watermelons grown in a farm located in Brgy. Wakat, Calauan–Nagcarlan Rd, Laguna. Machine learning models, including CNN, Random Forest (RF), Support Vector Regressor (SVR), and XGBoost Regressor, are implemented and evaluated using a 70% training, 15% validation, and 15% testing data split with R^2 score, MAE, and RMSE as performance metrics. The study is limited to post-harvest fruit analysis, along with a prototype and a demo interface, excluding the development system and the evaluation of non-visual traits such as aroma, internal texture, nutritional value, and environmental factors. In addition, 15 Sweet Gold watermelon samples collected from nearby markets are used to test the prototype system and assess its applicability in real-world scenarios and usefulness to intended beneficiaries.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature

This chapter presents a comprehensive review of related literature, both local and international, to establish the theoretical foundation of the study. This review highlights key ideas and findings from previous research to demonstrate the relevance of the study within a broader academic and practical context.

Fruit Ripeness and Sweetness Estimation

Several studies in recent years have explored non-destructive methods for estimating the internal quality and sweetness of fruits, focusing on technologies applicable in real-time, in-field conditions.

A systematic review of ripeness detection methods was conducted in oil palm fresh fruit bunches (FFB) and concluded that combining computer vision with convolutional neural networks (CNNs) offers the most feasible solution for field applications. The review revealed that artificial neural networks (ANNs), CNNs, and support vector machines (SVMs) are widely used for classifying maturity stages, with CNN models showing higher potential due to their ability to extract hierarchical features and perform real-time detection, especially when integrated with models like YOLOv3. While CNNs showed promising results in maturity classification, the study suggested that further improvements could involve integrating CNNs with IoT systems for geotagging ripe fruits and optimizing CNN architectures through hyperparameter tuning and dataset expansion (Lai et al., 2023).

Expanding of CNN's capability, Olisah et al. (2024) addressed the difficulty of detecting ripeness in blackberry fruits (Rubus fruticosus), which often do not display visible changes during maturation. It proposed a novel multi-input CNN ensemble model using hyperspectral imaging, trained on VGG16 architecture and optimized through stack generalization. The model achieved 95.1% accuracy on unseen data and 90.2% accuracy under real-world farm conditions. Interestingly, the study found that machine sensory outputs strongly correlated with human sensory evaluations, particularly texture. However, a limitation of this past study was that it was primarily conducted in a controlled environment, suggesting a gap in understanding how spectral reflectance changes under natural farm conditions. Future work was recommended to test the system in real environments and explore its integration with robotic harvesters or augmented reality systems to support human pickers.

A YOLOv8-based deep learning method was introduced for fruit ripeness detection using a custom dataset. The model was trained and validated on images of fruits at various ripeness levels under different conditions. It achieved a high level of precision (98.1%) and recall (98.0%) with a mean average precision (mAP) score of 99.1%, highlighting its reliability and accuracy. These findings showed that the model was robust across different fruit types and environmental conditions. However, the recent study also acknowledged the challenges of maintaining high accuracy across deep learning models and recommended future exploration of image augmentation techniques to further improve generalizability and real-world performance (Zhang, 2023).

In a broader context, both destructive and non-destructive methods were reviewed for measuring Brix, a key indicator of fruit sweetness. Traditional methods like hydrometers and refractometers, while widely used, are destructive, requiring juice samples. Refractometers offer better precision but still face limitations like sensitivity to temperature and manual error. The review emphasized the need for faster, low-cost, and environmentally friendly alternatives that minimize sample preparation. This underscores the demand for reliable non-destructive technologies in modern agricultural and food industries and signals a shift toward digital and optical solutions that can be implemented on-site (Jaywant et al., 2022).

Wang et al. (2021) focused on estimating the soluble solids content (SSC) of bagged and non-bagged apples using spectral data. The study developed and compared six models, concluding that the ACO-PLS model was the most accurate, achieving correlation coefficients (R) of up to 0.96 for bagged apples. This approach provided a quick and non-destructive way to assess apple sweetness and showed that non-bagged apples tend to have higher SSC. These findings suggest that spectral techniques can serve as effective tools for grading fruit quality, and highlight the role of data preprocessing and feature extraction (e.g., PCA and ant colony optimization) in enhancing model performance.

Lastly, a non-destructive method was introduced for classifying the sweetness of Khao Tang Kwa pomelos using machine learning with acoustic and image processing. The LightGBM model achieved a 93.86% accuracy and 0.98 AUC score. Interestingly, gland density and peel thickness were found to have a negative correlation with sweetness. Even when using a reduced feature

set, the model maintained strong performance, suggesting its suitability for real-world use. However, future studies could investigate whether this method can be adapted for other pomelo varieties or integrated with mobile sensing systems for broader agricultural applications (Sarakum and Sukpancharoen, 2025).

Aesthetic and Physical Features of Sweet Gold Watermelon

Numerous studies have explored the correlation between visual and morphological traits of fruits with internal quality indicators such as sweetness, ripeness, and market value. Image processing and machine learning have been used more and more to study visual characteristics, such as color, texture, shape, and size, in order to provide easy-to-use non-invasive evaluation techniques.

As stated by Zhu et al. (2022), the high precision of a deep learning-based image analysis model (Mask R-CNN) was demonstrated in extracting tomato fruit phenotypes. The method successfully detected features like fruit color, shape (vertical/horizontal diameters), locule number, and pericarp thickness, achieving a precision of over 0.95. However, while the model showed strong performance in visual trait extraction, this past study did not directly correlate these features with sweetness or other internal quality measures, indicating a gap in linking appearance with internal content.

A real-time solution was contributed through the development of a citrus fruit grading machine powered by a lightweight CNN model called SortNet. This system classified fruits into “accept” or “reject” categories based on visual surface features and weight. It presents a practical advancement in automated

sorting but focuses primarily on surface defects and shape rather than sweetness or internal attributes. While its success in real-time deployment is notable, there is an opportunity to integrate internal quality indicators such as sugar content or firmness using hybrid models or sensor fusion (Dubey et al., 2023). Further attempts to categorize fruits were invested according to the external morphology using methods like GLCM in conjunction with SVM and Discrete Wavelet Transform (DWT). The results showed improved accuracy (up to 97.2% for pomegranates) when color and texture features were combined. Interestingly, this approach uniquely excluded color in some experiments and still achieved high accuracy using only the hue component. While this past work reinforces the importance of external traits, it again lacks direct association with measurable internal qualities like sugar levels or firmness (Chandra et al., 2024).

Through the combination of SVM and Vision Transformer (ViT) feature extraction, the study was able to obtain an impressive classification accuracy of banana types to 99.86%. The work stands out for its advanced model that captures fine-grained features, even at early fruit development stages. However, like other image-based studies, it does not establish a direct relationship between these features and sweetness. There is a clear research gap in integrating these high-accuracy models with physiological traits such as sugar content or antioxidant levels, which are vital for quality assessment (Ergun, 2025). Compare that to, a study that directly correlated skin color and textural traits with sweetness in pawpaw. Using traditional instruments such as the Hunter LAB color meter and digital refractometer, the past study discovered strong positive correlations between color parameters and total soluble solids

(TSS), a proxy for sweetness. Skin a* values and hue angles were notably predictive of ripeness and sweetness, showing that visual cues can relate to internal traits. However, this study relied on manual instruments and statistical analysis, leaving room for future research to automate this process using image processing and machine learning techniques (Adainoo et al., 2023). Similarly, digital image processing and SVM classification was applied to analyze the quality of Manila mangoes based on maturity and peel damage. The method achieved 86% accuracy and demonstrated how texture and color can be used to detect fruit defects and maturity. Still, the model did not include TSS or direct measures of sweetness. The integration of such internal quality indicators with visual traits using non-invasive techniques remains an open opportunity for future work (Valverde et al., 2022).

Lastly, Alenazi et al. (2020) demonstrated that non-destructive near-infrared (NIR) spectroscopy can predict internal tomato quality traits such as lycopene, β-carotene, total phenolics, and flesh firmness with moderately high accuracy. Although the approach does not use visual traits, the findings suggest that combining NIR spectral data with image-based morphological features can enhance overall fruit quality assessment, particularly for internal attributes like sweetness.

Image Processing Techniques in Agriculture

Recent studies have shown great interest in computer vision for assessing fruit quality, using advanced image processing and machine learning methods to support agricultural actions.

Convolutional Neural Networks (CNNs) were applied using TensorFlow and Keras to classify fruits from images. The outcome suggests CNNs can help improve fruit quality inspection, reduce waste, and enhance compliance with food standards, all while supporting environmental and economic goals. However, the study also highlighted challenges such as the need for model optimization and the limitations of hardware used in embedded systems. The past study emphasized the importance of refining CNN architectures and deploying more efficient systems for real-time sorting and grading in agricultural environments (Boukli and Jebbor, 2024). An alternative to CNNs was proposed by using Vision Transformers (ViT), especially suited for cases with limited data and computing power. This past study demonstrated that ViTs can reach similar performance levels as CNNs while requiring fewer training samples, achieving 90% accuracy with only one-third the data needed by CNNs. This makes ViTs highly promising for remote or low-resource agricultural settings. This approach simplifies the model development process, enabling its use even on standard computers without GPU access. The future work focuses on applying ViTs for harder classification tasks and for object detection or segmentation to locate defects more precisely in fruit images (Knott et al., 2022).

Another advancement, RGB images and eight machine learning models were used to assess blueberry quality, focusing on firmness and soluble solids. The study found that RGB images moderately correlated with quality traits, but integrating ML significantly improved accuracy (R^2 up to 0.99). The study stands out for using mobile devices to collect data, offering a low-cost, non-destructive solution. This shows strong potential for supporting real-world decisions in harvesting and quality grading. However, the work is limited to two quality traits,

and the study recommends expanding to other fruit types and qualities in future studies (Barbosa Jr. et al., 2025). This past study explored how computer vision supports agricultural automation in six major areas, including fruit quality testing. The review stressed that the integration of computer vision with deep learning and spectral analysis holds the key to making agricultural systems more intelligent and coordinated. However, it emphasized the need for large, annotated datasets and more robust AI models. The recommendations include combining computer vision with traditional imaging and exploring more intelligent equipment to solve real-time monitoring and quality testing issues on farms (Tian et al., 2020).

Kumar et al. (2024) offered a technical review of various image-processing methods, such as morphological operations, texture analysis, and deep neural networks. These findings highlight the value of combining multiple vision techniques to address real-life farming challenges. The past study showed how image pre-processing and feature extraction can help build accurate classification systems. It recommends continued research on the real-world deployment of these techniques, particularly for smallholder farmers in developing countries, to ensure the solutions are accessible and sustainable.

In another recent study, oil palm leaf disease detection was studied using PCA and an Artificial Neural Network (ANN) classifier. With a high accuracy of 99.67%, the system demonstrated the potential of using color histograms and ANN to distinguish between healthy and infected leaves. The experiments showed ANN was more effective than other classifiers like SVM or decision trees. While the results are promising, the model relies heavily on carefully segmented input, indicating the need for automation in image pre-processing

in future research (Hamdani et al., 2021). Mask R-CNN and 3D photogrammetry were used to extract detailed fruit traits like maturity and volume for blueberries. The model achieved 97.3% accuracy in detecting berry count per cluster and a strong R^2 score for estimating maturity. This approach combines 2D and 3D data for more precise fruit monitoring, making it ideal for estimating harvest timing. It recommends expanding the system to other fruits and improving the 3D reconstruction pipeline for large-scale use (Jiang et al., 2021).

Lastly, Sheng et al. (2023) addressed the challenge of segmenting fruits in complex environments using an edge-guided model. The EdgeSegNet model integrates boundary-aware modules to sharpen object edges and achieves high segmentation accuracy (MIoU over 0.90). This is crucial for dealing with real orchard conditions like fruit overlap and leaf shading. Future work should improve generalization across fruit types and lighting conditions.

Machine Learning in Fruit Quality Prediction

In recent years, various machine learning and deep learning approaches have been developed to predict the ripeness or sweetness of fruits based on visual data. These methods aim to provide accurate, fast, and non-destructive techniques for fruit quality assessment, especially important in agriculture, supply chains, and retail.

One of the widely used methods is the Convolutional Neural Network (CNN), which can classify fruit images into categories such as fresh or rotten, or identify different stages of ripeness. CNN with OpenCV assesses both fruit type and ripeness through color detection. This method can identify fresh and

spoiled fruits and evaluate ripeness levels, providing a full picture of fruit quality for both farmers and retailers. However, the recent study did not go into real-time application or scalability, especially under different lighting conditions or for different fruit types (Sangeetha et al., 2024).

Aherwadi et al. (2022) focused specifically on banana ripeness classification using CNN and AlexNet. The study showed that CNN performed better across all datasets, achieving up to 99.36% accuracy with image augmentation. This highlights the importance of dataset preparation in visual models. While the model gave excellent results, the study was limited to bananas. It suggested applying the model to other fruits like mango, orange, and papaya in the future. The recommendation shows that while CNN is strong in feature extraction, its generalizability still needs testing on a wider range of fruit types. Similarly, modern deep learning models such as CNN, GAN, and Vision Transformers (ViT) were reviewed for smart agriculture tasks, including fruit classification. The paper highlighted that CNN remains the most reliable and accurate model across agricultural applications. However, the recent study emphasized that success depends heavily on quality datasets and real-time implementation. This points to a recurring gap in the field: many studies achieve high accuracy in controlled environments but struggle with real-world deployment where variables such as lighting, fruit orientation, and background can reduce model performance (Dhanya et al. 2022).

A hybrid approach was proposed by combining acoustic signals and image data to classify the sweetness of watermelons. Using multiple machine learning models, they found Gradient Boosted Trees to be the most effective with an accuracy of 92%. The inclusion of rind patterns and weight shows that

adding multi-modal inputs may improve predictions. However, the reliance on tapping sound requires manual input, which limits full automation. This past study suggests that future systems might benefit from merging visual and non-visual features for more robust sweetness prediction (Chawgien and Kiattisin, 2021).

In another past study, XGBoost and Artificial Neural Networks (ANN) were used to predict tomato fruit quality traits such as Brix content (a sweetness indicator), lycopene levels, and color values using environmental and visual data. XGBoost consistently outperformed ANN, achieving a very high R^2 of 0.98 for Brix predictions. The method included time-series analysis and feature engineering, showing how integrating temporal climate patterns with visual features can lead to more accurate predictions. However, it also found that ANN struggled with color predictions (a/b ratio), highlighting the importance of choosing the right model depending on the type of quality trait being analyzed (Helyes et al., 2024).

Synthesis and Relevance of the Study

The synthesis and relevance of this study lie in its comprehensive review of current machine learning and computer vision applications for fruit quality assessment, particularly through image-based analysis.

Literature widely supports deep learning, particularly CNNs, for accurately classifying visual traits like ripeness and detecting spoiled fruits. However, most existing models overlook internal qualities like sweetness, which are equally important for consumer satisfaction and market value (Sangeetha et al., 2024; Aherwadi et al., 2022; Dhanya et al., 2022). Some

studies explored sweetness using non-image data like acoustic signals (Chawgien & Kiattisin, 2021) or environmental inputs (Helyes et al., 2024), but these lacked full automation or real-time visual analysis. To address these gaps, this research proposes a hybrid machine learning approach that uses image-based features to estimate sweetness levels in Citrullus Lanatus (Sweet Gold watermelon). This approach builds upon current models by emphasizing internal quality prediction without destructive testing in addition to visual ripeness.

This research offers a practical, non-invasive solution for farmers, retailers, and quality management researchers by using an image-based system to relate visual features to sweetness. As demand for automated and precise agricultural technologies increases, this examination contributes to the development of greater intelligence and efficient fruit quality evaluation systems, supporting modern farming practices and reducing post-harvest losses.

Table 1. Literature Matrix

Authors (Year)	Title	Algorithm/Methods	Key Findings
Olisah et al, 2024	Convolutional neural network ensemble learning for hyperspectral imaging-based blackberry fruit ripeness detection in uncontrolled farm environment	Convolutional Neural Network (CNN)	The proposed model achieved 95.1 % accuracy on unseen sets and 90.2 % accuracy with in-field conditions.
Sarakum and Sukpancharoen, 2025	Non-destructive sweetness classification of Khao Tang Kwa pomelos using machine learning with acoustic and image processing	Light Gradient Boosting Machine (LGBM) 10-fold cross-validation	LightGBM model achieved 93.86 % accuracy and 0.98 AUC score for classification.
Chandra et al., 2024	Fruit Classification based on Shape,	Discrete Wavelet Transform (DWT), Hue	Classification performance

	Color and Texture using Image Processing Techniques	Saturation Value (HSV), Gray Level Co-occurrence Matrix (GLCM), SVM	improved significantly, reaching 94% accuracy for guava and 97.2% for pomegranate, with higher accuracy achieved when color and texture features were combined. Notably, using only the Hue component from the HSI image achieved 100% classification accuracy.
Ergun, 2025	High Precision Banana Variety Identification using Vision Transformer-based Feature Extraction and Support Vector Machine	ViT model, 5-FCV, SVM, GBM, MLP and CNN, metrics including CAR, F1 score and RCL	The study achieved very high classification accuracy, reaching 99.86% on the BananaSet and 99.70% on the BananaImageBD, surpassing previous benchmarks by 1.77%.
Adainoo et al., 2023	Correlations between color, textural properties and ripening of the North American pawpaw (Asimina triloba) fruit	Hunter LAB color meter (Chroma Meter CR-410, Konica Minolta), digital refractometer (HI96800, Hanna Instruments) at ~25 °C, texture profile analysis (TPA), Principal component analysis (PCA), hierarchical cluster analysis (HCA) and Pearson's correlation analysis	The total soluble solids content (Brix) of the pulp of the ripe fruits was 21.52 ± 0.94.
Knott et al., 2022	Facilitated machine learning for image-based fruit quality assessment	Vision Transformers (ViT), Convolutional Neural Networks (CNNs)	The study achieves a classification accuracy within 1% of the top-performing CNN while requiring three times fewer training samples to reach 90% accuracy.
Hamdani et al., 2021	Detection of oil palm leaf disease based on color histogram and supervised classifier	Principal Component Analysis (PCA), KNN, Naïve Bayes, C.45, Decision tree, SVM, and ANN	The evaluation achieved high performance, with sensitivity of 99.3%, specificity of 100%, and overall accuracy of 99.67%.

Jiang et at., 2021	Three-dimensional photogrammetry with deep learning instance segmentation to extract berry fruit harvestability traits	Mask R-CNN	The model achieves 97.3% accuracy in determining fruit numbers per cluster. Cluster maturity prediction shows strong performance with an R-squared of 0.908 and an RMSE of 0.068. For clusters with over 10 berries, the berry volume estimation has a low error, with an RMSE of 2.92 cm^3, translating to less than 0.292 cm^3 error per berry.

Conceptual Framework of the Study

This section presents the conceptual framework of the study on analyzing the complete process of developing a regression model for Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') using visual data and machine learning techniques. It is divided into three main phases: Data Collection/Acquisition, Model Development/Training, and Model Evaluation and Testing.

In the first phase, data are collected using a high-quality camera to capture external features such as color, texture, and shape, and a Brix refractometer to measure the Brix value representing internal sweetness. The collected data are divided into training, validation, and test sets, followed by annotation and image preprocessing to ensure data quality. The second phase focuses on developing predictive models using several machine learning algorithms, including Convolutional Neural Network (CNN), Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). A hybrid approach is implemented using EfficientNetB-0, where

annotated image features serve as inputs and measured Brix values as target outputs. In the final phase, the trained models are evaluated using standard performance metrics such as R^2 Score, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The best-performing model is selected as the final regression model, and the results are demonstrated through a prototype and visual plots. The overall framework provides a fast, non-destructive solution for fruit quality assessment for farmers, retailers.

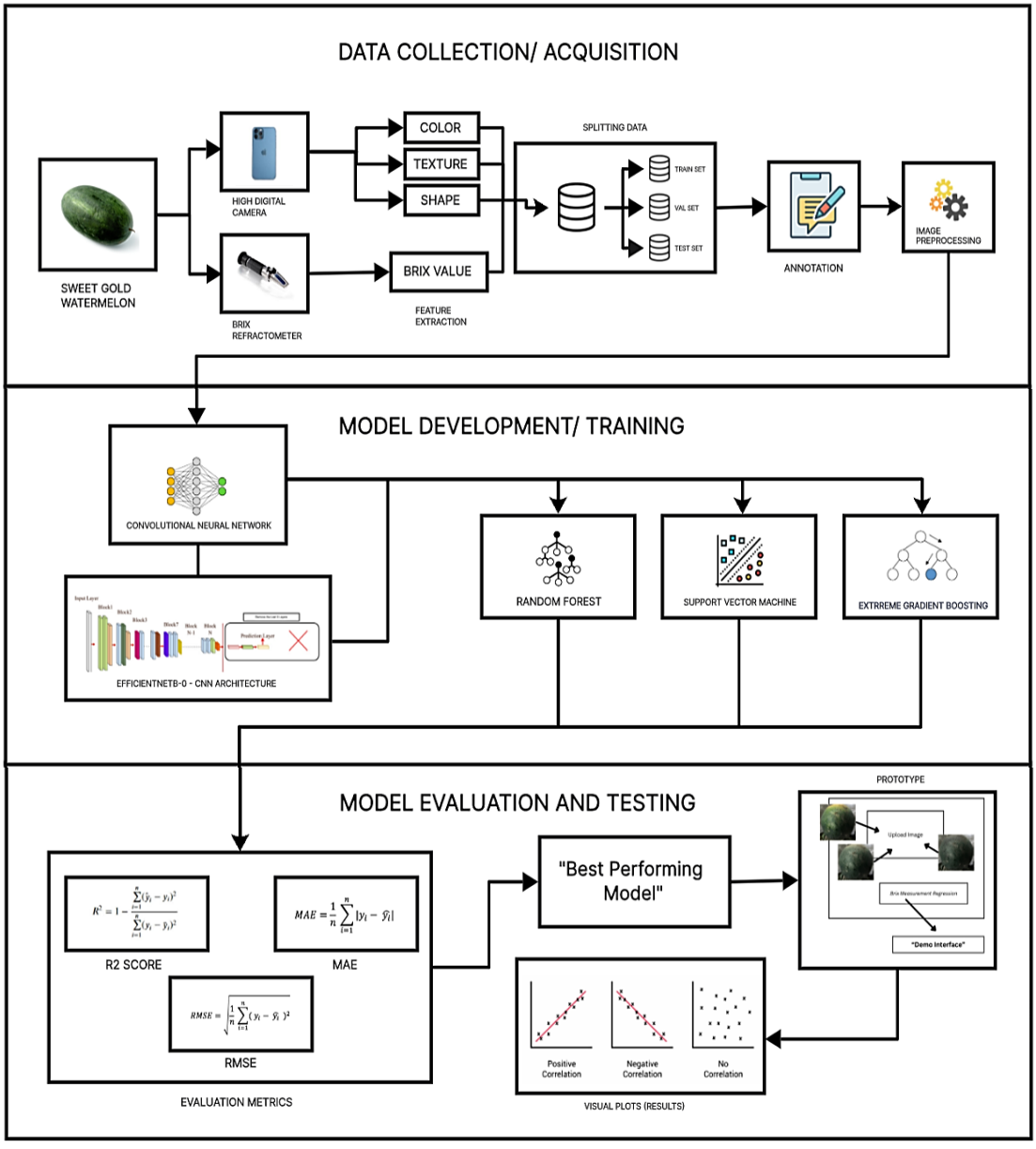


Figure 1. Concept of Developing Model for Studying Sweet Gold Watermelon

CHAPTER 3

METHODOLOGY

This chapter outlines the systematic steps undertaken to investigate the correlation between the aesthetic traits and the sweetness level of Sweet Gold watermelons (*Citrullus lanatus* 'Sweet Gold') using machine learning and image processing. The process is designed to provide a practical, replicable methodology rooted in non-destructive testing and visual data analysis.

Experimental Design

This section adopts a quantitative-experimental research design, specifically focused on supervised machine learning, to develop a regression model that estimates the sweetness level of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') based on visual features. This study involves directly comparing multiple hybrid machine-learning models to determine the most effective algorithm for this regression analysis.

The experiment involves capturing 900 high-resolution images from 150 harvested Sweet Gold watermelons, with each fruit photographed from 6 angles under consistent lighting conditions. Alongside the images, the sweetness level of each fruit is objectively measured using a manual Brix refractometer, producing continuous target values used for regression analysis. To train machine learning models and assess their generalization, the dataset is split into training, validation, and test sets at 70:15:15. Convolutional Neural Networks (CNNs), Random Forest (RF), Support Vector Regression (SVR), and XGBoost Regressor (XGB) belong to the machine learning techniques that are used and evaluated. Feature extraction focuses on visual traits like color,

shape, and texture. Image preprocessing techniques and data augmentation are applied to improve model robustness. The models are assessed using standard evaluation metrics such as R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) to determine their predictive accuracy and reliability. This experimental approach enables a non-destructive, efficient, and reproducible method for estimating internal fruit quality solely from external appearance, aligning with industry needs and sustainable agricultural practices.

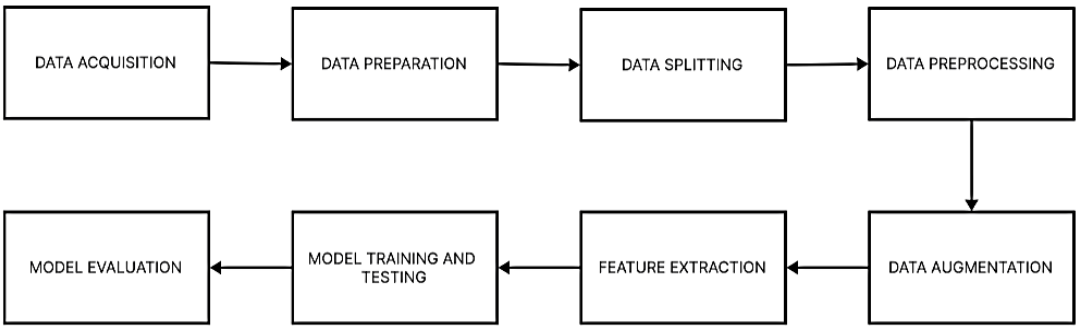


Figure 2. Workflow for Sweetness Prediction of Sweet Gold Watermelon

Study Area

The study was conducted at a local farm in Barangay Wakat, Nagcarlan, Laguna, at an elevation of approximately 31 meters above sea level, home to around 1,700 residents as of the 2020 census (PhilAtlas, 2020). The Calauan–Nagcarlan Road extends through this agro-rural area, which benefits from a highland environment and irrigation from the nearby Santa Cruz River basin, allowing a variety of agricultural activities. Agricultural fruits like watermelon, which need well-drained soils and consistent moisture levels, depend heavily on the local climate and water supplies (Philippine Morning Post, 2021).

This local farm is managed by a farmer from Victoria, Laguna, which is commonly known as the "Duck Capital" of Laguna because of its solid duck and poultry fields. Victoria and Nagcarlan share a tropical climate, which is suitable

for growing watermelon and other crops (Laguna Living, 2007). The region's long history of horticulture is highlighted by insights from farmer interviews in Victoria, which also provide essential background information on watermelon sweetness deviations associated with environmental and seasonal factors (Farmer Interview, 2025).

Regionally, this study is further contextualized by agricultural support from neighboring municipalities, such as Calauan, which actively promotes fruit production and innovation through institutions such as the Laguna Agricultural Trading Center and Bayer's demonstration farms in Barangay Dayap (PNA, 2023; UPLB, 2022). These areas host seasonal agri-training and sustainable farming initiatives, making them conducive to implementing non-destructive, image-based technologies in quality assessment.

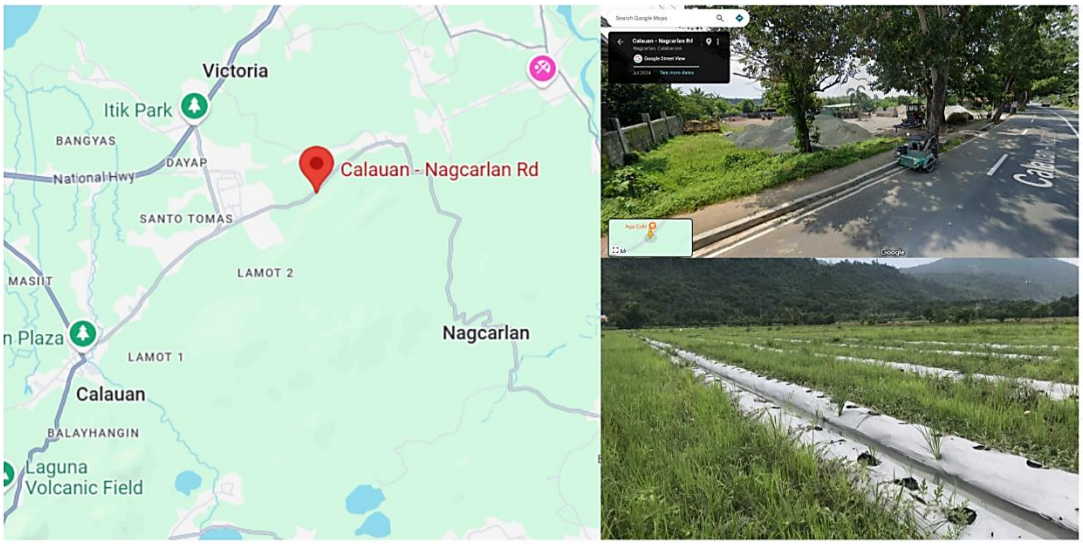


Figure 3. Brgy. Wakat, Calauan - Nagcarlan Rd, Laguna, Maps: Google Maps

Materials

This study utilizes various materials to capture and measure the aesthetic quality of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold'). These materials are categorized into hardware and software specifications.

Hardware Specifications. The hardware tools were chosen to ensure accurate data collection, precise measurement of fruit attributes, and reliable computational support for model training. High-resolution imaging captured detailed visual features, while instruments like the Brix refractometer, weighing scale, and measuring tape provided precise physical and sweetness measurements. A high-performance laptop enabled efficient image processing, data management, and training of hybrid machine learning models.

Table 2. Hardware Specifications

Tool	Purpose
High-Resolution Camera	An iOS camera with enough resolution is used to capture high-quality images of Sweet Gold watermelons.
Brix Refractometer	The Sugar Refractometer with ATC (IC300010 model), a manual refractometer, is a simple yet reliable instrument for estimating the percentage of soluble sweetness.
Weighing Scale	This tool is used to measure the weight of each Sweet Gold watermelon.
Measuring Tape	This tool is used to measure the height and width of the Sweet Gold watermelons.
Laptop Device	The Acer laptop, equipped with an AMD Ryzen 5 7520U with Radeon Graphics (2.80 GHz) processor and 16.0 GB RAM (15.2 GB usable), is used for documentation, image processing, data analysis, and model training.

Software Specifications. The software tools were selected to support image preprocessing, machine learning model development, and evaluation. Libraries and frameworks enabled tasks such as data cleaning, augmentation, model training, and performance assessment to ensure accurate and reliable predictions.

Table 3. Software Specifications

Tool	Purpose
Python	This tool serves as the primary programming language for coding and data processing.
Visual Studio Code (VSC)	This tool is used as the IDE for writing and testing data preparation code.
Google Colab	This tool is used for model training and execution with visual outputs.
Microsoft Excel	This tool is used for data recording and organization, including Brix measurements and image metadata.
Microsoft Word	This tool is used for documentation tasks.
iLoveIMG	This online tool is used to convert image formats from HEIC to JPEG for compatibility.
OS	This library is used to handle directory and file path operations.
OpenCV	This library is used for image preprocessing and feature extraction, such as resizing, cropping, and enhancement.
NumPy	This library is used for numerical computations and handling arrays.
Pandas	This library is used for organizing and processing tabular data, including image metadata.
Matplotlib	This library is used for creating visualizations.
Random & re	These libraries are used for generating random values and handling string operations.
Scikit-learn	This library is used for dataset splitting, evaluation metrics, and applying classical machine learning algorithms like Random Forest and SVR.
TensorFlow & Keras	These libraries are used to build the Convolutional Neural Network (CNN) model.
XGBoost	This library is used for implementing gradient boosting regression.
Math	This library is used for mathematical calculations.
Joblib	This is used for saving and loading trained models.

Data Acquisition Implementation

Data acquisition begins with an extensive inspection to capture all visual attributes of each Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold'). Six angles are used to capture high-resolution images that show the differences in color, texture, and shape of each crop. In addition to data imaging, precise measurements of weight, height, and width are taken using calibrated instruments to provide detailed metadata for each sample. To evaluate sweetness, each watermelon is carefully pierced, and the juice is extracted and measured using a manual refractometer to determine the Brix value. Every observation and measurement is systematically documented, creating detailed and usable data for subsequent analysis.



Figure 4: Data Collection Procedures of Sweet Gold Watermelon

Sweetness values (target variable) are measured using a handheld refractometer (Instrument Choice, *n.d.*). A manual Brix refractometer was then used to assess each sample's sweetness level and calculate its sugar amount. The table (Table 4) below shows the Brix scale used to determine sugar content of a Sweet Gold watermelon.

Table 4. Sweetness/Brix Measurement Basis (Santos, 2010)

Brix Scale	Description
8-10	Moderately sweet
11-13	Acceptably sweet
14-16	Very sweet

After data collection, high-resolution images are first annotated and standardized to ensure consistency across all samples. Each image was mapped to its corresponding metadata, including weight, height, and width, to maintain strong correlations between physical and visual data. All images captured from the same watermelon share a common Fruit ID and identical physical metadata. The first 20 Fruit ID results of connecting each image to its corresponding metadata, which is taken from a spreadsheet, are displayed in the figure (Figure 5) below.

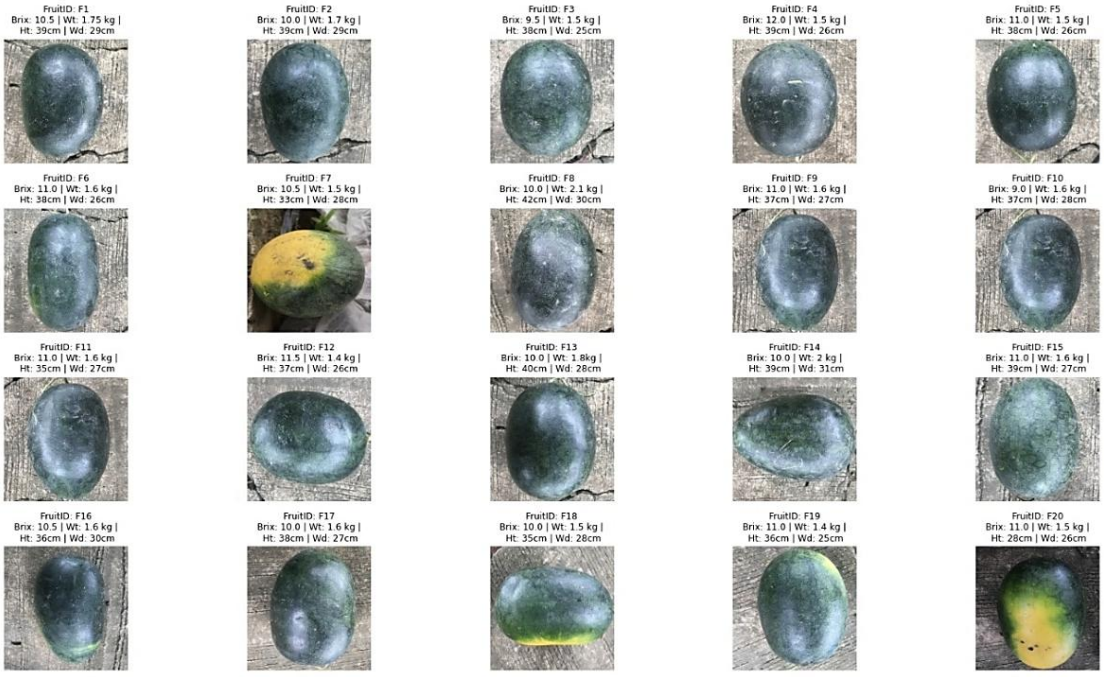


Figure 5. Data Mapping Process

Data Splitting

The dataset was split at the fruit level rather than the image level, since images were initially mapped to their corresponding fruit IDs, along with their Brix levels and metadata. This approach ensures that all images of the same fruit are allocated exclusively to either the training (70%), validation (15%), or test set (15%), preventing data leakage caused by similar images being distributed across different sets. A total of 105 fruits were used for training, 22 for validation, and 23 for testing, yielding 630 training images, 132 validation images, and 138 test images. This fruit-level split allows the model to learn generalizable patterns in fruit characteristics and sweetness (Brix) rather than memorizing specific fruits

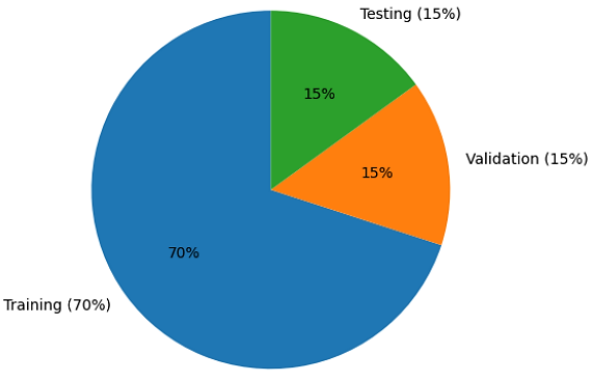


Figure 6. Data Splitting, Pie Chart

Data Preprocessing

Preprocessing was then applied to enhance data quality and reduce computational complexity, addressing common issues such as noise, inconsistency, and class imbalance in the raw dataset (Ultralytics, 2025). Background removal is performed using the GrabCut algorithm, which segments the foreground from the background via iterative graph-cut

optimization, allowing the model to focus on the fruit itself (Rother et al., 2004). This method models the image as a graph and iteratively refines pixel classification by minimizing an energy function, allowing accurate isolation of the watermelon from complex or uneven backgrounds. The figure (Figure 7) below shows that each image is subsequently resized to 224×224 pixels and normalized to a [0,1] range.

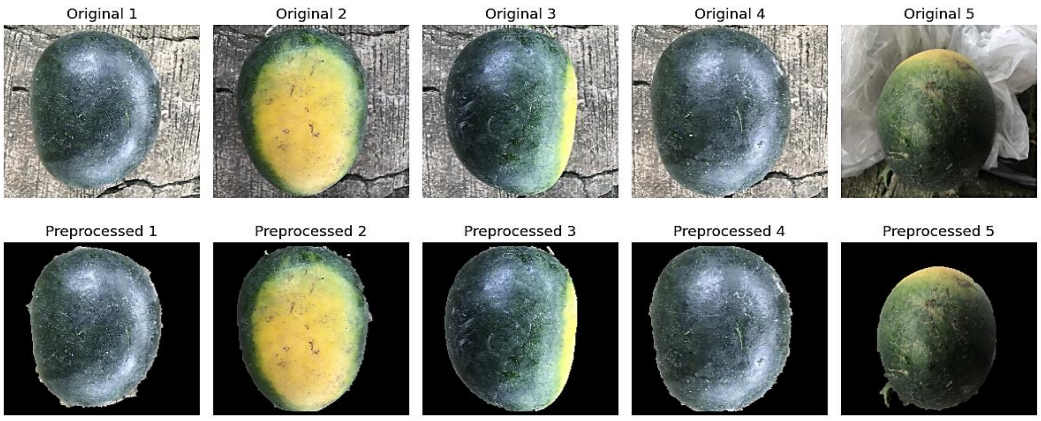


Figure 7. Image Preprocessing, Grabcut

Data Augmentation

However, data augmentation techniques were applied only to the training set to increase dataset diversity while preserving the integrity of the validation and test sets. For each original training image, four additional samples were generated, resulting in a fivefold expansion of the training data. With 105 training fruits corresponding to 630 original images, this process increased the total number of training images to 3,150 images (630 × 5).

Formula: $$N_aug = N_orig \times (1 + A) \quad \textbf{(1)}$$

The applied data augmentation techniques included horizontal flipping, vertical flipping, and rotations of 90° and 180°. These transformations generated additional training samples while preserving the original visual

characteristics of each fruit. New fruit IDs were assigned when necessary to maintain proper data organization and traceability. This augmentation strategy increased dataset diversity, improved model generalization, and minimized overfitting without introducing data leakage between training, validation, and test sets. The figure (Figure 8) below shows examples of training images after applying data augmentation techniques.

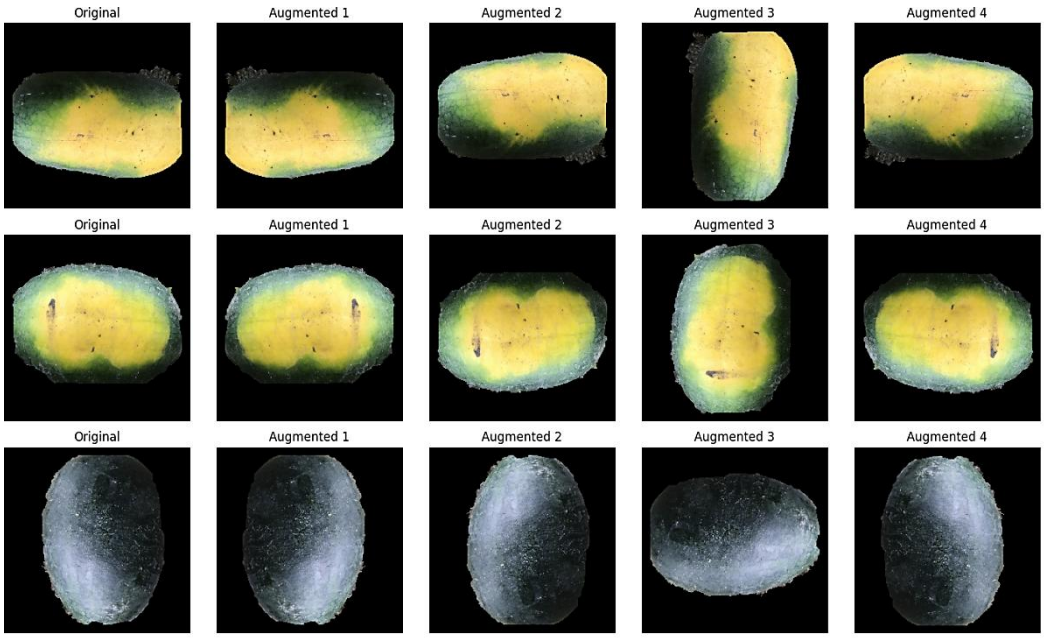


Figure 8. Data Augmentation Process, “Original vs. Augmented”

Model Development

Machine learning algorithms are essential for determining the correlation between the sweetness level in Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') and its aesthetic features. These algorithms enable the system to discover patterns and connections between image attributes, including color, shape, and texture, and their corresponding Brix values, because sweetness is a subjective attribute that cannot be seen directly. Comparing multiple algorithms is necessary to identify which model provides the most accurate and

reliable regression. Each algorithm has different strengths in handling image features and learning complex patterns, so a fair comparison ensures that the best-performing model is reasonable for practical use.

A model architecture is used for feature extraction to obtain visual patterns from images, which are then used to predict sweetness levels. These extracted features serve not only for direct prediction but also as inputs for hybrid models, combining them with multiple algorithms to enhance learning from complex patterns. This solution maximizes predictive performance and provides a solid basis for comparing model configurations to choose the best approach for valid sweetness regression by leveraging feature extraction with hybrid approaches that benefit from significant visual information from different algorithms. While the validation set supported hyperparameter configuration and model selection, including optimization of CNN architectural parameters (such as dropout rates and dense layer sizes) and tuning of hybrid regression models, the training set served to learn model parameters. The test set was only used for the final assessment of performance on unseen samples. All models can be trained and assessed under uniform conditions due to this standardized data split, enabling a fair comparison of regression performance. Additional models incorporating metadata, including weight, height, and width, were trained to investigate the correlation between external fruit characteristics and sweetness (Brix) values. Although these models constitute hybrid approaches, metadata was not used to select the optimal model for the deployed system, which relies exclusively on image input. These models use metadata only for statistical purposes, offering insights into the relationships between physical attributes and Brix.

Machine Learning Algorithms

This section describes the approaches used to analyze the data and the algorithms applied to determine the correlation between the sweetness level and the visual features of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold'). It provides a basis for comparing model performance and for determining the most suitable approach to achieving the study's objectives.

Convolutional Neural Network. An algorithm identified as a powerful deep learning architecture useful for computer vision tasks, capable of extracting layered visual data from raw images. In past research, Convolutional Neural Networks (CNNs) were commonly structured into two main parts: a feature extractor composed of convolution and pooling layers, and a classifier composed of fully connected layers. The feature extractor turns raw images into smaller feature representations, while the classifier uses these to make predictions, such as identifying objects or detecting diseases. CNNs are utilized in this study as a hybrid approach, with a focus on regression rather than classification. The final layers are built to generate continuous Brix regression, rather than different categories. The architecture leverages a pre-trained model, EfficientNetB0, as the backbone for feature extraction, followed by dense layers for predicting sweetness levels.

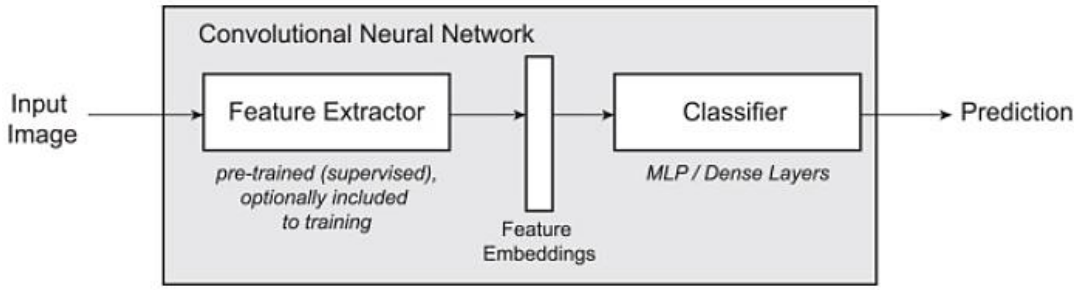


Figure 9. Convolutional Neural Network (Vashwani et al., 2017 & Knott et al., 2023)

Random Forest. An algorithm identified as a non-parametric ensemble method that combines numerous decision trees to identify complex patterns in data and has proven solid forecasting performance. In past research, CNN–RF hybrid models were used, in which Random Forests replaced conventional fully connected layers to produce final predictions, while Convolutional Neural Networks (CNNs) automatically extracted high-dimensional image features. Because the RF classifier performed better with the extracted features than dense layers alone, this method reduced overfitting and increased accuracy, especially when the dataset was small. In this study, the same hybrid approach is applied, with EfficientNetB0 used as the feature extractor to capture layered visual data from Sweet Gold watermelon images, and Random Forest utilized to model the correlation between external features and continuous Brix regression.

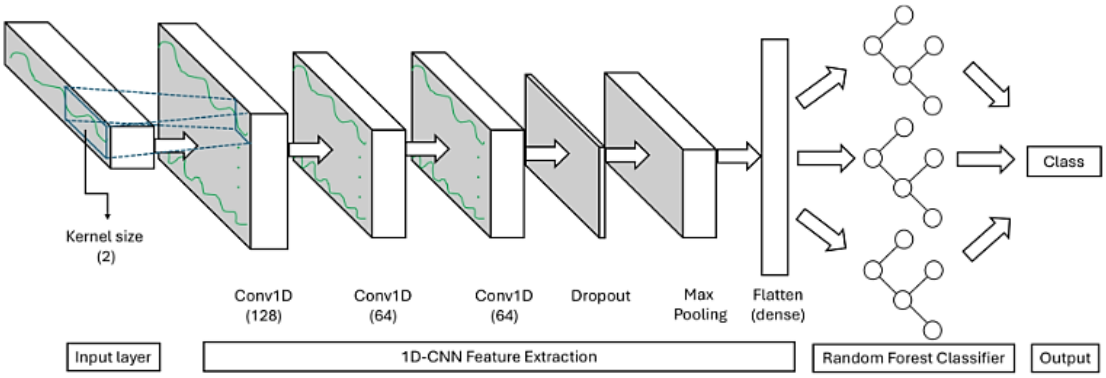


Figure 10. Random Forest, Hybrid CNN feature extraction (Sudiana et al., 2025)

Support Vector Regression. This algorithm is a statistical learning theory-based technique that builds on the ideas of Support Vector Machines (SVM), which Vapnik originally developed for two-class classification problems. The SVR, an upgraded SVM, performs well for time-series prediction. In past

research, SVR and CNNs were combined by using CNNs as feature extractors, as always, and replacing the fully connected layers with an SVR classifier for final predictions. Several formulas were used to extract and refine features before they reached the SVR classifier. Formula (1) calculated the dimension of each convolutional layer's output, ensuring that the feature map sizes remained consistent as the network deepened. Formula (2) describes the pooling process, where max-pooling is applied to reduce the dimensions of the feature maps while preserving the most important values. Formula (3) defined the relationship between kernel counts and output dimensions, helping determine how many filters were applied at each stage. A sparse regularization penalty, represented in formulas (4) and (5), was added to the error function to reduce the number of parameters and improve computational efficiency by encouraging sparsity in the learned weights (You et al., 2017). In this study, while EfficientNetB0 is adopted as the CNN backbone, the same hybrid approach is applied across models: CNN handles feature extraction, and alternative machine learning methods (SVR) replace the fully connected layers to improve generalization and prediction accuracy.

Formula 1: $$D_{j_1} = D_{j_1-1} - \mathrm{C}_i + 1,\ j_1 = 2,4,6;\ \ i = 1,2,3 \quad \textbf{(2)}$$

Formula 2: $$D_{j_2} = \frac{D_{j_2-1}}{P_i},\ j_2 = 3,5,7;\ \ i = 1,2,3 \quad \textbf{(3)}$$

Formula 3: $$D_8 = n_k \times D_7 \times D_7 \quad \textbf{(4)}$$

Formula 4: $$\widetilde{E}^n = E^n + \sum_{i,j} |\sigma_{\mathrm{ij}}| \quad \textbf{(5)}$$

Formula 5: $$\frac{\partial \widetilde{E}}{\partial W^l} = \frac{\partial E}{\partial W^l} + \frac{\partial \varphi}{\partial W^l} \quad \textbf{(6)}$$

Extreme Gradient Boosting. A machine learning algorithm that improves prediction accuracy by combining decision trees with regularization

to prevent overfitting. It has been widely used in structured data tasks due to its efficiency and robustness. In past research, the Gradient Boosted Decision Tree (GBDT) method builds multiple decision trees in sequence, where each tree learns from the errors of the previous one. XGBoost enhances this approach by optimizing tree construction and adding regularization to achieve better accuracy and prevent overfitting. In this study, the same principle of regularized boosting is applied, but in a new context. Instead of working only with structured tabular data, this research integrates XGBoost into a hybrid framework that uses visual features extracted by EfficientNetB0 as inputs to predict continuous sweetness levels of post-harvest watermelons.

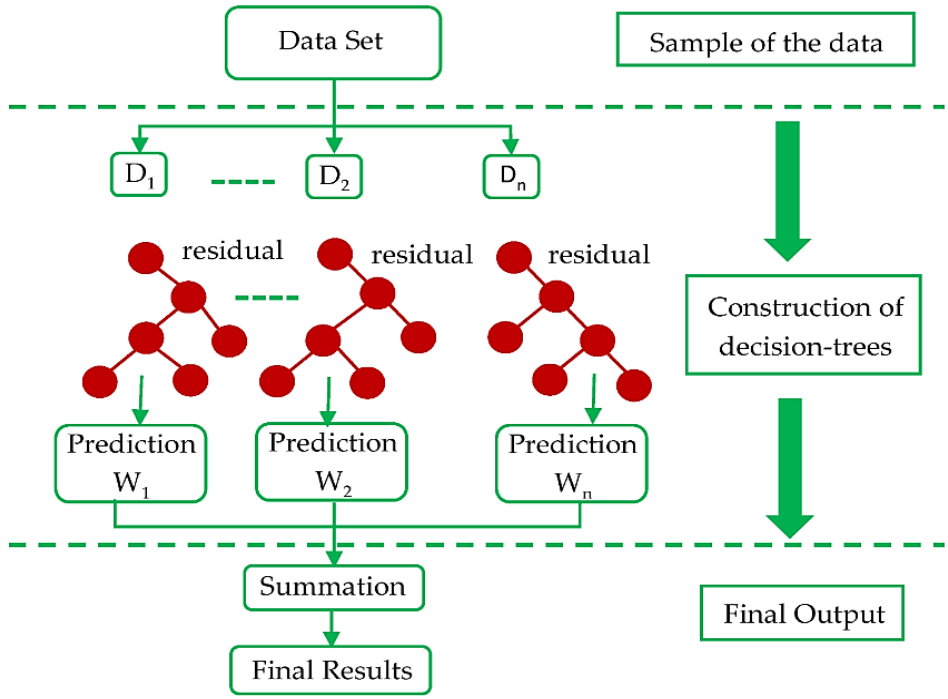


Figure 11. Structure of XGBoost Algorithm (Amjad et al., 2022 & Chen & Guestrin)

Theorem Foundations

This study is grounded in Statistical Learning Theory (SLT), which provides a mathematical framework for understanding how models learn

patterns from data and generalize to unseen samples (Vapnik, 1998). SLT supports the use of supervised learning methods employed in this research, including CNNs, Random Forests, SVR, and XGBoost, by emphasizing error minimization and generalization.

In this study, SLT supports the development and optimization of regression models for predicting watermelon sweetness from visual features, ensuring reliability in agricultural quality assessment.

Model Evaluation

To evaluate the model, these metrics are used: R-squared (R^2) Score to determine how well the regression explains the variance in actual sweetness measurements, Mean Absolute Error (MAE) to determine the average magnitude of prediction errors in Brix regression, and Root Mean Squared Error (RMSE) to assess how far predicted sweetness levels deviate from the actual measurements. These metrics provide a comprehensive evaluation of model accuracy and reliability and are used to compare the performance of different machine learning algorithms and to help choose the best model for sweetness regression.

Formula: $$R^2 = 1 - \frac{\sum_{1=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{1=1}^{N} (y_i - \bar{y})^2} \quad \textbf{(7)}$$

Formula: $$MAE = \frac{1}{N} \sum_{i=1}^{N} \left| y_i - \hat{y}_i \right| \quad \textbf{(8)}$$

Formula: $$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - p_i)^2} \quad \textbf{(9)}$$

Ethical Considerations

The study anticipates the sweetness regression of Sweet Gold watermelon using machine learning and image analysis techniques. All watermelons used in the study come from local farms with proper authorization, ensuring that agricultural products are obtained legally and respectfully. After image collection and sweetness measurement, the samples remain intact, and no invasive procedures are carried out.

The dataset is carefully selected to reflect a variety of watermelon appearances, including differences in size, shape, color, lighting, and natural distortions, to promote ethical AI development. This diversity helps reduce prediction bias and supports fairer model performance. Potential limitations, such as prediction errors caused by overlapping visual characteristics, are acknowledged, and these challenges are reported transparently during the analysis and interpretation of results. This approach provides a supportive tool to help farmers and post-harvest workers assess watermelon quality more efficiently, reducing reliance on manual inspection. It aims to save time and labor while remaining simple, affordable, and accessible, ultimately improving productivity and promoting sustainable farming practices.

Overall, this approach provides a practical tool for farmers and post-harvest workers, enabling more consistent assessment of watermelon quality while reducing dependence on manual inspection. By offering a simple, cost-effective, and accessible solution, the method aims to improve operational efficiency, save labor, and promote sustainable agricultural practices, ultimately supporting decision-making in both field and post-harvest environments.

CHAPTER 4

RESULT AND DISCUSSION

This chapter presents the experimental findings of the study, focusing on the evaluation of machine learning models in predicting the sweetness level (Brix value) of Sweet Gold watermelons (*Citrullus lanatus* 'Sweet Gold') using external visual features. The results highlight both individual models and hybrid approaches that integrate deep feature extraction with algorithms such as Convolutional Neural Network (CNN), Random Forest (RF), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGBoost).

Analyses are organized around the research objectives, particularly the correlation between fruit aesthetics such as color, texture, and shape with measured sweetness levels. The discussion further interprets these results to assess regression performance, identify the most reasonable hybrid model, and provide insights into the potential of non-destructive, image-based methods for fruit quality assessment.

High-resolution Images of Sweet Gold Watermelon

A total of 150 Sweet Gold watermelons were imaged from six angles per fruit, yielding a diverse and comprehensive visual dataset that captures the full range of external characteristics. All images were captured under a controlled setup, with a fixed camera distance, a uniform background, and natural sunlight as the primary lighting source. This setup minimized unwanted noise such as harsh shadows, glare, uneven illumination, and background distractions. The figure (Figure 12) below shows the dataset organization, where each folder

represents a single Sweet Gold watermelon and contains six images captured from different angles to comprehensively capture its external aesthetic traits.

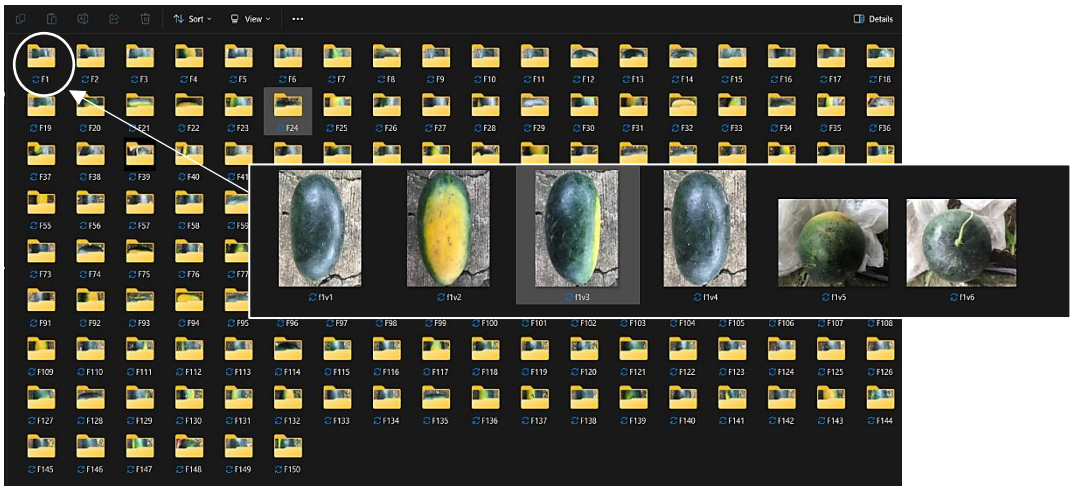


Figure 12. Folder Structure and Multi-Angle Images of Sweet Gold Watermelons

As a result, all samples exhibited consistent external aesthetic traits, particularly color distribution, surface texture, and overall shape. To further enhance the robustness of the dataset, several preprocessing steps were applied. The dataset was expanded using image augmentation, yielding 3,150 training images from the original captures. This included transformations such as rotation, flipping, and slight color adjustments, which helped the model generalize better to unseen variations. To ensure that the visual characteristics evaluated were exclusively those of the Sweet Gold watermelon and not due to environmental effects, all images were also preprocessed to remove the background and normalize its appearance.

Aesthetic Traits and Sweetness Level of Sweet Gold Watermelon

A manual Brix refractometer was used to assess each watermelon's Brix value in order to quantify its sweetness. This provided a trustworthy baseline for establishing a relationship between internal sugar concentration

and visual characteristics. Figure 13 shows the procedure of measuring the Brix value, from extracting juice from the fruit to reading the refractometer scale. The process ensured that the juice samples were handled consistently, minimizing any contamination or dilution that could affect the sugar concentration measurement. Each sample was measured individually, and the refractometer reading was recorded as the Brix value, which quantifies the fruit's sugar content. These measured Brix values served as the target variable for the subsequent predictive modeling, providing a reliable ground truth for training and validating models designed to estimate sweetness based on visual and other measurable fruit characteristics.



Figure 13. Sweet Gold Watermelon Measurement Procedure

Alongside the image data, comprehensive metadata for each watermelon was recorded and stored in an Excel file. This metadata included identifiers that map directly to each image, capturing key attributes and measurements of the fruit. By organizing this information systematically, it became possible to link visual features with quantitative data for each sample, supporting more accurate analyses and modeling. The structured metadata also facilitates

filtering, sorting, and cross-referencing within the dataset, which is crucial for subsequent machine learning tasks or statistical evaluations. The figure (Figure 14) below shows the first 20 Sweet Gold watermelon metadata entries in the Excel file: *SweetGoldData.xlsx*, illustrating the mapping between the images and their corresponding recorded attributes.

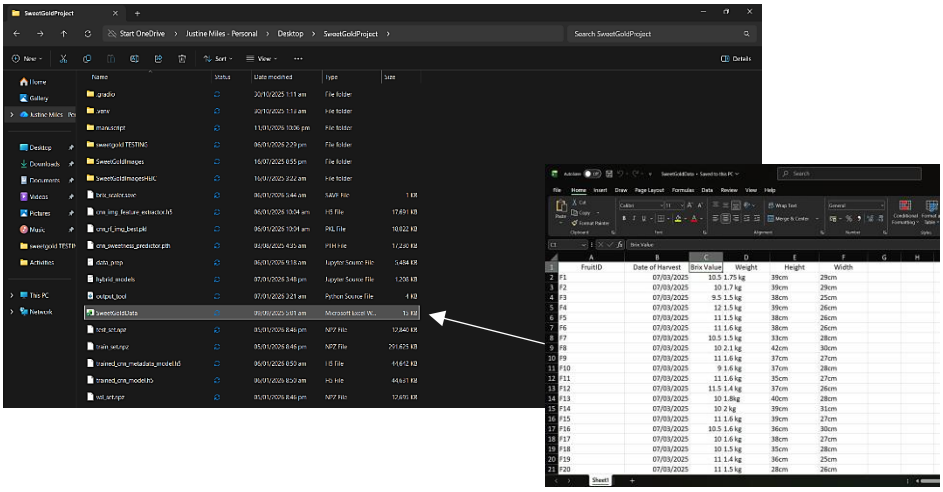


Figure 14. Sweet Gold Watermelon Metadata

The dataset was split at the fruit level to ensure that all images of the same fruit were assigned exclusively to either the training, validation, or test set, preventing data leakage caused by visually similar images appearing across different subsets. Using this strategy, 105 fruits were allocated for training, and their original images were augmented to produce a total of 3,150 training images, while 22 fruits (660 images) and 23 fruits (690 images) were used for validation and testing, respectively. This fruit-level split allowed the model to learn generalizable patterns linking fruit characteristics to sweetness (Brix) rather than memorizing individual fruits. The test set, which only included unseen fruits, was utilized to produce the final performance results, whereas the validation set was used for overfitting control and model tuning. Thus, the results fairly reflect the model's ability to predict Brix levels for new fruits under

realistic conditions. A Convolutional Neural Network (CNN) architecture, as a feature extractor, was used for learns expressive visual representations that are fed into subsequent classification tasks. Specifically, EfficientNetB0, a member of the EfficientNet family of CNN models, is employed for this purpose due to its efficient, scalable design that balances network depth, width, and input resolution via compound scaling (Mustafa and Wardhara, 2025). Each input image was converted into a high-level numerical representation by EfficientNetB0, which kept the convolutional backbone while eliminating the original classification layers. These representations encode external aesthetic attributes such as rind color variation, surface texture, and overall shape, which are visually observable characteristics hypothesized to influence or indicate internal sweetness. This demonstrates that complicated visual patterns can be effectively translated by convolutional neural networks into structured feature vectors appropriate for regression tasks. The performance is significantly benefited by the architectural design of EfficientNetB0. It was able to learn rich visual features while retaining computational efficiency by utilizing Mobile Inverted Bottleneck Convolution (MBConv) blocks and compound scaling. Additionally, the inclusion of Squeeze-and-Excitation mechanisms enabled the model to emphasize informative feature channels, improving sensitivity to subtle visual differences in watermelon appearance. These design elements help explain the strong predictive capability observed in the regression results, as the model was able to focus on specific visual cues rather than irrelevant background information (Saeed, 2023; Tan & Le, 2019).

The primary aspect of the feature extraction process was the application of global average pooling at the end of the EfficientNetB0 neural network. This

operation compressed those spatial feature maps into a 1280-dimensional feature vector by averaging activations across each channel. Mathematically, this pooling operation computes the mean activation of each feature map, enabling the model to summarize the presence of learned visual patterns across the entire image. This process reduces spatial bias while preserving semantic meaning, resulting in robust and compact image representations that contributed to stable regression performance across validation and test sets (Lin et al., 2013). These extracted deep features were further refined by the fully connected regression head, particularly the penultimate dense layer with 128 neurons, which served as a latent representation closely aligned with sweetness prediction. These formed embeddings integrate multiple visual characteristics at once, allowing the model to map external aesthetics to continuous Brix values. Higher-level features were able to particularly adapt to the unique features of Sweet Gold watermelon due to the further fine-tuning of the EfficientNetB0's highest layers. This alteration explains an observed improvement in model performance after fine-tuning, as the network learned task-specific visual patterns rather than relying just on generic ImageNet features (Yosinski et al., 2014).

Formula: $$F_{\text{GAP}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x(i,j) \quad \textbf{(10)}$$

Hybrid Machine Learning Algorithms for Best Performing Model

Table 5 presents the performance results of the CNN-only and hybrid machine learning models evaluated using R^2, MAE, and RMSE. These models combine EfficientNetB0-based image feature extraction with different

regression approaches, including Random Forest (RF), Support Vector Regression (SVR), and XGBoost, with and without additional metadata. The findings provide a direct comparison of model performance to achieve the objectives of the study and provide insights into how well external appearances taken from Sweet Gold watermelon images can be employed to predict sweetness levels.

Among all evaluated models, CNN + Random Forest (CNN + RF) achieved the best overall performance, obtaining the highest R^2 score (−0.0295) and the lowest MAE (0.1623) and RMSE (0.0388). Although the R^2 value remains negative, this model consistently produced the smallest prediction errors, indicating relatively more stable and reliable predictions compared to the other approaches. The better performance of the Random Forest approach can be due to its ensemble-based structure, which is effective at understanding non-linear patterns and reducing variance when operating on high-dimensional CNN-extracted feature vectors (Alex, 2024). The CNN-only model and CNN + XGBoost showed moderate performance, with R^2 scores of −0.1648 and −0.1331, respectively. These results reflect that the direct end-to-end regression and boosting-based technique were less accurate in predicting the underlying relationship between visual aesthetics and sweetness, even if the CNN was able to extract significant visual features.

The CNN + Support Vector Regression (CNN + SVR) model yielded poorer performance, with an R^2 score of −0.3280 and higher MAE and RMSE values compared to the CNN-only and CNN + RF models. This outcome indicates that SVR struggled to generalize from the extracted feature space, likely due to its sensitivity to feature scaling and kernel selection when handling

complex, high-dimensional data. Similarly, the CNN + Metadata model did not improve performance, producing an R^2 score of −0.2399.

The lowest-performing models were those that combined CNN features, metadata, and classical regressors, particularly CNN + Metadata + SVR, which recorded the lowest R^2 score (−1.7725) and the highest MAE and RMSE values. Additionally, the CNN + Metadata + RF and CNN + Metadata + XGBoost models showed significantly worse performance. This implies that the additional metadata did not provide sufficient useful data to enhance prediction accuracy when combined with visual features alone. Overall, a constant negative R^2 scores across all models indicate that the learned relationships between external visual features and sweetness levels were weaker than a simple mean-based baseline predictor. The findings reveal empirical proof that, even when powerful CNN-based feature extraction and hybrid machine learning approaches are used, external aesthetic factors alone do not have enough reasoning potential for predicting internal sweetness under these experimental settings.

Table 5. Hybrid Machine Learning Models Performance Results

Model	R^2 score	MAE	RMSE
CNN-only	-0.1648	0.1743	0.0439
CNN + RF	-0.0295	0.1623	0.0388
CNN + SVR	-0.3280	0.1884	0.0500
CNN + XGB	-0.1331	0.1690	0.0427
CNN + Metadata	-0.2399	0.1786	0.0467
CNN + Metadata + RF	-1.4112	0.2243	0.0908
CNN + Metadata + SVR	-1.7725	0.2478	0.1044
CNN + Metadata + XGB	-1.4735	0.2270	0.0931

Evaluation for Hybrid Machine Learning Model Performance

The R^2 score was used as the primary metric for selecting the best-performing regression model because it indicates how well the model explains the variability in the target output. A higher R^2 value reflects a better fit between predicted and actual values, making it a reliable indicator of overall model performance.

To support this evaluation, MAE and RMSE were also considered, as they measure the average prediction error and penalize larger errors, respectively. Lower MAE and RMSE values indicate improved prediction accuracy. While all three metrics provide a comprehensive assessment of model performance, R^2 remains the main basis for comparison because it directly reflects the model's ability to explain the true data trend (scikit-learn, n.d.).

Figure 15 above visualizes the comparative performance of all hybrid machine learning models for predicting the sweetness (Brix value) of Sweet Gold watermelons. The figure combines R^2 scores (blue bars) with MAE (orange line) and RMSE (green line) values to provide a holistic view of model accuracy. From the figure, it is evident that the CNN + RF model achieved the highest R^2 and the lowest MAE and RMSE, indicating the most reliable prediction among the tested models. Conversely, models combining CNN features with metadata and regressors, particularly CNN + Metadata + SVR, exhibit the lowest R^2 values and the highest error metrics, reflecting poor predictive performance. Same thought as earlier, this trend demonstrates that while CNN-extracted features alone can capture some predictive information

about sweetness, adding metadata in this dataset increases error, and overall, all models show negative R^2 values, indicating that predictions explain less variance than a mean-based baseline. The line plots, therefore, provide a clear visual representation of model error trends, complementing the R^2 performance shown by the bars.

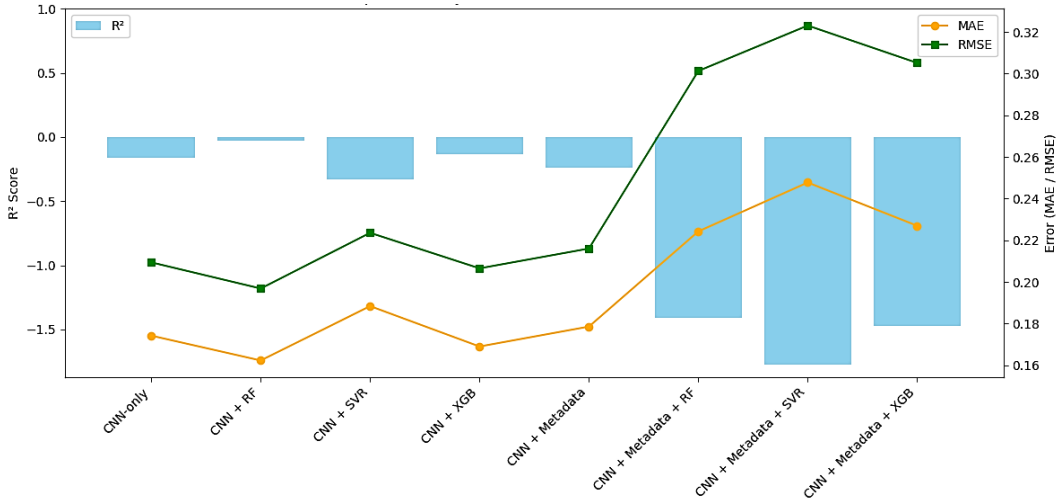


Figure 15. Comparison of Hybrid Models for Sweetness Prediction

Since the focus of this study is the visual features of Sweet Gold watermelon images rather than the fruit's physical traits, the analysis emphasizes models trained exclusively on image data. Therefore, the evaluation compared CNN-only models with hybrid models that combine CNNs with other techniques. The goal is to identify the best-performing model based purely on visual features, which can then serve as a reliable trained model for the predictive regression tool.

When comparing models using only visual features from Sweet Gold watermelon images and excluding metadata, the results indicate that the regression task remains challenging, as reflected in generally low to negative R^2 values across all models. This indicates that the target parameter's range

cannot be fully determined from image data alone. This doesn't imply there is no connection at all, but it does explain a weak, negative correlation between the visual features and the target variable. CNN + Random Forest (CNN + RF) consistently outperforms the other image-based and hybrid image-only models across all evaluation metrics. When using only visual data, it yields the most accurate, steady, and trustworthy predictions, as evidenced by its highest R2 score (least negative), and lowest RMSE and MAE.

Overall, the comparative evaluation shows clear performance differences among the models, demonstrating that some approaches are more effective at extracting meaningful visual patterns than others. When the analysis is extended to include metadata alongside image features, the CNN + Random Forest (CNN + RF) model consistently performs best, achieving lower prediction errors and more stable estimates than the other approaches. This consistent superiority across both image-only and image-plus-metadata comparisons confirms that CNN + RF is reasonable and suitable for achieving the study’s goal, as it effectively integrates visual information with additional contextual data to improve regression performance and support a reliable predictive tool.

Hyperparameter Configurations Across Machine Learning Models

Hyperparameter configurations are used across the machine learning models to evaluate and ensure accessibility, consistency, and fair performance comparison in this study. These settings directly influence model convergence, generalization, and predictive accuracy, making their documentation essential. (Goodfellow, Bengio, & Courville, 2016; Bergstra & Bengio, 2012).

Table 6 presents the hyperparameters used to train the EfficientNetB0 CNN using image data only, carefully selected to achieve an effective balance between predictive accuracy, training stability, and computational efficiency. All images were resized to 224 × 224 × 3 and preprocessed using the *EfficientNet preprocess_input* function to ensure consistency with pretrained ImageNet weights and to normalize input distributions. A batch size of 32 was chosen to provide stable gradient updates while avoiding excessive memory consumption during training. Training was divided into two stages: head training and fine-tuning, allowing the model to first learn task-specific features from the newly added layers before gradually adapting selected layers of the pretrained backbone.

For optimization, the Adam optimizer was employed due to its adaptive learning rate and strong performance across deep learning regression tasks, while Mean Squared Error (MSE) was selected as the loss function because it effectively penalizes larger prediction errors and is well suited for continuous-valued outputs such as Brix levels. During head training, a higher learning rate of 1×10^{-3} was applied to accelerate convergence of the newly initialized layers. This was followed by a reduced learning rate of 1×10^{-5} during fine-tuning to prevent disruption of pretrained feature representations. Approximately 70% of the base layers were frozen to preserve general visual features, while the remaining 30% were unfrozen to allow limited but meaningful adaptation to the visual characteristics of Sweet Gold watermelon.

To further enhance model robustness and generalization, training callbacks such as early stopping, learning rate reduction on plateau, and model checkpointing were incorporated. These mechanisms help mitigate overfitting,

dynamically adjust learning behavior, and ensure that the best-performing model configuration is retained.

Table 6. Hyperparameters used in EfficientNetB0 CNN-only (Tan & Le, 2019; Chollet, 2017)

Category	Hyperparameters	Value
Input & Preprocessing	Image size	224 x 224 x 3
	Image preprocessing	EfficientNet *preprocess_input*
Training Configuration	Batch size	32
	Epochs (head training)	30
	Epochs (fine-tuning)	40
	Shuffle	True
Optimization	Optimizer	Adam
	Learning Rate (head)	1×10^{-3}
	Learning Rate (fine-tuning)	1×10^{-5}
	Loss function	Mean Squared Error (MSE)
Fine-tuning Strategy	Frozen base layers	70%
	Trainable base layers	30%
Callbacks	Early stopping patience	8
	Restore best weights	True
	LR reduction factor	0.3
	LR reduction patience	3
	Model checkpoint	Best only
Model Saving	Head model file	*cnn_baseline_head_best.h5*
	Fine-tuned model file	*cnn_baseline_finetuned_best.h5*

Table 7 summarizes the hyperparameters used for training the EfficientNetB0 CNN with integrated metadata, with most settings retained from Table 6 to ensure a fair and consistent comparison between the image-only and image-plus-metadata models. Image inputs were resized to 224 × 224 × 3 and preprocessed using the *EfficientNet preprocess_input* function, while scaled metadata features were introduced through a separate input layer whose dimensions matched the metadata feature set. A batch size of 32 was maintained to ensure stable training and comparable computational requirements across experiments.

Training followed the same two-stage strategy of head training and fine-tuning, enabling the newly added fusion layers to learn task-specific relationships before selectively adapting deeper pretrained layers. The Adam optimizer with MSE loss was used for regression, applying learning rates of 1×10^{-3} for head training and 1×10^{-5} for fine-tuning to ensure stable learning. Freezing 70% of the base layers preserved general visual features while allowing controlled adaptation of the remaining layers to image and metadata inputs. Early stopping with best-weight restoration was applied to prevent overfitting and ensure optimal model selection, following best practices in multimodal transfer learning.

Table 7. Hyperparameters used in EfficientNetB0 CNN + Metadata

Category	Hyperparameters	Value
Input & Preprocessing	Image size	224 × 224 × 3
	Image preprocessing	EfficientNet *preprocess_input*
Metadata Input	Metadata input dimension	*X_meta_train_scaled.shape[1]*
Training Configuration	Batch size	32
	Epochs (head training)	30
	Epochs (fine-tuning)	40
	Shuffle	True
Optimization	Optimizer	Adam
	Learning rate (head)	1×10^{-3}
	Learning rate (fine-tuning)	1×10^{-5}
	Loss function	Mean Squared Error (MSE)
Fine-Tuning Strategy	Frozen base layers	70%
	Trainable base layers	30%
Callbacks	Early stopping patience	8
	Restore best weights	True

However, the Random Forest regressor used in the hybrid models was configured with the same hyperparameters for both CNN (images only) + RF and CNN + metadata + RF approaches. Using the same hyperparameters ensures a fair comparison, so any performance differences can be attributed to

the input features rather than the regressor. The number of estimators (300) was set to create a sufficiently large ensemble of decision trees, thereby improving prediction stability and reducing variance. A maximum tree depth of 20 allows the model to capture complex patterns in the CNN-extracted features while preventing overfitting, and a minimum sample split of 5 ensures that splits occur only when there is enough data to make them meaningful, supporting generalization to unseen samples. The random state (42) was set to ensure reproducibility across runs. The complete configuration is summarized in Table 8 below.

Table 8. Hyperparameters used in Hybrid Random Forest.

Category	Hyperparameter	Value
Model	Regressor	*RandomForestRegressor*
Training Configuration	Number of estimators	300
	Maximum tree depth	20
	Minimum samples split	5
Optimization / Control	Random state	42

Following a similar approach, the Support Vector Regressor (SVR) was used to model the relationship between the CNN-extracted features and the target Brix values, both for the image-only and image-plus-metadata hybrid approaches to maintain a fair comparison between the two models. The RBF kernel was chosen to capture non-linear patterns in the CNN features, while the C parameter of 10 controls the balance between model complexity and the penalty for mispredictions. The epsilon value of 0.1 defines a tolerance margin, allowing the model to ignore minor deviations and focus on more significant errors, and the gamma parameter, set to "scale," adapts automatically to the

data distribution, ensuring proper mapping of input features into the kernel space. To support stable training, feature values were standardized using *StandardScaler*, and the target Brix values were *MinMax-scaled*. These hyperparameters were determined through trial and error to best fit the extracted features, ensuring reliable and consistent regression outputs across both image-only and image-plus-metadata hybrid models. The complete configuration is summarized in Table 9 below.

Table 9. Hyperparameters used in Hybrid Support Vector Regressor.

Category	Hyperparameter	Value
Model	Regressor	Support Vector Regressor (SVR)
Kernel Configuration	Kernel	RBF
	C	10
	Epsilon	0.1
	Gamma	Scale
Feature Scaling	Feature scaler	*StandardScaler*
Target Scaling	Target values	*MinMax-scaled Brix*

Lastly, for the hybrid models incorporating Extreme Gradient Boosting (XGBoost), the XGBRegressor was used to capture complex, non-linear relationships in both CNN-extracted features and combined CNN + metadata features. The model was configured with 500 estimators and a learning rate of 0.05 for stable, gradual updates. A maximum tree depth of 8 and a subsample/column sample by tree of 0.8 were set to balance feature interactions with generalization. L2 regularization (lambda = 1.0) was applied to prevent overfitting, and the objective was set to *reg:squarederror* with RMSE as the evaluation metric. The tree method was set to hist for faster computation,

random state = 42 ensured reproducibility, and number of jobs = −1 allowed parallel processing for efficiency. As with the other hybrid models, the target Brix values were *MinMax-scaled* to support model stability and consistent learning. These hyperparameters were tuned through trial and error to ensure the *XGBRegressor* could effectively leverage the features from both image-only and image-plus-metadata models, producing accurate and reliable results. The full configuration is summarized in Table 10.

Table 10. Hyperparameters used in Hybrid Extreme Gradient Boosting.

Category	Hyperparameter	Value
Model	Regressor	XGBRegressor
Training Configuration	Number of estimators	500
	Learning rate	0.05
	Maximum tree depth	8
	Subsample	0.8
	Column sample by tree	0.8
Regularization	Lambda (L2)	1.0
Optimization / Control	Objective	reg:squarederror
	Evaluation metric	RMSE
	Tree method	Hist
	Random state	42
	Number of jobs	−1
Target Scaling	Target values	MinMax-scaled Brix

Principal Component Analysis with CNN Features

The PCA visualization shows how the CNN-extracted features are distributed in a lower-dimensional space and how these distributions relate to sweetness values. Since CNN features are high-dimensional, PCA reduces them to two main components, allowing them to be visualized in 2D. The color of each point indicates its corresponding sweetness level *(brix_scaled)*, with darker shades representing lower sweetness and brighter shades representing

higher sweetness. The figure (Figure 16) below shows that many samples are clustered closely together, indicating that the CNN learned consistent visual patterns from the images. Fruits with similar external appearances are clustered together in the feature space, indicating that the model successfully captured visual characteristics such as color patterns, surface texture, and shape.

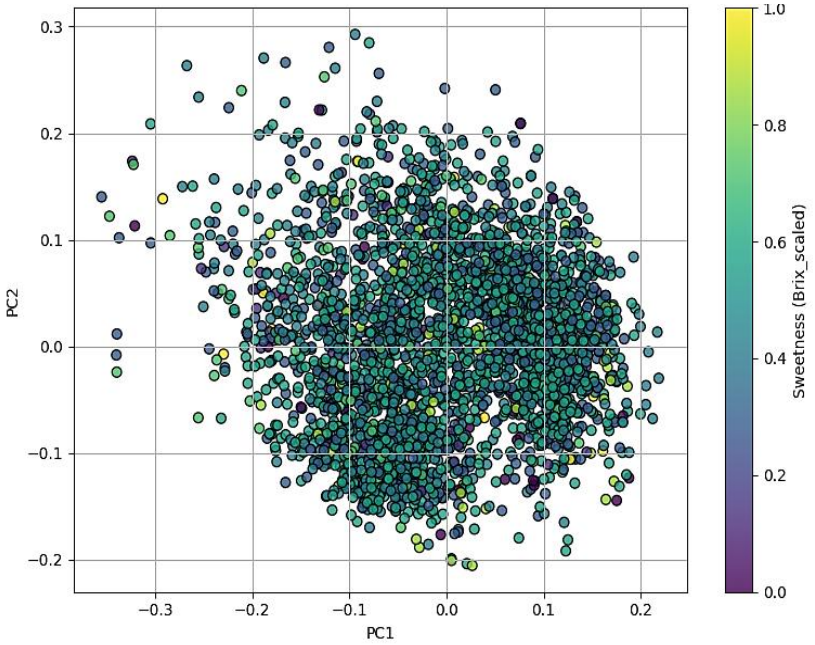


Figure 16. CNN Feature (Images only) Colored by Brix

However, the sweetness values within these clusters are highly mixed. There is no clear separation or trend where sweeter fruits are grouped apart from less sweet ones. It means there is only a weak correlation between sweetness and the visual characteristics that the CNN has learned, including shape, texture, and size-related indicators. This points to a low rather than no correlation between external appearance and internal sweetness. If these features were strongly related to sweetness, clearer groupings or color patterns would be visible in the plot. The lack of this foundation helps explain why the model showed low or even negative performance correlations (Al Hammami & Hassan, 2025).

Feature Importance for CNN + Random Forest

Figure 17 shows the top 20 most important features identified by the Random Forest (RF) model when combined with CNN-extracted image features. The height of each bar, representing a CNN feature, shows how much that feature influenced the model's sweetness prediction. Higher bars indicate that the feature was used extensively or had greater implications for the model's development. The results show that all the top features come from CNN image embeddings, indicating that the model relied primarily on visual information rather than metadata. However, the importance values are relatively close to each other, and none of the features stand out as strongly dominant. This means there is no single visual feature that provides much data on sweetness. Instead, the model is using many weak visual signals, each contributing only slightly to the outcome. The absence of strong, interpretable features further explains why the model showed low, negative correlation. Although the model extracts visual patterns, internal sweetness is not significantly reflected in these patterns (Niu et al., 2022).

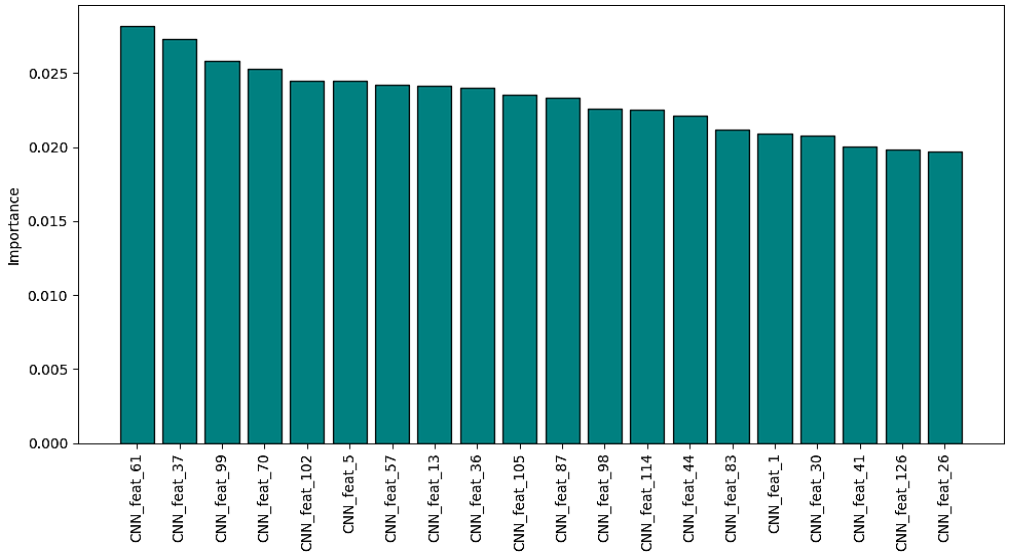


Figure 17. Top 20 Important Feature (CNN + RF)

Correlation Heatmap

The correlation heatmap shows the strength and direction of relationships between all input features and the target variable (Brix). It helps identify which features have positive, negative, or weak associations with sweetness. Figure 18 shows the correlation heatmap between CNN-extracted image features and Brix (sweetness). Each row represents a CNN feature, and the color indicates the strength of its correlation with sweetness. Warmer colors represent higher positive correlation, while cooler colors represent weak or negative correlation (Buyrukoglu & Akbas, 2022).

The figure (Figure 18) below shows that most CNN features have very weak correlation with sweetness. The dominant blue tones across the heatmap indicate that the majority of visual features extracted by the CNN do not have a strong linear relationship with Brix. Only a few features show slightly higher correlation values, and even these are not strong. This means that while the CNN can extract many visual patterns from the fruit images, most of these patterns are not directly related to sweetness.

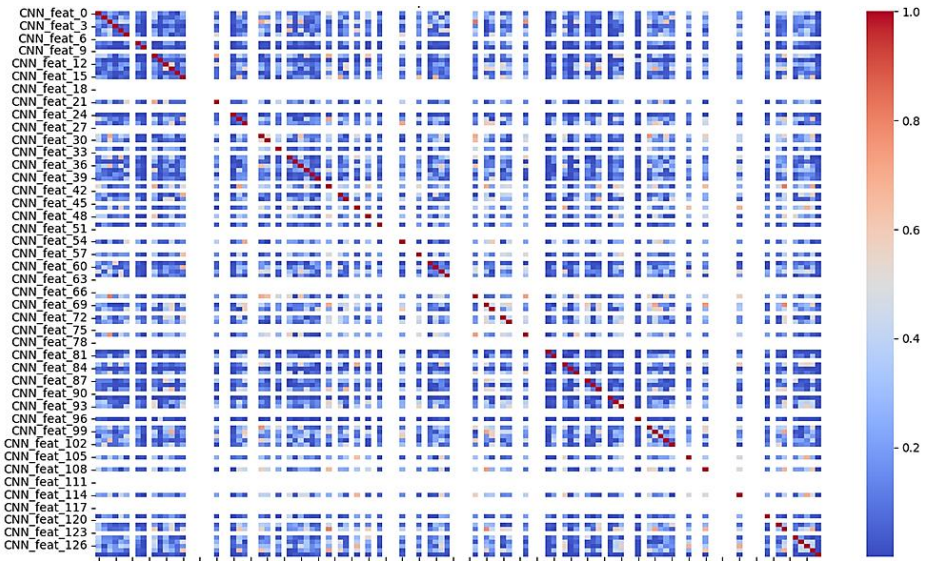


Figure 18. CNN Feature vs. Sweetness

Figure 19 shows that physical measurements such as weight, height, and width have a very weak correlation with sweetness. The dominant blue tones across the bottom row and rightmost column of the heatmap indicate that these physical dimensions do not have a strong linear relationship with the scaled sweetness values, with coefficients ranging only from -0.12 to -0.15. While there are strong internal correlations between the physical traits themselves (such as the 0.82 correlation between weight and width shown in orange), these do not translate to sweetness. This means that while a fruit may be larger or heavier, those patterns are not directly related to sweetness. This supports the conclusion that external physical metrics, much like generic visual patterns, do not strongly explain the internal Brix levels of the fruit.

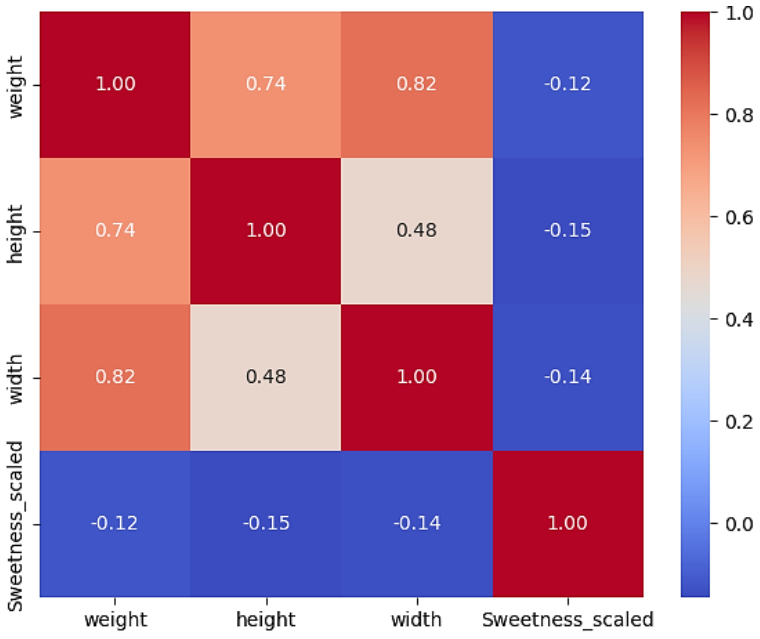


Figure 19. Physical Traits vs. Sweetness

Overall, the heatmaps reveal that aesthetic features, as captured by a standard CNN, are largely disconnected from the fruit's physical dimensions. It suggests that the "look" of the fruit (its aesthetics) does not provide a reliable

guide to its size or sweetness, reinforcing the idea that the fruit's most important qualities are internal and not easily captured by simple visual cues.

Prototype for Predictive Tool

A prototype predictive tool is built around the best-performing trained model, which integrates a Convolutional Neural Network (CNN) with a Random Forest (RF) regressor using image-based features only. This hybrid approach was selected due to its strong predictive performance and its ability to capture complex, non-linear relationships within high-dimensional visual data. The CNN serves as an automatic feature extractor, learning discriminative visual patterns related to fruit appearance, while the Random Forest effectively models interactions among these learned features to estimate sweetness.

To demonstrate the functionality of the system, a user-facing application titled "*SweetGold – AI Sweetness Prediction System*" was developed using the Gradio framework. The interface provides a simple web-based environment where users can upload images of Sweet Gold fruit. Once an image is submitted, it is processed by the CNN to extract visual features, which are then passed directly to the Random Forest model to generate a predicted °Brix value in real time. This prototype translates a complex deep learning and ensemble learning pipeline into a practical and accessible tool for non-destructive sweetness estimation. By relying solely on visual data, the approach demonstrates the feasibility of image-based fruit quality assessment, enabling agricultural monitoring and quality control without physical or invasive measurements. The figure (Figure 20) below presents the full-screen screenshot of the system, including its integrated file ecosystem.

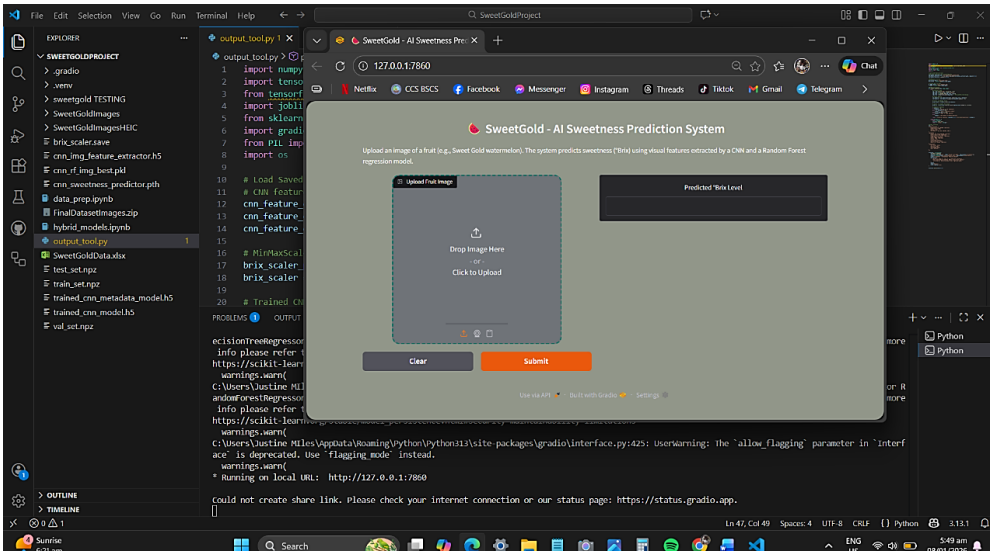


Figure 20. Predictive Tool for Sweet Gold Watermelon, Prototype

As shown in Figure 20 above, the file explorer in the development environment (VS Code) reveals the specific components that make this integrated system functional. Each file plays a crucial role in the "learned patterns" mentioned in this study:

***output_tool.py*:** The main script that launches the web interface and coordinates the model's logic.

***cnn_img_feature_extractor.h5*.** The trained CNN architecture responsible for extracting the aesthetic features from uploaded images.

***cnn_rf_img_best.pkl*.** The finalized Random Forest model that performs the ultimate sweetness prediction.

***brix_scaler.save*.** A saved scaling object (likely *MinMaxScaler*) used to ensure the output matches the original °Brix units.

***hybrid_models.ipynb*.** The research environment where the CNN+RF approach was tested against other methods.

***train/test/val_set.npz*.** Compressed data files containing the features used to train and validate the selected configuration.

The testing phase evaluated the predictive accuracy of the proposed system by comparing model-generated sweetness predictions with manually measured sweetness values. A total of 15 Sweet Gold watermelons were used in this phase, with sweetness measured using a standard °Brix method over 5 consecutive days. To ensure consistency and a fair evaluation, both image capture and manual sweetness measurement were performed at exactly 8:00 PM each day throughout the five-day testing period. This controlled timing minimizes variability caused by diurnal changes in fruit physiology and environmental conditions, enabling a more reliable comparison between predicted and actual sweetness values. The Sweet Gold images used for testing are shown in Figure 21, which illustrates the visual input data used during the evaluation process.

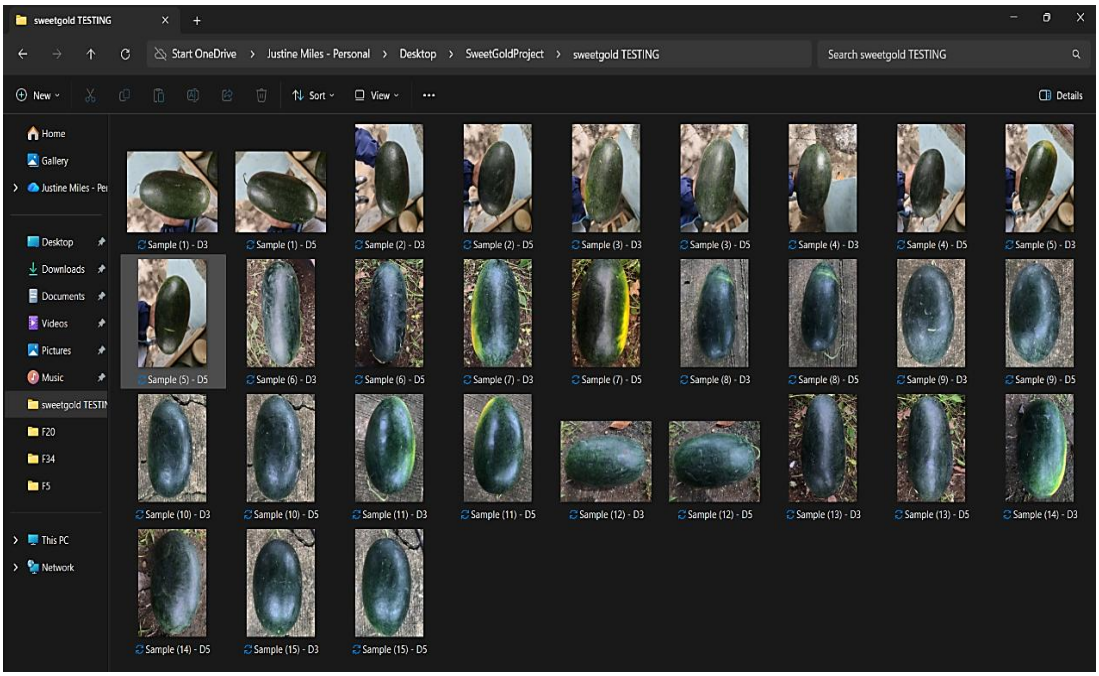


Figure 21. Sweet Gold Watermelon, Testing Images

For each fruit, sweetness measurements were recorded across the five-day period (December 12 to 16), allowing observation of changes in sweetness over time and comparison with the system's predictions. In addition, two representative images per fruit were captured on Day 3 and Day 5 for documentation and analysis. Day 3 was selected instead of Day 1 because, according to the fruit supplier, the watermelons had already been harvested approximately one week prior to collection. As a result, Day 1 measurements may not accurately represent a consistent ripeness baseline across samples. Focusing on Day 3 and Day 5 ensures that the evaluation reflects a more stable stage of fruit ripeness, reducing variability caused by differences in harvest timing and storage conditions. This setup provides a fairer and more reliable assessment of the system's ability to predict sweetness based on visual information, while aligning the testing protocol with real-world handling and distribution practices. The figure (Figure 22) below shows the corresponding recorded °Brix measurements used during the evaluation process.

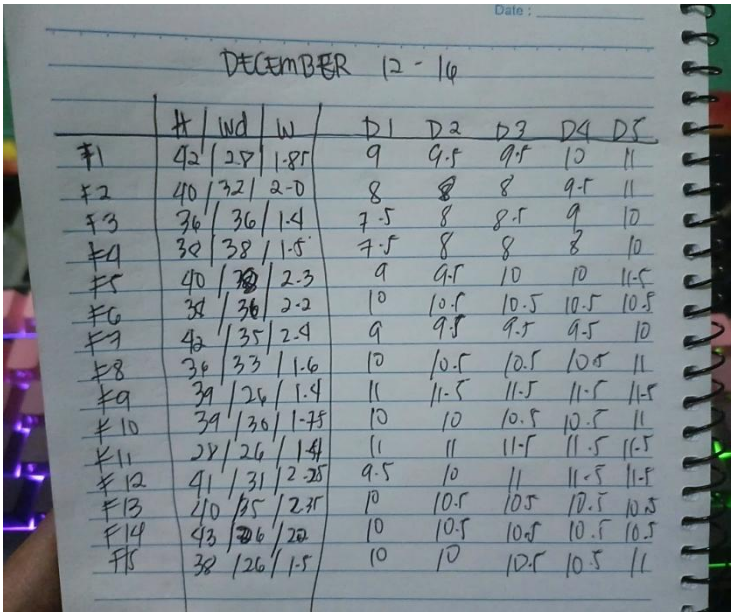


Figure 22. Reference Data for Sweet Gold Watermelon

As a result of the testing phase, these results indicate that the trained model provides reliable but not highly accurate predictions of sweetness. The predicted °Brix values generally fall within an expected sweetness range rather than matching the exact measured values. This tendency supports the seen negative correlation, showing that while visual features contain some informative indicators, these factors are not enough for precise sweetness estimation. Importantly, this result shows that there is a slight correlation between external appearance and internal sweetness rather than zero at all. The figure (Figure 23) below presents an example of uploading images in the developed demonstration system, illustrating its effectiveness in a real-world scenario.

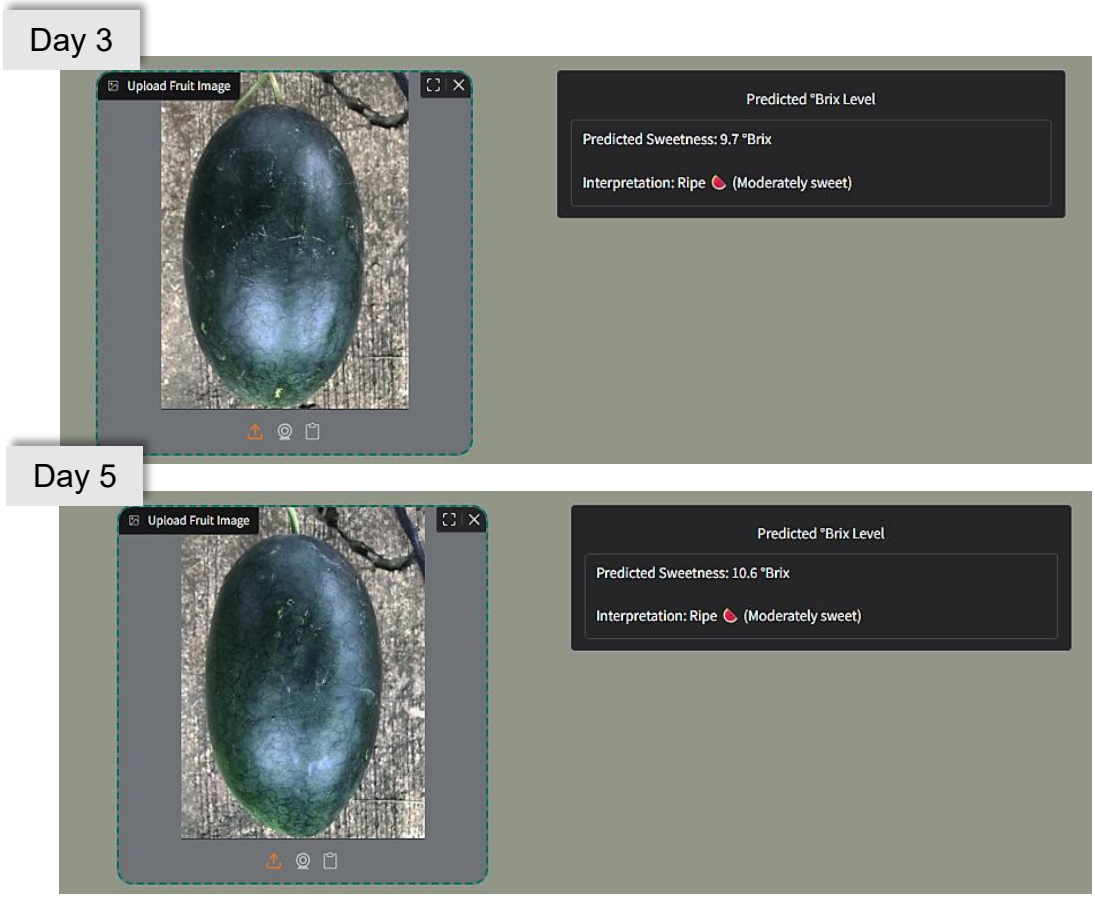


Figure 23. Sample Input/Output, Day 3 Image and Day 5 Image

Testing observations showed that fruit color strongly influenced model predictions, with uniformly yellow fruits generally predicted as moderately sweet, while darker green or green-yellow areas were linked to higher sweetness estimates. This suggests that the CNN primarily depends on color-related cues, particularly overall skin coloration, rather than detailed texture or structural features, indicating that ripeness-related visual traits are more informative for sweetness prediction. The actual °Brix measurements of the testing images captured on Day 3 and Day 5 are summarized in Table 11.

Table 11. Actual Brix Measurement for Testing Images of Sweet Gold Watermelon with Day 3 and Day 5

Testing Images	Day 3	Day 5
Testing Image 1	10.6	10.6
Testing Image 2	10.9	10.9
Testing Image 3	10.6	10.6
Testing Image 4	10.6	10.8
Testing Image 5	11	11.9
Testing Image 6	11.1	11.6
Testing Image 7	10.6	10.5
Testing Image 8	9.7	9.7
Testing Image 9	10.6	10.8
Testing Image 10	10.6	10.7
Testing Image 11	9.6	10.5
Testing Image 12	10.6	10.5
Testing Image 13	10.6	10.6
Testing Image 14	10.5	10.5
Testing Image 15	9.7	10.6

Despite its capabilities, the model was not expected to predict exact °Brix values, as sweetness is determined by internal biochemical processes that cannot be fully observed from the fruit's surface. This limitation explains why some predictions showed negative correlations during testing and highlights the challenges of relying solely on visual information for precise sweetness estimation.

The results show that the model relies mainly on color as the primary visual cue for sweetness prediction, with certain color shades providing stronger signals than shape, size, or texture. However, visual features alone, even when combined with basic physical measurements, are insufficient to accurately capture internal sweetness. While CNN-based analysis is effective for identifying relative sweetness trends and supporting non-destructive screening and quality control, precise °Brix estimation still requires direct biochemical testing.

Overall, the study demonstrates that the proposed CNN + Random Forest system is effective as a rapid, non-destructive screening tool, capable of monitoring sweetness progression, flagging fruits likely to be sweeter or less sweet, and supporting agricultural monitoring, quality control, and research applications. The model can track general trends in fruit sweetness over time, helping farmers and post-harvest workers make informed decisions without the need for destructive testing. While the model provides actionable guidance, its predictions should be interpreted as indicative rather than exact, highlighting both the potential and limitations of predictive modeling for internal fruit traits.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the findings gathered throughout the experimentation and model evaluation phases of the study. The key outcomes of the model evaluation are consolidated in this chapter, highlighting the model with the highest predictive performance and its potential applicability in practical fruit quality assessment. Based on these results, conclusions are drawn regarding model effectiveness and operational relevance. Recommendations are also provided to guide future improvement, scalability, and application of AI-driven sweetness estimation in agricultural practice.

Summary

This study investigated the correlation between external aesthetic traits and internal sweetness (°Brix) of Sweet Gold watermelons (*Citrullus lanatus* 'Sweet Gold') using image processing and hybrid machine learning techniques. A total of 150 harvested watermelons were collected and imaged under controlled conditions, resulting in 900 high-resolution images captured from multiple angles. To improve model robustness, data augmentation expanded the dataset to approximately 3,150 training images. Each fruit's sweetness was measured using a handheld Brix refractometer, while physical characteristics such as weight, height, and width were recorded as supplementary metadata. Images were processed using EfficientNet-B0 as a convolutional neural network (CNN) feature extractor to obtain deep visual representations. Multiple regression models were developed and compared, including CNN-only models, CNN with metadata, and hybrid approaches combining CNN-extracted features

with traditional machine learning algorithms such as Random Forest (RF), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGBoost). Model performance was evaluated using standard regression metrics such as R^2, MAE, and RMSE, to assess prediction accuracy.

Experimental results showed that visual features alone were insufficient for accurate sweetness prediction, while the inclusion of physical metadata provided modest improvements. Among all tested configurations, the CNN + Random Forest model using image features achieved the best overall performance, demonstrating reliable predictions and improved stability. Further analysis using principal component analysis, correlation heatmaps, and feature importance indicated that fruit color was the most influential visual trait, while shape, size, and texture were less predictive. A prototype prediction tool was developed using the best-performing model to demonstrate a practical post-harvest application. Overall, the study highlights the potential and limitations of non-destructive, image-based methods for estimating watermelon sweetness, providing a foundation for AI-driven fruit quality assessment in agriculture.

Conclusions

Based on the objectives and findings of the study, the following conclusions are drawn to:

1. **Capture high-resolution images of Sweet Gold watermelons using a controlled imaging setup.**

   The controlled acquisition of high-resolution images successfully captured consistent external aesthetic traits of Sweet Gold watermelons. Although the number of raw samples was limited,

applying data augmentation increased variability and improved the model's ability to learn general visual patterns of fruit appearance.

2. **Analyze and extract visual features from the images to investigate their correlation with sweetness levels measured using a manual Brix refractometer.**

   The actual sweetness regression (Brix value) of each sampled watermelon was successfully measured manually using a handheld refractometer. The analysis proves that visual features alone are insufficient for accurately estimating internal fruit quality. Even when included alongside basic physical measurements such as weight, height, and width, non-invasive methods cannot fully capture the complex mechanisms underlying sweetness accumulation.

3. **Train and compare multiple hybrid machine learning algorithms to predict sweetness levels based on image processing techniques.**

   Several image processing techniques were applied before training the models, including image annotation, background removal, resizing, normalization, and metadata alignment, to ensure that both image features and physical measurements were prepared in a usable, structured format. After preprocessing, different regression architectures were trained and compared, including CNN-only, CNN + Random Forest (RF), CNN + SVR, CNN + XGBoost, CNN + metadata, CNN + metadata + RF, CNN + metadata + SVR, and CNN + metadata + XGBoost. These hybrid approaches were included simply because these allow traditional machine learning algorithms

to learn patterns from both CNN-extracted deep features and physical traits at the same time, which is important to determine how much visual features and numerical attributes contribute to sweetness prediction. The purpose of comparing all these model variations was to identify which combination of feature representation and algorithm produced the most consistent and generalizable outcome. Through these comparative experiments, the researcher determined that the model configuration with the best performance and most stable predictive behavior was the one tested.

4. **Assess model performance using evaluation metrics.**

   The CNN + Random Forest configuration achieved the highest performance among all model variations. Specifically, this hybrid approach achieved an R^2 score of -1.4112, a Mean Absolute Error (MAE) of 0.2243, and a Root Mean Squared Error (RMSE) of 0.0908. These values indicate that the model not only explained a large portion of the sweetness variance but also produced low prediction errors relative to the actual measured Brix values. Because of this strong predictive performance and generalization stability, the CNN + RF model was selected as the primary model for deployment and served as the basis of the developed prototype prediction tool.

5. **Determine the extent to which external aesthetic features can reliably predict the sweetness level of Sweet Gold watermelons, and present the results through a visual demonstration of the trained model.**

Findings indicated that fruit skin color is the most influential aesthetic feature correlated with sweetness, providing a stronger predictive signal than shape, size, or texture. The final results were visualized using PCA feature clustering, correlation heatmaps, and feature importance plots to clearly illustrate this behavior. A working prediction prototype was also demonstrated using the best model to show actual performance in real-world fruit samples not included in the dataset.

Recommendations

Based on the conclusions, the following recommendations are made:

1. **Expand the dataset:** Expanding the dataset with more samples and greater diversity is essential because it will expose the model to broader visual patterns. The recommendation on more samples & diversity will reduce overfitting and allow the system to learn variations in fruit appearance across different farms and conditions. This will improve the model's ability to generalize beyond the limited dataset used in the study.
2. **Include non-harvested / field images:** Capturing images directly from the field will provide the model with more raw environment samples that represent watermelons in natural conditions before harvesting. This strengthens robustness in unpredictable outdoor lighting and surroundings. It will also expand the use case of the model from post-harvest evaluation to early field assessment.
3. **Integrate with a physical device for deployment:** Building a real-world prototype using an Arduino / microcontroller-based prototype ensures

that the system can run beyond laboratory testing. A device with a camera module and an embedded predictive model can be used in markets or farms. This allows validation of system practicality and real deployment performance.

4. **Collect richer metadata and standardize measurement protocols:** Increasing metadata dimensions and applying standardized measurement protocols would improve label accuracy for sweetness. More structured variables, such as farm origin, planting method, or environment, can improve model understanding. This will reduce noise and improve the reliability of regression outputs.
5. **Extend the method to other fruit types and cross-crop evaluation:** Applying this method to other fruit crops will lead to cross-crop evaluation and broaden agricultural value. If the hybrid CNN + RF approach succeeds across multiple fruit types, the technique becomes industry-flexible. This improves the research impact beyond the Sweet Gold watermelon alone.
6. **Incorporate additional non-invasive sensing techniques**: Adding sensors such as near-infrared (NIR) spectroscopy or hyperspectral imaging could complement visual data and improve the accuracy of internal quality predictions. Combining multiple non-destructive sensing methods may better capture biochemical properties, leading to more reliable sweetness estimation.

REFERENCES

(2025). Proquest.com. https://www.proquest.com/openview/694f561925131e0f42741a48de690f54/1?pq-origsite=gscholar&cbl=2069459&fbclid=IwY2xjawN0mTZleHRuA2FlbQIxMABicmlkETE0NUdEeFZicjd3YkFRSTRVAR4L67Zk7Iml_epCt3FfeBpOAF6dn1Mp5pXkIXgOBnANJIIjwg3E1G3WuuYhww_aem_o1qDQs4g0_MUNxsIUZH3RQ

3.3. Metrics and scoring: quantifying the quality of predictions. (n.d.). Scikit-Learn. https://scikit-learn.org/stable/modules/model_evaluation.html#r2-score

A Fruit Ripeness Detection Method using Adapted Deep Learning-based Approach - ProQuest. (2023). Proquest.com. https://www.proquest.com/openview/4474b00348fc6513dddadf424eb69bdc/1?pq-origsite=gscholar&cbl=5444811

Achour, B., Belkadi, M., Filali, I., Laghrouche, M., & Lahdir, M. (2020). Image analysis for individual identification and feeding behaviour monitoring of dairy cows based on Convolutional Neural Networks (CNN). Biosystems Engineering, 198, 31–49. https://doi.org/10.1016/j.biosystemseng.2020.07.019

Adainoo, B., Thomas, A. L., & Krishnaswamy, K. (2023). Correlations between color, textural properties and ripening of the North American pawpaw (Asimina triloba) fruit. Sustainable Food Technology. https://doi.org/10.1039/d2fb00008c

Aguirre-Radilla, J., De, E., José Luis Hernández-Hernández, Carranza-Gómez, J., José Antonio Montero-Valverde, & Martínez-Arroyo, M. (2022). Texture and Color-Based Analysis to Determine the Quality of the Manila Mango Using Digital Image Processing Techniques. Communications in Computer and Information Science, 93–106. https://doi.org/10.1007/978-3-031-19961-5_7

Aherwadi, N., Mittal, U., Singla, J., Jhanjhi, N. Z., Yassine, A., & Hossain, M. S. (2022). Prediction of Fruit Maturity, Quality, and Its Life Using Deep Learning Algorithms. Electronics, 11(24), 4100. https://doi.org/10.3390/electronics11244100

Ajay, A. (2024, September 30). *What are the Advantages and Disadvantages of Random Forest?* Pickl.AI. https://www.pickl.ai/blog/advantages-and-disadvantages-random-forest

Alenazi, M. M., Shafiq, M., Alsadon, A. A., Alhelal, I. M., Alhamdan, A. M., Solieman, Talaat. H. I., Ibrahim, A. A., Shady, M. R., & Saad, M. A. O. (2020). Non-destructive assessment of flesh firmness and dietary antioxidants of greenhouse-grown tomato (Solanum lycopersicum L.) at different fruit maturity stages. Saudi Journal of Biological Sciences, 27(10), 2839–2846. https://doi.org/10.1016/j.sjbs.2020.07.004

Amjad, M., Ahmad, I., Ahmad, M., Wróblewski, P., Kamiński, P., & Amjad, U. (2022). Prediction of Pile Bearing Capacity Using XGBoost Algorithm: Modeling and Performance Evaluation. Applied Sciences, 12(4), 2126. https://doi.org/10.3390/app12042126

Barbosa Júnior, M. R., Santos, R. G. dos, Sales, L. de A., Vargas, R. B. S., Deltsidis, A., & Oliveira, L. P. de. (2025). Image-based and ML-driven analysis for assessing blueberry fruit quality. Heliyon, 11(3), e42288. https://doi.org/10.1016/j.heliyon.2025.e42288

BD. (2025, March 5). Watermelon Production Guide: How to Grow High-Yield and Disease-Resistant Watermelons. Business Diary. https://businessdiary.com.ph/33117/watermelon-production-guide-how-to-grow-high-yield-and-disease-resistant-watermelons/

Bergstra, J., & Bengio, Y. (2012). Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research*, *13*, 281–305. https://www.jmlr.org/papers/volume13/bergstra12a/bergstra12a.pdf

Boukili, M., & Nawfal Jebbor. (2024). Performance Optimization of Convolutional Neural Networks (CNN) in Multilabel Fruit Classification. 182, 1–6. https://doi.org/10.1109/iraset60544.2024.10548403

Calauan agri-trading post to boost farmers' income. (2019). Pna.gov.ph. https://www.pna.gov.ph/articles/1057983

Cesar. (2025). My Laguna. Blogspot.com. https://laguna-living.blogspot.com/2007/06/my-laguna.html

Chakraborty, S. K., A., S., Dubey, K., Jat, D., Chandel, N. S., Potdar, R., Rao, N. R. N. V. G., & Kumar, D. (2023). Development of an optimally designed real-time automatic citrus fruit grading–sorting machine leveraging computer vision-based adaptive deep learning model. Engineering Applications of Artificial Intelligence, 120, 105826. https://doi.org/10.1016/j.engappai.2023.105826

Chandra, S. S., Devarapalli Dharmika, Guntupalli Vijayadurgarao, & V, S. S. (2024). Fruit Classification based on Shape, Color and Texture using Image Processing Techniques. ResearchGate, 10(02), 100–107. https://www.researchgate.net/publication/389597736_Fruit_Classification_based_on_Shape_Color_and_Texture_using_Image_Processing_Techniques

Chawgien, K., & Kiattisin, S. (2021). Machine learning techniques for classifying the sweetness of watermelon using acoustic signal and image processing. Computers and Electronics in Agriculture, 181, 105938. https://doi.org/10.1016/j.compag.2020.105938

Citrullus lanatus (Watermelon, Watermelons) | North Carolina Extension Gardener Plant Toolbox. (2025). Ncsu.edu. https://plants.ces.ncsu.edu/plants/citrullus-lanatus/

Cui, H., Ding, Z., Zhu, Q., Wu, Y., & Gao, P. (2020). Population structure and genetic diversity of watermelon (Citrullus lanatus) based on SNP of

chloroplast genome. 3 Biotech, 10(8). https://doi.org/10.1007/s13205-020-02372-5

Deepika Bairwa, & Gaurav Sharma. (2017). Classification of Fruits Based on Shape, Color and Texture using Image Processing Techniques. International Journal of Engineering Research And, V6(12). https://doi.org/10.17577/ijertv6is120057

Dhanya, V. G., Subeesh, A., Kushwaha, N. L., Vishwakarma, D. K., Nagesh Kumar, T., Ritika, G., & Singh, A. N. (2022). Deep learning-based computer vision approaches for smart agricultural applications. Artificial Intelligence in Agriculture, 6. https://doi.org/10.1016/j.aiia.2022.09.007

Dhiman, P., Kaur, A., Balasaraswathi, V. R., Gulzar, Y., Alwan, A. A., & Hamid, Y. (2023). Image Acquisition, Preprocessing and Classification of Citrus Fruit Diseases: A Systematic Literature Review. Sustainability, 15(12), 9643. https://doi.org/10.3390/su15129643

Ergün, E. (2025). High precision banana variety identification using vision transformer-based feature extraction and support vector machine. Scientific Reports, 15(1). https://doi.org/10.1038/s41598-025-95466-0

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. Deeplearningbook.org; MIT Press. https://www.deeplearningbook.org/

Gumiran, A. E. (2020, December 9). Organic Farms in CALABARZON that You Should Know About - The Queen's Escape. The Queen's Escape. https://thequeensescape.com/organic-farms-in-calabarzon-that-you-should-know-about/

Hamdani, H., Septiarini, A., Sunyoto, A., Suyanto, S., & Utaminingrum, F. (2021). Detection of oil palm leaf disease based on color histogram and supervised classifier. Optik, 245, 167753. https://doi.org/10.1016/j.ijleo.2021.167753

https://www.facebook.com/specialtyproduceapp, & https://www.facebook.com/specialtyproduceapp. (2025). Yellow Watermelons. Specialtyproduce.com. https://specialtyproduce.com/produce/melon/watermelon/yellow_7663.php

Jayanthi J, Arun Kumar K, & Verma, S. (2024). Harnessing Computer Vision for Agricultural Transformation: Insights, Techniques, and Applications. 111–131. https://doi.org/10.1007/978-981-99-8684-2_8

Jaywant, S. A., Singh, H., & Arif, K. M. (2022). Sensors and instruments for brix measurement: A review. Sensors, 22(6), 2290. https://doi.org/10.3390/s22062290

K. Sangeetha, P. Vishnu Raja, Siranjeevi S, Suman J, & Rohith S. (2024). Classification of Fruits and its Quality Prediction using Deep Learning. https://doi.org/10.1109/icicv62344.2024.00059

Kavitha, J. C., & Suruliandi, A. (2016, January 1). Texture and color feature extraction for classification of melanoma using SVM. IEEE Xplore. https://doi.org/10.1109/ICCTIDE.2016.7725347

Knott, M., Perez-Cruz, F., & Defraeye, T. (2022). Facilitated machine learning for image-based fruit quality assessment. Journal of Food Engineering, 111401. https://doi.org/10.1016/j.jfoodeng.2022.111401

Lai, J. W., Ramli, H. R., Ismail, L. I., & Wan Hasan, W. Z. (2023). Oil Palm Fresh Fruit Bunch Ripeness Detection Methods: A Systematic Review. Agriculture, 13(1), 156. https://doi.org/10.3390/agriculture13010156

Lin, M., Chen, Q., & Yan, S. (2013). *Network In Network*. ArXiv.org. https://arxiv.org/abs/1312.4400

M'hamdi, O., Takács, S., Palotás, G., Ilahy, R., Helyes, L., & Pék, Z. (2024). A Comparative Analysis of XGBoost and Neural Network Models for Predicting Some Tomato Fruit Quality Traits from Environmental and Meteorological Data. Plants, 13(5), 746. https://doi.org/10.3390/plants13050746

Mohamed, K. (2025, March 30). *Demystifying CNNs: A Deep Dive into Convolutional Neural Network Fundamentals*. Medium. https://medium.com/@karemsaeed2468/demystifying-cnns-a-deep-dive-into-convolutional-neural-network-fundamentals-0b9ed1d1d2fa

Moreda, G. P., Avendaño, A., Ruiz-Altisent, M., & Perdigones, A. (2012). Shape determination of horticultural produce using two-dimensional computer vision – A review. Journal of Food Engineering, 108(2), 245–261. https://doi.org/10.1016/j.jfoodeng.2011.08.011

Mustafa, M. F. R., & Wardhana, A. K. (2025). Performance Comparison of Machine Learning Algorithms Using EfficientNetB0 Feature Extraction on Dental Disease Classification. *Journal of Applied Informatics and Computing*, *9*(5), 2131–2142. https://doi.org/10.30871/jaic.v9i5.10286

Ni, X., Li, C., Jiang, H., & Takeda, F. (2021). Three-dimensional photogrammetry with deep learning instance segmentation to extract berry fruit harvestability traits. ISPRS Journal of Photogrammetry and Remote Sensing, 171, 297–309. https://doi.org/10.1016/j.isprsjprs.2020.11.010

Niu, T., Li, J., Wei, W., & Yue, H. (2022). A hybrid deep learning framework integrating feature selection and transfer learning for multi-step global horizontal irradiation forecasting. Applied Energy, 326, 119964. https://doi.org/10.1016/j.apenergy.2022.119964

Olisah, C. C., Trewhella, B., Li, B., Smith, M. L., Winstone, B., E. Charles Whitfield, Felicidad Fernández Fernández, & Duncalfe, H. (2024). Convolutional neural network ensemble learning for hyperspectral imaging-based blackberry fruit ripeness detection in uncontrolled farm environment. Engineering Applications of Artificial Intelligence, 132, 107945–107945. https://doi.org/10.1016/j.engappai.2024.107945

Palengke Boy - Wet Market. (2017). Palengkeboy.com. https://www.palengkeboy.com/wet-market-shop-detail/419843-yellow-watermelon-cantaloupe

Panigrahi, B., Kathala, K. C. R., & Sujatha, M. (2023). A Machine Learning-Based Comparative Approach to Predict the Crop Yield Using Supervised Learning With Regression Models. Procedia Computer Science, 218, 2684–2693. https://doi.org/10.1016/j.procs.2023.01.241

Paris, H. S. (2015). Origin and emergence of the sweet dessert watermelon, Citrullus lanatus. Annals of Botany, 116(2), 133–148. https://doi.org/10.1093/aob/mcv077

Park, G., Durre Shahwar, Jang, G., Shin, J., Kwon, G., Kim, Y., Chang Oh Hong, Jin, B., Kim, H., Lee, O., & Park, Y. (2023). Identification of a novel locus C2 controlling canary yellow flesh color in watermelons. Frontiers in Genetics, 14. https://doi.org/10.3389/fgene.2023.1256627

Patel, K., & Patel, H. B. (2021, September 1). A Comparative Analysis of Supervised Machine Learning Algorithm for Agriculture Crop Prediction. IEEE Xplore. https://doi.org/10.1109/ICECCT52121.2021.9616731

Prijeesh. (2023, December 15). Image Segmentation Using GrabCut - Scaler Topics. Scaler Topics. https://www.scaler.com/topics/image-segmentation-using-grabcut/

Sarakum, T., & Sukpancharoen, S. (2025). Non-destructive sweetness classification of Khao Tang Kwa pomelos using machine learning with acoustic and image processing. Journal of Food Composition and Analysis, 142, 107385. https://doi.org/10.1016/j.jfca.2025.107385

Sheng, X., Kang, C., Zheng, J., & Lyu, C. (2023). An edge-guided method to fruit segmentation in complex environments. Computers and Electronics in Agriculture, 208, 107788. https://doi.org/10.1016/j.compag.2023.107788

Sudiana, D., Putri, S. H., Kushardono, D., Prabuwono, A. S., Sri Sumantyo, J. T., & Rizkinia, M. (2025). CNN-Random Forest Hybrid Method for Phenology-Based Paddy Rice Mapping Using Sentinel-2 and Landsat-8 Satellite Images. Computers, 14(8), 336. https://doi.org/10.3390/computers14080336

SWEET GOLD WATERMELON F1. (2025, November 3). Flickr; SWEET GOLD WATERMELON F1 | VERY PROLIFIC HIGH YIELDING VARIE… | Flickr. https://www.flickr.com/photos/52749860@N02/4865380506/in/photostream/

Tan, M., & Le, Q. V. (2019, May 28). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ArXiv.org. https://arxiv.org/abs/1905.11946

Tian, H., Wang, T., Liu, Y., Qiao, X., & Li, Y. (2020). Computer vision technology in agricultural automation —A review. Information Processing in Agriculture, 7(1), 1–19. https://doi.org/10.1016/j.inpa.2019.09.006

Trends of Fresh Watermelon production in Philippines. (2024). Tridge. https://www.tridge.com/intelligences/watermelon/PH/production

Ultralytics. (n.d.). Preprocessing Annotated Data. Docs.ultralytics.com. https://docs.ultralytics.com/guides/preprocessing_annotated_data/

wadii. (2025, May 3). Watermelon Cultivation: Complete Guide for Sowing, Growing, and Profit. PermaTerra Farm. https://permaterrafarm.com/en/watermelon-cultivation-guide/

Wakat, Nagcarlan, Laguna Profile – PhilAtlas. (1990, May). Philatlas.com. https://www.philatlas.com/luzon/r04a/laguna/nagcarlan/wakat.html

Wang, F., Zhao, Y., Zheng, J., Qi, K., Fan, Y., Yuan, X., & Ruan, H. (2021). Estimation model of soluble solids content in bagged and non-bagged apple fruits based on spectral data. Computers and Electronics in Agriculture, 191, 106492. https://doi.org/10.1016/j.compag.2021.106492

Welcome To Zscaler Directory Authentication. (2025). Dergipark.org.tr. https://dergipark.org.tr/en/pub/bajece/article/973129?fbclid=IwY2xjawN0mXpleHRuA2FlbQIxMABicmlkETE0NUdEeFZicjd3YkFRSTRVAR5_spv02D0AwGzKhxbVNCDO1L344tXA0C4S1pHLoA8RavR3pLhHUTFnenCZEw_aem_pWBXc1mUF8Eo0ijhy_y-rQ

Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). *How transferable are features in deep neural networks?* ArXiv.org. https://arxiv.org/abs/1411.1792

You, W., Shen, C., Guo, X., Jiang, X., Shi, J., & Zhu, Z. (2017). A hybrid technique based on convolutional neural network and support vector regression for intelligent diagnosis of rotating machinery. 9(6), 168781401770414-168781401770414. https://doi.org/10.1177/1687814017704146

Zhu, Y., Gu, Q., Zhao, Y., Wan, H., Wang, R., Zhang, X., & Cheng, Y. (2022). Quantitative Extraction and Evaluation of Tomato Fruit Phenotypes Based on Image Recognition. Frontiers in Plant Science, 13, 859290. https://doi.org/10.3389/fpls.2022.859290

APPENDICES

APPENDIX A

LETTER OF APPROVAL

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the thesis entitled:

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

JUSTINE MILES DEL ROSARIO, LOUISE ANDREI SAPIN, and JOHN DAVE SILVA have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

Rowan N. Elomina, PhD
Thesis Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The thesis entitled " **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING**" prepared and submitted by **Del Rosario, Justine Miles, Sapin, Louise Andrei** and **Silva, John Dave** in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent System, is hereby recommended for approval and acceptance.

ROWAN N. ELOMINA, PhD
Thesis Adviser

APPENDIX B

CERTIFICATE OF ACCEPTANCE

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

April 21,2025

Eddie Custudio Mesina

Local Farmer

Barangay Wakat, Nagcarlan, Laguna

Subject: Request to gather data of Sweet Gold Watermelon for Academic Thesis

Dear Mr. Mesina,

We are third-year Bachelor of Science in Computer Science students from Laguna State Polytechnic University - Los Baños Campus (LSPU-LBC), conducting research for our thesis titled **"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"**

To support our study, we respectfully request to gather some data of sweet gold Watermelon with their images to visualize the fruit, and take note the external features like height and weight. Specifically, we are seeking sample images to assist us in developing a machine learning model to predict if there is a correlation between aesthetic traits and sweetness level of Sweet Gold Watermelon

Rest assured that the gathered data will be used strictly for academic purposes and will be properly cited in our thesis. We will fully adhere to any guidelines provided by your local farm regarding to gathered data, use, and confidentiality. Should there be any required procedures or formalities, we are willing to comply accordingly.

Thank you very much for your time and kind consideration. We are hoping for a favorable response.

Respectfully,

JUSTINE MILES DEL ROSARIO
justinemiles26.lspu@gmail.com
Researchers, BS Computer Science

LOUISE ANDREI SAPIN
louise.sapin5@gmail.com
Researchers, BS Computer Science

JOHN DAVE SILVA
johndavesilva00@gmail.com
Researchers, BS Computer Science

Noted by:

ROWAN N. ELOMINA, PhD
Thesis Adviser

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

Eddie Custudio Mesina
Local Farmer





Figure B1: Data Gathering Process

APPENDIX C

SURVEY QUESTIONAIRRE

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Interview (Survey) for Victoria, Laguna - Farmers

Melon Farming in General

1. How long have you been planting melons?	We've been planting melons for many years. Before planting, we soak the seeds in a wet cloth overnight. Once they sprout, we transfer them to the soil. It usually takes about 3 days for the seedlings to grow
2. What types of melons have you planted before or are currently planting?	We've planted Sweet Gold watermelon and Sweet Melon since 2018. Before that, from 2005 to 2012, we also planted Honey Dew.
3. Are there melon types you stopped planting? Why?	Yes, we stopped planting Honey Dew because we took a break from farming and later changed to a different variety that was easier to sell.
4. Which type of melon is most popular or wanted by buyers?	Both Sweet Gold and Sweet Melon are popular. Buyers get them right after harvest and bring them to markets like Manila or Sta. Cruz. But Sweet Gold gives better profit because we can harvest more of it, and it sells at ₱35 per kilo. Sweet Melon sells at ₱40 per kilo, but the earnings are not as high.
5. Do buyers look for a specific look, like skin color, size, or shape?	Buyers accept all as long as there are no defects or signs of over ripeness. Fruits with damage are sold at half the price or sometimes not bought at all.

"Sweet Gold Watermelon" Focused

6. Does a good-looking Sweet Gold watermelon always mean it's sweet?	Not always. Even if the fruit looks ripe or reaches 60 days, it may not be sweet, especially if the weather is rainy. Too much water and not enough sun affect its sweetness.
7. How do you know when Sweet Gold is sweet and ready to harvest?	When it reaches 60 days, we will check its small stem. If it's soft and easy to pull,

	it usually means the fruit is ready and sweet.
8. Can you tell if it's sweet just by looking at it?	Sometimes yes. By 60 days, the inside is already yellow, which shows it's ripe. But again, if the soil is too wet or the area has too much shade, it may not be that sweet.
9. Does the season or month affect the sweetness of Sweet Gold?	Yes. During dry months like January to March, fruits grow bigger and sweeter. But during rainy months like May to July, the harvest is lower and not as sweet.
10. Do some Sweet Gold fruits have rough skin or spots? How often does that happen?	Yes, some fruits have rough skin, cracks, or dark spots. It's common. These are usually called "rejects" and are sold at a lower price. Sometimes, if the damage is bad, the fruit becomes sour.
11. What does a good-quality Sweet Gold watermelon look like?	The best ones have smooth skin, even color, and no damage. They also grow better if the soil is not too wet and gets enough sunlight.
12. How much Sweet Gold do you usually harvest? Does the season affect the amount?	Yes, the weather makes a big difference. From January to March, we harvest around 7,000 kilos. But from May to July, we only get about 3000 - 4,000 kilos because of the rain.
13. How many fruits do you get per plant?	It depends on the weather. In the dry season, more fruits grow. In the rainy season, the fruits are fewer and sometimes smaller.
14. Do you use any tools like a refractometer to check sweetness, or just by experience?	We just use experience. For Sweet Gold, if its small stem is soft at 60 days, then it's usually sweet and ready.
15. How can you tell if the fruit is overripe or damaged?	If there are cracks or bruises, the fruit might already be damaged and could taste sour. We try to sort these out before selling.

Eddie Custudio Mesina

Local Farmer / Owner

APPENDIX D

EVALUATION PROCESS & VALIDATION





Figure C1: Validation Process

The researchers conducted the system evaluation and data/model performance validation together with Mr. Eddie Mesina, a Sweetgold Watermelon expert from Wakay, Nagcarlan, and three Information Technology professionals from Calamba City: Mr. Eric Espiridion (IT / Data Analyst), Mr. Rendel Banasihan (IT), and Mr. Dionisio Hermedes II (IT). These evaluators were presented with the actual system outputs and model-generated results to review and assess whether the generated data, predictions, and performance metrics truly represented the conditions and classifications described in the study. Their review strengthened the study's overall validity and reliability by confirming that the system's output was reliable, consistent, and in keeping with reality.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CUCUMIS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

Name (optional): _____. Location: _____Date:_____

Instruction: Please read each statement carefully and choose the option that best represents your opinion or experience with the **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING.**

Use the scale below when answering:

4 – Strongly Agree
3 – Agree
2 – Disagree
1 – Strongly Disagree

Note: There are no right or wrong answers. We only want your honest feedback to help us improve the model.

	4	3	2	1
A. Predictive Accuracy:				
This refers to how well the model predicts sweetness levels.				
1. The prediction model can accurately estimate Brix value based on external watermelon features.				
2. Error values (MAE/RMSE) are within acceptable standards for agricultural produce quality assessment.				
3. The model's R^2 score indicates high performance in prediction.				
4. The model effectively analyzes visual traits to determine sweetness.				
B. PERFORMANCE EFFICIENCY:				
This refers to the speed and computational processing effectiveness.				
1. The model processes image features efficiently.				
2. The computational resources required are reasonable.				
3. The model remains responsive even with larger data input.				
4. The model performance is scalable for future system implementation.				
C. Reliability & Consistency:				
Refers to stability of performance across data variations				
1. The model produces consistent predictions across different tests.				
2. The model remains accurate even with varied sample conditions.				
3. Results show stable performance across training, validation, and testing datasets.				
D. Applicability in Agriculture:				
This refers to practical benefits for farmers and stakeholders				
1. The model can be applied effectively to fruit quality assessment.				
2. The approach supports decision-making for farmers and sellers.				
3. The model remains responsive even with larger data input.				
4. The model has potential for technological integration (e.g., automated sorting).				

“A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING”

Research Objectives:

The primary objective of this study is to develop and evaluate a hybrid machine learning model that demonstrates the correlation between the sweetness level (Brix value regression) of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') and its external aesthetics such as color, texture, and shape, and size using image processing techniques. Specifically, the study aims to:

1. Capture high-resolution images of Sweet Gold watermelons using a controlled imaging setup.
2. Analyze and extract visual features from the images to investigate their correlation with sweetness levels measured using a manual Brix refractometer.
3. Train and compare multiple machine learning algorithms to predict sweetness levels based on image processing techniques.
    a. Convolutional Neural Networks (CNNs)
    b. Random Forest (RF)
    c. Support Vector Regression (SVR)
    d. Extreme Gradient Boosting (XGBoost Regressor)
4. Assess model performance using evaluation metrics.
    a. Coefficient of determination (R^2 score)
    b. Mean Absolute Error (MAE)
    c. Root Mean Squared Error (RMSE)
5. Determine the extent to which external aesthetic features can reliably predict the sweetness level of Sweet Gold watermelons, and present the results through a visual demonstration of the trained model.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

Name (optional): Dionisio N. Hemedez II Location: calamba city Date: 11/03/2025

Instruction: Please read each statement carefully and choose the option that best represents your opinion or experience with the **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING.** Use the scale below when answering:

4 – Strongly Agree
3 – Agree
2 – Disagree
1 – Strongly Disagree

Note: There are no right or wrong answers. We only want your honest feedback to help us improve the model.

A. Predictive Accuracy:

This refers to how well the model predicts sweetness levels.

	4	3	2	1
1. The prediction model can accurately estimate Brix value based on external watermelon features.			/	
2. Error values (MAE/RMSE) are within acceptable standards for agricultural produce quality assessment.		/		
3. The model's R^2 score indicates high performance in prediction.		/		
4. The model effectively analyzes visual traits to determine sweetness.			/	

B. Performance Efficiency:

This refers to the speed and computational processing effectiveness.

	4	3	2	1
1. The model processes image features efficiently.	/			
2. The computational resources required are reasonable.	/			
3. The model remains responsive even with larger data input.	/			
4. The model performance is scalable for future system implementation.	/			

C. Reliability & Consistency:
Refers to stability of performance across data variations

	4	3	2	1
1. The model produces consistent predictions across different tests.	/			
2. The model remains accurate even with varied sample conditions.		/		
3. Results show stable performance across training, validation, and testing datasets.	/			

D. Applicability in Agriculture:
This refers to practical benefits for farmers and stakeholders

	4	3	2	1
1. The model can be applied effectively to fruit quality assessment.			/	
2. The approach supports decision-making for farmers and sellers.		/		
3. The model remains responsive even with larger data input.	/			
4. The model has potential for technological integration (e.g., automated sorting).	/			

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

Name (optional): REDEL BANASIHAN Location: CALAMBA Date: 11-03-25

Instruction: Please read each statement carefully and choose the option that best represents your opinion or experience with the **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING.** Use the scale below when answering:

4 – Strongly Agree
3 – Agree
2 – Disagree
1 – Strongly Disagree

Note: There are no right or wrong answers. We only want your honest feedback to help us improve the model.

A. Predictive Accuracy:

This refers to how well the model predicts sweetness levels.

	4	3	2	1
1. The prediction model can accurately estimate Brix value based on external watermelon features.	✓			
2. Error values (MAE/RMSE) are within acceptable standards for agricultural produce quality assessment.		✓		
3. The model's R^2 score indicates high performance in prediction.		✓		
4. The model effectively analyzes visual traits to determine sweetness.	✓			

B. Performance Efficiency:

This refers to the speed and computational processing effectiveness.

	4	3	2	1
1. The model processes image features efficiently.	✓			
2. The computational resources required are reasonable.	✓			
3. The model remains responsive even with larger data input.		✓		
4. The model performance is scalable for future system implementation.	✓			

C. Reliability & Consistency:

Refers to stability of performance across data variations

	4	3	2	1
1. The model produces consistent predictions across different tests.	✓			
2. The model remains accurate even with varied sample conditions.		✓		
3. Results show stable performance across training, validation, and testing datasets.	✓			

D. Applicability in Agriculture:

This refers to practical benefits for farmers and stakeholders

	4	3	2	1
1. The model can be applied effectively to fruit quality assessment.		✓		
2. The approach supports decision-making for farmers and sellers.	✓			
3. The model remains responsive even with larger data input.	✓			
4. The model has potential for technological integration (e.g., automated sorting).	✓			

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

Name (optional): Eddie Cuytudo MeSiva Location: Wakat, Nagcalan Date: 10-30-25

Instruction: Please read each statement carefully and choose the option that best represents your opinion or experience with the **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING.** Use the scale below when answering:

4 – Strongly Agree
3 – Agree
2 – Disagree
1 – Strongly Disagree

Note: There are no right or wrong answers. We only want your honest feedback to help us improve the model.

A. Predictive Accuracy:

This refers to how well the model predicts sweetness levels.

1. The prediction model can accurately estimate Brix value based on external watermelon features.
2. Error values (MAE/RMSE) are within acceptable standards for agricultural produce quality assessment.
3. The model's R^2 score indicates high performance in prediction.
4. The model effectively analyzes visual traits to determine sweetness.

4	3	2	1
	/		
	/		
	/		
/			

B. Performance Efficiency:

This refers to the speed and computational processing effectiveness.

1. The model processes image features efficiently.
2. The computational resources required are reasonable.
3. The model remains responsive even with larger data input.
4. The model performance is scalable for future system implementation.

4	3	2	1
	/		
/			
	/		
/			

C. Reliability & Consistency:

Refers to stability of performance across data variations

1. The model produces consistent predictions across different tests.
2. The model remains accurate even with varied sample conditions.
3. Results show stable performance across training, validation, and testing datasets.

4	3	2	1
/			
		/	
	/		

D. Applicability in Agriculture:

This refers to practical benefits for farmers and stakeholders

1. The model can be applied effectively to fruit quality assessment.
2. The approach supports decision-making for farmers and sellers.
3. The model remains responsive even with larger data input.
4. The model has potential for technological integration (e.g., automated sorting).

4	3	2	1
	/		
/			
	/		
	/		

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

"A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING"

Name (optional): ERIC V. ESPIRIDION Location: CALAMBA CITY Date: 11-03-25

Instruction: Please read each statement carefully and choose the option that best represents your opinion or experience with the **A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN CUCUMIS LANATUS 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING.** Use the scale below when answering:

4 – Strongly Agree
3 – Agree
2 – Disagree
1 – Strongly Disagree

Note: There are no right or wrong answers. We only want your honest feedback to help us improve the model.

A. Predictive Accuracy:

This refers to how well the model predicts sweetness levels.

	4	3	2	1
1. The prediction model can accurately estimate Brix value based on external watermelon features.			/	
2. Error values (MAE/RMSE) are within acceptable standards for agricultural produce quality assessment.		/		
3. The model's R^2 score indicates high performance in prediction.		/		
4. The model effectively analyzes visual traits to determine sweetness.			/	

B. Performance Efficiency:

This refers to the speed and computational processing effectiveness.

	4	3	2	1
1. The model processes image features efficiently.	/			
2. The computational resources required are reasonable.	/			
3. The model remains responsive even with larger data input.	/			
4. The model performance is scalable for future system implementation.	/			

C. Reliability & Consistency:
Refers to stability of performance across data variations

	4	3	2	1
1. The model produces consistent predictions across different tests.	/			
2. The model remains accurate even with varied sample conditions.		/		
3. Results show stable performance across training, validation, and testing datasets.	/			

D. Applicability in Agriculture:
This refers to practical benefits for farmers and stakeholders

	4	3	2	1
1. The model can be applied effectively to fruit quality assessment.			/	
2. The approach supports decision-making for farmers and sellers.		/		
3. The model remains responsive even with larger data input.	/			
4. The model has potential for technological integration (e.g., automated sorting).	/			

APPENDIX C

CODE SCREENSHOTS

Data Preparation (Visual Studio Code)

```python
# Storage for images + metadata
images, fruit_ids, brix_values, weights, heights, widths = [], [], [], [], [], []
preview_images, preview_info = [], []

for _, row in df.iterrows():
    fruit_id = str(row['FruitID']).strip()
    folder_path = os.path.join(image_dir, fruit_id)

    if os.path.isdir(folder_path):
        added_to_preview = False
        for file in sorted(os.listdir(folder_path)):
            if file.lower().endswith(('.jpg', '.jpeg', '.png')):
                img_path = os.path.join(folder_path, file)
                img = cv2.imread(img_path)

                # ✅ Safety check for corrupted files
                if img is None:
                    continue

                img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
                img = cv2.resize(img, image_size)

                images.append(img)
                fruit_ids.append(fruit_id)
                brix_values.append(row['Brix Value'])
                weights.append(row['Weight'])
                heights.append(row['Height'])
                widths.append(row['Width'])

                if not added_to_preview:
                    preview_images.append(img)
                    preview_info.append((fruit_id, row['Brix Value'], row['Weight'], row['Height'], row['Width']))
                    added_to_preview = True
```

```python
X = np.array(images, dtype=np.uint8)
# ✅ Free memory
images = None

y_fruit = np.array(fruit_ids)
y_brix = np.array(brix_values)
y_weight = np.array(weights)
y_height = np.array(heights)
y_width = np.array(widths)

print(f"✅ Loaded {len(X)} images. Preview samples: {len(preview_images)}")
print(f"\nShow 25 Image Mapping Samples : {len(preview_images)} total sample fruits.")

# Show 25 Sample Fruit
fig, axs = plt.subplots(5, 5, figsize=(18, 12))
axs = axs.ravel()

for i in range(25):
    img = preview_images[i]
    fid, brix, wt, ht, wd = preview_info[i]
    axs[i].imshow(img)
    axs[i].set_title(f"FruitID: {fid}\nBrix: {brix} | Wt: {wt} |\n Ht: {ht} | Wd: {wd}", fontsize=9)
    axs[i].axis('off')

plt.tight_layout()
plt.show()
```

Figure D1: Image Mapping

```python
from sklearn.model_selection import train_test_split

unique_fruits = np.unique(y_fruit)
train_fruits, temp_fruits = train_test_split(unique_fruits, test_size=0.30, random_state=42)
val_fruits, test_fruits = train_test_split(temp_fruits, test_size=0.5, random_state=42)

# ✅ Optimization: Use set for faster indexing
def get_indices(target_fruits, all_fruit_ids):
    target_set = set(target_fruits)
    return [i for i, fid in enumerate(all_fruit_ids) if fid in target_set]

train_idx = get_indices(train_fruits, y_fruit)
val_idx = get_indices(val_fruits, y_fruit)
test_idx = get_indices(test_fruits, y_fruit)

X_train, y_fruit_train = X[train_idx], y_fruit[train_idx]
X_val, y_fruit_val = X[val_idx], y_fruit[val_idx]
X_test, y_fruit_test = X[test_idx], y_fruit[test_idx]

y_brix_train, y_brix_val, y_brix_test = y_brix[train_idx], y_brix[val_idx], y_brix[test_idx]
y_weight_train, y_weight_val, y_weight_test = y_weight[train_idx], y_weight[val_idx], y_weight[test_idx]
y_height_train, y_height_val, y_height_test = y_height[train_idx], y_height[val_idx], y_height[test_idx]
y_width_train, y_width_val, y_width_test = y_width[train_idx], y_width[val_idx], y_width[test_idx]

print(f"✅ Split done: {len(X_train)} train, {len(X_val)} val, {len(X_test)} test images")
```

Figure D2: Data Splitting

```python
import cv2
import numpy as np

def remove_background(img):
    mask = np.zeros(img.shape[:2], np.uint8)
    # ✅ Fix: 2px margin is safer for full-frame fruits
    rect = (2, 2, img.shape[1]-4, img.shape[0]-4)
    bgdModel = np.zeros((1,65), np.float64)
    fgdModel = np.zeros((1,65), np.float64)

    try:
        cv2.grabCut(img, mask, rect, bgdModel, fgdModel, 5, cv2.GC_INIT_WITH_RECT)
        mask2 = np.where((mask==2)|(mask==0), 0, 1).astype('uint8')
        return img * mask2[:, :, np.newaxis]
    except:
        return img

def preprocess_images(images_array):
    X_prep = []
    for img in images_array:
        img_nobg = remove_background(img)
        X_prep.append(img_nobg.astype(np.float32) / 255.0)
    return np.array(X_prep, dtype=np.float32)

X_train = preprocess_images(X_train)
X_val = preprocess_images(X_val)
X_test = preprocess_images(X_test)

print(f"✅ Preprocessing done: images background removed and normalized")
```

```python
X_train_orig = X[train_idx]  # original images corresponding to X_train

num_samples = 5
plt.figure(figsize=(12, 6))

for i in range(num_samples):
    # Original image (before preprocessing)
    plt.subplot(2, num_samples, i+1)
    plt.imshow(X_train_orig[i])
    plt.axis("off")
    plt.title(f"Original {i+1}")

    # Preprocessed image
    plt.subplot(2, num_samples, i+1+num_samples)
    plt.imshow(X_train[i])  # normalized + background removed
    plt.axis("off")
    plt.title(f"Preprocessed {i+1}")

plt.tight_layout()
plt.show()
```

Figure D3 Image Preprocessing

```python
import cv2
import numpy as np

def augment_image(img):
    augmented = [img] # Original
    augmented.append(cv2.flip(img, 1)) # Horizontal
    augmented.append(cv2.flip(img, 0)) # Vertical

    # ✅ Fix: Rotation and Brightness for better variety
    M270 = cv2.getRotationMatrix2D((img.shape[1]//2, img.shape[0]//2), 270, 1)
    augmented.append(cv2.warpAffine(img, M270, (img.shape[1], img.shape[0])))

    bright = np.clip(img * 1.15, 0, 1.0)
    augmented.append(bright)

    return augmented

X_train_aug, y_brix_aug, y_weight_aug, y_height_aug, y_width_aug, y_fruit_aug = [], [], [], [], [], []

for i in range(len(X_train)):
    imgs = augment_image(X_train[i])
    for aug_img in imgs:
        X_train_aug.append(aug_img)
        y_brix_aug.append(y_brix_train[i])
        y_weight_aug.append(y_weight_train[i])
        y_height_aug.append(y_height_train[i])
        y_width_aug.append(y_width_train[i])
        y_fruit_aug.append(y_fruit_train[i])

X_train_aug = np.array(X_train_aug, dtype=np.float32)

print(f"✅ Augmentation done: {len(X_train_aug)} training images")
```

```python
import matplotlib.pyplot as plt
import random
import numpy as np

# Number of original images to display
num_samples = 4
indices = random.sample(range(len(X_train)), num_samples)

# Set up the figure: rows = num_samples, columns = 1 original + 4 augmented = 5
plt.figure(figsize=(15, 3 * num_samples))

for i, idx in enumerate(indices):
    # Compute start index in augmented array
    start_idx = idx * 5  # each original generates 5 images

    # Get the 5 images (original + 4 augmentations)
    images_to_show = X_train_aug[start_idx:start_idx+5]

    for j, img in enumerate(images_to_show):
        if img.dtype != np.uint8:
            img_display = np.clip(img * 255, 0, 255).astype(np.uint8)
        else:
            img_display = img

        plt.subplot(num_samples, 5, i*5 + j + 1)
        if img_display.ndim == 2:
            plt.imshow(img_display, cmap='gray')
        else:
            plt.imshow(img_display)
        plt.axis('off')

        # Set titles
        if j == 0:
            plt.title("Original")
        else:
            plt.title(f"Augmented {j}")

plt.tight_layout()
plt.show()
```

Figure D4: Data Augmentation

```python
np.savez_compressed('train_set.npz', X=X_train_aug, y_fruit=np.array(y_fruit_aug),
                    y_brix=np.array(y_brix_aug), y_weight=np.array(y_weight_aug),
                    y_height=np.array(y_height_aug), y_width=np.array(y_width_aug))

np.savez_compressed('val_set.npz', X=X_val, y_fruit=y_fruit_val, y_brix=y_brix_val,
                    y_weight=y_weight_val, y_height=y_height_val, y_width=y_width_val)

np.savez_compressed('test_set.npz', X=X_test, y_fruit=y_fruit_test, y_brix=y_brix_test,
                    y_weight=y_weight_test, y_height=y_height_test, y_width=y_width_test)

print("✅ Data preparation complete and files saved.")
```

Figure D5: Saving to NPZ File

Model Development (Google Colab)

```python
import numpy as np
from google.colab import files

# Upload the .npz files
uploaded = files.upload()
```

```
Choose Files  No file chosen    Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving test_set.npz to test_set.npz
Saving train_set.npz to train_set.npz
Saving val_set.npz to val_set.npz
```

```python
# Load training set
train_data = np.load('train_set.npz')
X_train = train_data['X']
y_fruit_train = train_data['y_fruit']
y_brix_train = train_data['y_brix']
y_weight_train = train_data['y_weight']
y_height_train = train_data['y_height']
y_width_train = train_data['y_width']

# Load validation set
val_data = np.load('val_set.npz')
X_val = val_data['X']
y_fruit_val = val_data['y_fruit']
y_brix_val = val_data['y_brix']
y_weight_val = val_data['y_weight']
y_height_val = val_data['y_height']
y_width_val = val_data['y_width']

# Load test set
test_data = np.load('test_set.npz')
X_test = test_data['X']
y_fruit_test = test_data['y_fruit']
y_brix_test = test_data['y_brix']
y_weight_test = test_data['y_weight']
y_height_test = test_data['y_height']
```

```
y_width_test = test_data['y_width']

# Convert weight, height, width to float (if not already numeric)
y_weight_train = np.array([w.replace("kg", "") for w in y_weight_train]).astype(float)
y_height_train = np.array([h.replace("cm", "") for h in y_height_train]).astype(float)
y_width_train  = np.array([w.replace("cm", "") for w in y_width_train]).astype(float)

y_weight_val = np.array([w.replace("kg", "") for w in y_weight_val]).astype(float)
y_height_val = np.array([h.replace("cm", "") for h in y_height_val]).astype(float)
y_width_val  = np.array([w.replace("cm", "") for w in y_width_val]).astype(float)

y_weight_test = np.array([w.replace("kg", "") for w in y_weight_test]).astype(float)
y_height_test = np.array([h.replace("cm", "") for h in y_height_test]).astype(float)
y_width_test  = np.array([w.replace("cm", "") for w in y_width_test]).astype(float)
```

Figure D6: Loading CSV File & Final Dataset Images

```
import joblib
import numpy as np
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Brix Scaling (MinMax is usually best for target regression)
brix_scaler = MinMaxScaler()
y_train_scaled = brix_scaler.fit_transform(y_brix_train.reshape(-1, 1)).ravel()
y_val_scaled = brix_scaler.transform(y_brix_val.reshape(-1, 1)).ravel()
y_test_scaled = brix_scaler.transform(y_brix_test.reshape(-1, 1)).ravel()

# Metadata Scaling
X_meta_train = np.stack([y_weight_train, y_height_train, y_width_train], axis=1)
X_meta_val = np.stack([y_weight_val, y_height_val, y_width_val], axis=1)
X_meta_test = np.stack([y_weight_test, y_height_test, y_width_test], axis=1)

meta_scaler = StandardScaler()
X_meta_train_scaled = meta_scaler.fit_transform(X_meta_train)
X_meta_val_scaled = meta_scaler.transform(X_meta_val)
X_meta_test_scaled = meta_scaler.transform(X_meta_test)

# Save scalers immediately so the Output Tool can use them later
joblib.dump(brix_scaler, "brix_scaler.save")
```

Figure D7: Model Preparation

```
import tensorflow as tf
import numpy as np
import math
from tensorflow.keras import layers, models, optimizers
from tensorflow.keras.applications import EfficientNetB0
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
from tensorflow.keras.applications.efficientnet import preprocess_input

# Reproducibility
SEED = 42
tf.random.set_seed(SEED)
np.random.seed(SEED)

# Training Hyperparameters
IMG_SIZE = (224, 224, 3)
BATCH_SIZE = 32
EPOCHS_HEAD = 30          # Phase 1
EPOCHS_FINE = 40          # Phase 2
LR_HEAD = 1e-3            # Head training
LR_FINE = 1e-5            # Fine-tuning

# Regression Metrics
def report_regression_metrics(y_true_scaled, y_pred_scaled, scaler, prefix=""):
    y_true = scaler.inverse_transform(
        np.array(y_true_scaled).reshape(-1, 1)
    ).ravel()
    y_pred = scaler.inverse_transform(
        np.array(y_pred_scaled).reshape(-1, 1)
    ).ravel()

    r2 = r2_score(y_true, y_pred)
    mae = mean_absolute_error(y_true, y_pred)
    rmse = math.sqrt(mean_squared_error(y_true, y_pred))

    print(f"{prefix}R²: {r2:.4f} | MAE: {mae:.4f} | RMSE: {rmse:.4f}")
    return {"r2": r2, "mae": mae, "rmse": rmse}

# Callbacks
def make_callbacks(filepath):
    return [
        EarlyStopping(monitor="val_loss", patience=8, restore_best_weights=True),
        ReduceLROnPlateau(monitor="val_loss", factor=0.3, patience=3, verbose=1),
        ModelCheckpoint(filepath, monitor="val_loss", save_best_only=True, verbose=1)
    ]

# EfficientNet Base
def create_cnn_base(input_shape=IMG_SIZE, trainable=False):
    """
    EfficientNetB0 base for feature extraction
    - include_top=False: removes classification head
    - pooling='avg': outputs 1280-dimensional vector per image
    - trainable=False: frozen for head training
    """
    base_model = EfficientNetB0(
        include_top=False,
        weights="imagenet",
        input_shape=input_shape,
        pooling="avg"
    )
    base_model.trainable = trainable
    return base_model

# CNN Regressor (Images Only)
def create_cnn_regressor(input_shape=IMG_SIZE):
    inputs = layers.Input(shape=input_shape)

    # Base CNN
    base_model = create_cnn_base(input_shape, trainable=False)
    x = base_model(inputs)

    # Regression Head
    x = layers.Dense(256, activation="relu")(x)
```

```
    x = layers.Dropout(0.4)(x)
    x = layers.Dense(128, activation="relu")(x)
    outputs = layers.Dense(1, activation="linear")(x)

    # Compile Model
    model = models.Model(inputs, outputs)
    model.compile(
        optimizer=optimizers.Adam(learning_rate=LR_HEAD),
        loss="mse"
    )

    return model, base_model

# CNN + Metadata Hybrid Regressor
def create_cnn_metadata_regressor(meta_input_dim=3, input_shape=IMG_SIZE):
    img_inputs = layers.Input(shape=input_shape)
    meta_inputs = layers.Input(shape=(meta_input_dim,))

    # Base CNN
    base_model = create_cnn_base(input_shape, trainable=False)
    cnn_features = base_model(img_inputs)

    # Concatenate CNN features + metadata
    concat = layers.Concatenate()([cnn_features, meta_inputs])

    # Regression Head
    x = layers.Dense(256, activation="relu")(concat)
    x = layers.Dropout(0.4)(x)
    x = layers.Dense(128, activation="relu")(x)
    outputs = layers.Dense(1, activation="linear")(x)

    # Compile Model
    model = models.Model([img_inputs, meta_inputs], outputs)
    model.compile(
        optimizer=optimizers.Adam(learning_rate=LR_HEAD),
        loss="mse"
    )
```

```
    return model, base_model

# Image Preprocessing Utility
def preprocess_images(X):
    """
    Preprocess images for EfficientNetB0
    - Converts to float32
    - Applies EfficientNet preprocessing (scaling, normalization)
    """
    return preprocess_input(X.astype(np.float32))
```

Figure D8: EfficientNetB0

```
import tensorflow as tf

# CNN (Images Only) feature extractor
# Output: penultimate layer (Dense(128)) before final Dense(1)
cnn_img_feature_extractor = tf.keras.Model(
    inputs=cnn_model.input,                # image input only
    outputs=cnn_model.layers[-2].output    # Dense(128)
)

cnn_img_feature_extractor.trainable = False

def extract_cnn_img_features(model, X_img, batch_size=8):
    """
    X_img must already be preprocessed using preprocess_images()
    """
    return model.predict(
        X_img,
        batch_size=batch_size,
        verbose=1
    )

X_train_img_features = extract_cnn_img_features(
    cnn_img_feature_extractor,
    X_train_proc,
    batch_size=8
)

X_val_img_features = extract_cnn_img_features(
    cnn_img_feature_extractor,
    X_val_proc,
    batch_size=8
)

X_test_img_features = extract_cnn_img_features(
    cnn_img_feature_extractor,
    X_test_proc,
    batch_size=8
```

Figure D9: Feature Extractor for Hybrids Models (Images Only)

Convolutional Neural Network (EfficientNetB0)

```python
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
from tensorflow.keras import optimizers

# Preprocess images for EfficientNet
X_train_proc = preprocess_images(X_train)
X_val_proc   = preprocess_images(X_val)
X_test_proc  = preprocess_images(X_test)

# Create CNN model (frozen base)
cnn_model, cnn_base = create_cnn_regressor(input_shape=IMG_SIZE)

cnn_model.compile(
    optimizer=optimizers.Adam(learning_rate=LR_HEAD),
    loss="mse"
)

cnn_model.summary()

# Callbacks for HEAD training
callbacks_head = make_callbacks("cnn_baseline_head_best.h5")

# Train HEAD ONLY
history_head = cnn_model.fit(
    X_train_proc,
    y_train_scaled,
    validation_data=(X_val_proc, y_val_scaled),
    batch_size=BATCH_SIZE,
    epochs=EPOCHS_HEAD,
    callbacks=callbacks_head,
    shuffle=True
)

# Fine-tuning EfficientNet top layers
cnn_base.trainable = True

# Freeze lower 70%, fine-tune top 30%
fine_tune_at = int(len(cnn_base.layers) * 0.7)
```

```python
for layer in cnn_base.layers[:fine_tune_at]:
    layer.trainable = False

# Recompile with small LR
cnn_model.compile(
    optimizer=optimizers.Adam(learning_rate=LR_FINE),
    loss="mse"
)

# Callbacks for fine-tuning
checkpoint_path_fine = "cnn_baseline_finetuned_best.h5"
callbacks_fine = make_callbacks(checkpoint_path_fine)

# Fine-tune CNN
history_fine = cnn_model.fit(
    X_train_proc,
    y_train_scaled,
    validation_data=(X_val_proc, y_val_scaled),
    batch_size=BATCH_SIZE,
    epochs=EPOCHS_FINE,
    callbacks=callbacks_fine,
    shuffle=True
)

# Predict on Test Set
y_pred_test_scaled = cnn_model.predict(
    X_test_proc,
    batch_size=BATCH_SIZE
)

# Evaluate and Report Metrics
metrics = report_regression_metrics(
    y_test_scaled,
    y_pred_test_scaled,
    brix_scaler,
    prefix="CNN Baseline - Test: "
)
```

Figure D10: CNN-only (Images only)

```python
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
from tensorflow.keras import optimizers

# Preprocess images for EfficientNet
X_train_proc = preprocess_images(X_train)
X_val_proc   = preprocess_images(X_val)
X_test_proc  = preprocess_images(X_test)

# Create CNN + Metadata model (frozen base)
cnn_meta_model, cnn_base = create_cnn_metadata_regressor(
    meta_input_dim=X_meta_train_scaled.shape[1],
    input_shape=IMG_SIZE
)

cnn_meta_model.compile(
    optimizer=optimizers.Adam(learning_rate=LR_HEAD),
    loss="mse"
)

cnn_meta_model.summary()

# Callbacks for training
checkpoint_path_meta = "cnn_metadata_best.h5"
callbacks_meta = make_callbacks(checkpoint_path_meta)

# Train CNN + Metadata model
history_meta = cnn_meta_model.fit(
    (X_train_proc, X_meta_train_scaled),        # images + metadata
    y_train_scaled,                             # scaled Brix
    validation_data=((X_val_proc, X_meta_val_scaled), y_val_scaled),
    batch_size=BATCH_SIZE,
    epochs=EPOCHS_HEAD,                         # can start with same epochs as CNN head
    callbacks=callbacks_meta,
    shuffle=True
)
```

```python
# Fine-tuning EfficientNet top layers (optional)
cnn_base.trainable = True

# Freeze lower 70%, fine-tune top 30%
fine_tune_at = int(len(cnn_base.layers) * 0.7)
for layer in cnn_base.layers[:fine_tune_at]:
    layer.trainable = False

# Recompile with smaller learning rate
cnn_meta_model.compile(
    optimizer=optimizers.Adam(learning_rate=LR_FINE),
    loss="mse"
)

# Callbacks for fine-tuning
checkpoint_path_fine = "cnn_metadata_finetuned_best.h5"
callbacks_fine = make_callbacks(checkpoint_path_fine)

# Fine-tune CNN + Metadata
history_fine = cnn_meta_model.fit(
    (X_train_proc, X_meta_train_scaled),
    y_train_scaled,
    validation_data=((X_val_proc, X_meta_val_scaled), y_val_scaled),
    batch_size=BATCH_SIZE,
    epochs=EPOCHS_FINE,
    callbacks=callbacks_fine,
    shuffle=True
)

# Predict on Test Set
y_test_pred_meta_scaled = cnn_meta_model.predict(
    (X_test_proc, X_meta_test_scaled),
    batch_size=BATCH_SIZE
)
```

Figure D11: CNN + metadata (End-to-End Fusion)

```python
# FEATURE EXTRACTION
def extract_deep_features(model, img_gen, meta_data=None, batch_size=32):
    extractor = models.Model(inputs=model.inputs, outputs=model.get_layer("deep_feature").output)
    if meta_data is not None:
        combined = metadata_generator(img_gen, meta_data)
        features = extractor.predict(combined, steps=math.ceil(img_gen.n / batch_size), verbose=1)
    else:
        features = extractor.predict(img_gen, steps=math.ceil(img_gen.n / batch_size), verbose=1)
    return features

# Extract deep features
train_features = extract_deep_features(meta_model, train_gen, X_meta_train)
val_features   = extract_deep_features(meta_model, val_gen, X_meta_val)
test_features  = extract_deep_features(meta_model, test_gen, X_meta_test)
```

```
99/99 ━━━━━━━━━━━━━━━━━━━━ 195s 2s/step
22/22 ━━━━━━━━━━━━━━━━━━━━ 49s 2s/step
21/21 ━━━━━━━━━━━━━━━━━━━━ 53s 2s/step
```

Figure D12: Extract Deep Features for ML Hybrids

Random Forest

```python
from sklearn.ensemble import RandomForestRegressor
import numpy as np

# Flatten targets for RF
# Use MinMax-scaled Brix values
y_train_rf_img = y_train_scaled.ravel()
y_val_rf_img   = y_val_scaled.ravel()
y_test_rf_img  = y_test_scaled.ravel()

# Initialize Random Forest Regressor (images only)
rf_img_model = RandomForestRegressor(
    n_estimators=300,
    max_depth=20,
    min_samples_split=5,
    random_state=SEED,
    n_jobs=-1  # Use all CPU cores
)

# Train RF using CNN images-only features
rf_img_model.fit(X_train_img_features, y_train_rf_img)

# Evaluate on Validation Set
y_val_rf_img_pred = rf_img_model.predict(X_val_img_features)

# Report metrics
report_regression_metrics(
    y_val_rf_img,
    y_val_rf_img_pred,
    brix_scaler,
    prefix="CNN + RF (Images Only) - Val: "
)

# Evaluate on Test Set
y_test_rf_img_pred = rf_img_model.predict(X_test_img_features)
```

```python
# Report metrics
report_regression_metrics(
    y_test_rf_img,
    y_test_rf_img_pred,
    brix_scaler,
    prefix="CNN + RF (Images Only) - Test: "
)
```

Figure D13: CNN-only + Random Forest

```python
from sklearn.ensemble import RandomForestRegressor
import numpy as np

# Flatten targets for RF (CNN + Metadata)
y_train_rf_meta = y_train_scaled.ravel()
y_val_rf_meta   = y_val_scaled.ravel()
y_test_rf_meta  = y_test_scaled.ravel()

# Initialize Random Forest Regressor (CNN + Metadata)
rf_meta_model = RandomForestRegressor(
    n_estimators=300,
    max_depth=20,
    min_samples_split=5,
    random_state=SEED,
    n_jobs=-1  # Use all CPU cores
)

# Train RF using CNN + Metadata features
rf_meta_model.fit(X_train_meta_features, y_train_rf_meta)

# Evaluate on Validation Set
y_val_rf_meta_pred = rf_meta_model.predict(X_val_meta_features)

report_regression_metrics(
    y_val_rf_meta,
    y_val_rf_meta_pred,
    brix_scaler,
    prefix="CNN + Metadata + RF - Val: "
)

# Evaluate on Test Set
y_test_rf_meta_pred = rf_meta_model.predict(X_test_meta_features)

report_regression_metrics(
    y_test_rf_meta,
    y_test_rf_meta_pred,
    brix_scaler,
```

Figure D14: CNN + metadata + Random Forest

Support Vector Regressor

```python
from sklearn.svm import SVR
from sklearn.preprocessing import StandardScaler
import numpy as np

# Flatten targets for SVR (images only)
y_train_svr_img = y_train_scaled.ravel()
y_val_svr_img   = y_val_scaled.ravel()
y_test_svr_img  = y_test_scaled.ravel()

# Scale CNN images-only features (required for SVR)
svr_img_feature_scaler = StandardScaler()

X_train_svr_img = svr_img_feature_scaler.fit_transform(X_train_img_features)
X_val_svr_img   = svr_img_feature_scaler.transform(X_val_img_features)
X_test_svr_img  = svr_img_feature_scaler.transform(X_test_img_features)

# Initialize Support Vector Regressor (images only)
svr_img_model = SVR(
    kernel="rbf",
    C=10,
    epsilon=0.1,
    gamma="scale"
)

# Train SVR using CNN images-only features
svr_img_model.fit(X_train_svr_img, y_train_svr_img)

# Evaluate on Validation Set
y_val_svr_img_pred = svr_img_model.predict(X_val_svr_img)

report_regression_metrics(
    y_val_svr_img,
    y_val_svr_img_pred,
    brix_scaler,
    prefix="CNN + SVR (Images Only) - Val: "
)
```

```python
# Evaluate on Test Set
y_test_svr_img_pred = svr_img_model.predict(X_test_svr_img)

report_regression_metrics(
    y_test_svr_img,
    y_test_svr_img_pred,
    brix_scaler,
    prefix="CNN + SVR (Images Only) - Test: "
)
```

Figure D15: CNN-only + Support Vector Regressor

```python
from sklearn.svm import SVR
from sklearn.preprocessing import StandardScaler
import numpy as np

# Flatten targets for SVR (CNN + Metadata)
y_train_svr_meta = y_train_scaled.ravel()
y_val_svr_meta   = y_val_scaled.ravel()
y_test_svr_meta  = y_test_scaled.ravel()

# Scale CNN + Metadata features (required for SVR)
feature_scaler_meta = StandardScaler()

X_train_svr_meta = feature_scaler_meta.fit_transform(X_train_meta_features)
X_val_svr_meta   = feature_scaler_meta.transform(X_val_meta_features)
X_test_svr_meta  = feature_scaler_meta.transform(X_test_meta_features)

# Initialize Support Vector Regressor (CNN + Metadata)
svr_meta_model = SVR(
    kernel="rbf",
    C=10,
    epsilon=0.1,
    gamma="scale"
)

# Train SVR using CNN + Metadata features
svr_meta_model.fit(X_train_svr_meta, y_train_svr_meta)

# Evaluate on Validation Set
y_val_svr_meta_pred = svr_meta_model.predict(X_val_svr_meta)

report_regression_metrics(
    y_val_svr_meta,
    y_val_svr_meta_pred,
    brix_scaler,
    prefix="CNN + Metadata + SVR - Val: "
)
```

Figure D16: CNN + metadata + Support Vector Machine

Extreme Gradient Boosting

```python
from xgboost import XGBRegressor
import numpy as np

# Flatten targets for XGBoost (images only)
y_train_xgb_img = y_train_scaled.ravel()
y_val_xgb_img   = y_val_scaled.ravel()
y_test_xgb_img  = y_test_scaled.ravel()

# Initialize XGBoost Regressor (images only)
xgb_img_model = XGBRegressor(
    n_estimators=500,
    learning_rate=0.05,
    max_depth=8,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_lambda=1.0,
    random_state=SEED,
    n_jobs=-1,
    objective="reg:squarederror",
    tree_method="hist",
    eval_metric="rmse"
)

# Train XGBoost using CNN images-only features
xgb_img_model.fit(X_train_img_features, y_train_xgb_img)

# Evaluate on Validation Set
y_val_xgb_img_pred = xgb_img_model.predict(X_val_img_features)

report_regression_metrics(
    y_val_xgb_img,
    y_val_xgb_img_pred,
    brix_scaler,
    prefix="CNN + XGB (Images Only) - Val: "
)
```

```python
# Evaluate on Test Set
y_test_xgb_img_pred = xgb_img_model.predict(X_test_img_features)

report_regression_metrics(
    y_test_xgb_img,
    y_test_xgb_img_pred,
    brix_scaler,
    prefix="CNN + XGB (Images Only) - Test: "
)
```

Figure D17: CNN-only + XGBoost

```python
from xgboost import XGBRegressor
import numpy as np

# Flatten targets for XGBoost (CNN + Metadata)
y_train_xgb_meta = y_train_scaled.ravel()
y_val_xgb_meta   = y_val_scaled.ravel()
y_test_xgb_meta  = y_test_scaled.ravel()

# Initialize XGBoost Regressor (CNN + Metadata)
xgb_meta_model = XGBRegressor(
    n_estimators=500,
    learning_rate=0.05,
    max_depth=8,
    subsample=0.8,
    colsample_bytree=0.8,
    reg_lambda=1.0,
    random_state=SEED,
    n_jobs=-1,
    objective="reg:squarederror",
    tree_method="hist",
    eval_metric="rmse"
)

# Train XGBoost using CNN + Metadata features
xgb_meta_model.fit(X_train_meta_features, y_train_xgb_meta)

# Evaluate on Validation Set
y_val_xgb_meta_pred = xgb_meta_model.predict(X_val_meta_features)

report_regression_metrics(
    y_val_xgb_meta,
    y_val_xgb_meta_pred,
    brix_scaler,
    prefix="CNN + Metadata + XGB - Val: "
)
```

Figure D18: CNN + metadata + XGBoost

Predictive Tool (Best Performing Model)

```python
import numpy as np
import tensorflow as tf
from tensorflow.keras.applications.efficientnet import preprocess_input
import joblib
from sklearn.ensemble import RandomForestRegressor
import gradio as gr
from PIL import Image
import os

# Load Saved Models & Scaler
# CNN feature extractor (EfficientNetB0 base)
cnn_feature_extractor_path = "cnn_img_feature_extractor.h5"
cnn_feature_extractor = tf.keras.models.load_model(cnn_feature_extractor_path, compile=False)
cnn_feature_extractor.trainable = False

# MinMaxScaler for Brix
brix_scaler_path = "brix_scaler.save"
brix_scaler = joblib.load(brix_scaler_path)

# Trained CNN + RF model
rf_model_path = "cnn_rf_img_best.pkl"
rf_model = joblib.load(rf_model_path)

# Prediction Function
def predict_sweetness(img: Image.Image):
    try:
        # Preprocess Image
        img = img.convert("RGB").resize((224, 224))
        img_array = np.array(img, dtype="float32")
        img_array = np.expand_dims(img_array, axis=0)
        img_array = preprocess_input(img_array)  # EfficientNet preprocessing

        # Extract Features
        cnn_features = cnn_feature_extractor.predict(img_array, verbose=0)
        cnn_features = cnn_features.reshape((cnn_features.shape[0], -1))

        # Predict with Random Forest
        pred_scaled = rf_model.predict(cnn_features)

        # Convert to original Brix values
        pred_brix = brix_scaler.inverse_transform(pred_scaled.reshape(-1, 1))[0][0]

        # Categorize Sweetness
        if pred_brix < 8:
            category = "Unripe 🍏 (Low sweetness)"
        elif 8 <= pred_brix <= 10.9:
            category = "Ripe 🍉 (Moderately sweet)"
        elif 11 <= pred_brix <= 13.9:
            category = "Sweet 🍓 (Good quality)"
        else:
            category = "Very Sweet 🍯 (Premium quality)"

        return f"Predicted Sweetness: {pred_brix:.1f} °Brix\n\nInterpretation: {category}"

    except Exception as e:
        import traceback
        traceback.print_exc()
        return f"⚠️ Error: {str(e)}"

# Gradio Interface
css = """
.gradio-container {
    background-color: #919689 !important;
    border-radius: 20px;
    padding: 25px;
    box-shadow: 0 4px 15px rgba(0,0,0,0.1);
}
.upload-box {
    border: 3px dashed #00796B !important;
    border-radius: 15px !important;
    background-color: #6F7378 !important;
    width: 350px !important;
```

```
    background-color: #6F7378 !important;
    width: 350px !important;
    height: 350px !important;
    display: flex;
    align-items: center;
    justify-content: center;
    object-fit: contain !important;
}
.output-text {
    text-align: center;
    font-size: 20px;
    font-weight: bold;
    color: #2E7D32;
}
"""

interface = gr.Interface(
    fn=predict_sweetness,
    inputs=gr.Image(type="pil", label="Upload Fruit Image", elem_classes=["upload-box"]),
    outputs=gr.Textbox(label="Predicted °Brix Level", elem_classes=["output-text"]),
    title="🍉 SweetGold - AI Sweetness Prediction System",
    description=(
        "Upload an image of a fruit (e.g., Sweet Gold watermelon). "
        "The system predicts sweetness (°Brix) using visual features "
        "extracted by a CNN and a Random Forest regression model."
    ),
    css=css,
    theme="default",
    allow_flagging="never"
)

interface.launch(share=True)
```

Figure D19: Output Tool

APPENDIX E

REVISION MATRIX

MATRIX FOR REVISIONS

Commentors	Consultants	Comments	Page No.	Actions	Page No.
Mike Philip T. Ramos, MIT	Subject Specialist	Re-execution of Data Splitting and Training	27-30	The researchers re-executed data preparation and model training development.	29-31
Mike Philip T. Ramos, MIT	Subject Specialist	Correction of Inconsistencies: The scientific name "*Cucumis lanatus*" must be corrected.	Title Page	The researchers renamed the scientific name to *"Citrullus lanatus."*	Title Page
Sherwin B. Sapin, EdD	Statistic	Chapter 4: Unmet Objectives, Objective: 1-3	39-58	The researchers achieved discussing the result for objective 1-3.	39-46
Jocelyn O. Padallan, MIT	Technical	Do not cut Tables	35-51	The researchers achieved not cutting the tables.	25-66
Sherwin B. Sapin, EdD	Statistic	No Testing Phase *(include 15 more samples for testing)*	57-58	The researchers achieved testing 15 more samples in a developed prototype.	61-67

APPENDIX F

GRAMMARLY & TURNITIN REPORT

Grammarly Report

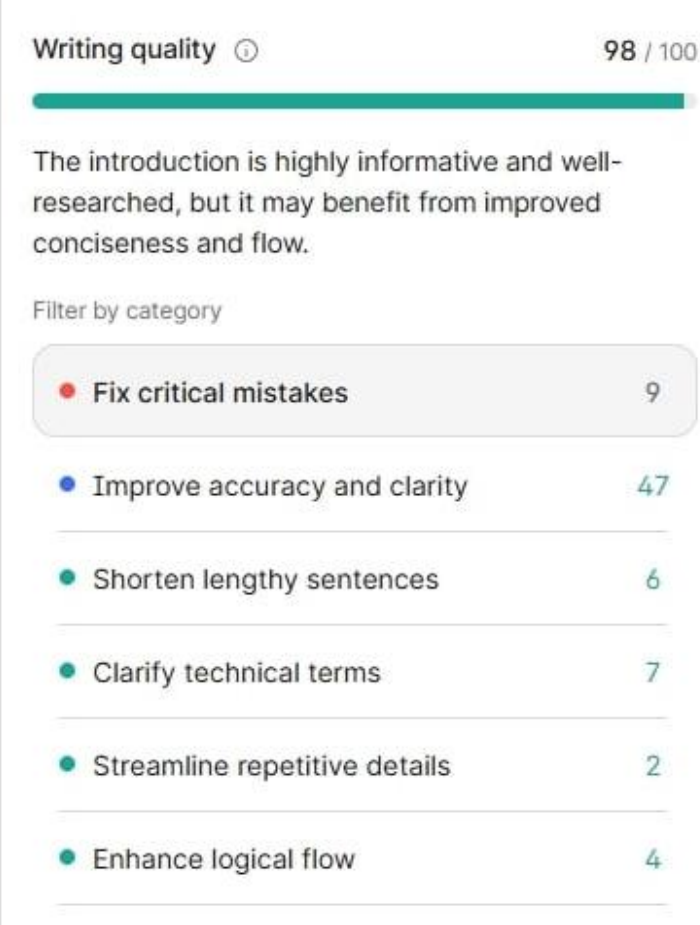


Figure E1: Grammarly Proofreader

Turnitin Report

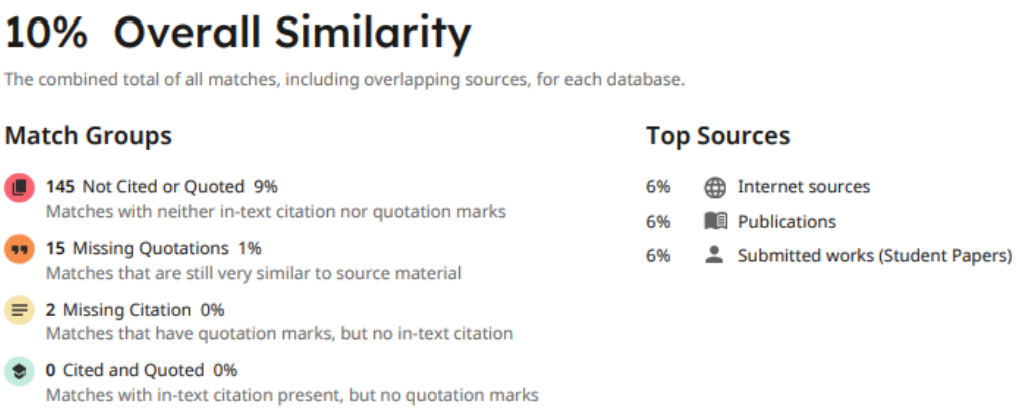


Figure E2: Plagiarism Checker

APPENDIX G

PUBLISHABLE PAPER (IEEE FORMAT)

A HYBRID MACHINE LEARNING APPROACH: CORRELATION BETWEEN AESTHETICS TRAITS AND SWEETNESS LEVEL IN *CITRULLUS LANATUS* 'SWEET GOLD' WATERMELON THROUGH IMAGE PROCESSING

JUSTINE MILES DEL ROSARIO, LOUISE ANDREI SAPIN, JOHN DAVE SILVA

College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Article History:
Received by 000000, 2026
Revised by Unknown 0 2026
Accepted by January 2026

Keywords: *Sweet Gold watermelon, °Brix prediction, Image processing, Convolutional neural network, Random Forest, Non-destructive fruit quality assessment*

Corresponding Author:

Justine Miles E. Del Rosario
College of Computer Studies,
Laguna State Polytechnic University,
Malinta, Los Baños, Laguna, Philippines 4030.
Email: justinemiles26.lspu@gmail.com

Abstract –

Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') is a high-value yellow-fleshed cultivar whose market price is strongly influenced by sweetness. However, post-harvest sweetness evaluation remains subjective and inconsistent, particularly under variable environmental conditions. This study investigates the correlation between external aesthetic traits and internal sweetness (°Brix) using image processing and hybrid machine learning techniques. A total of 150 harvested watermelons were imaged under controlled conditions, producing 900 high-resolution images that were expanded to 3,150 samples through data augmentation. Sweetness levels were measured using a handheld Brix refractometer, while physical attributes were recorded as supplementary metadata. Deep visual features were extracted using **EfficientNet-B0**, and multiple regression models were evaluated, including **CNN-only** and **hybrid CNN–machine learning approaches**. Model performance was assessed using **R^2, MAE, and RMSE**. Results indicate that visual features alone are insufficient for accurate sweetness prediction, though they enable reliable estimation within a general range. Among all configurations, the **CNN + Random Forest model** demonstrated the most stable performance **(R^2 score = -1.4112, MAE = 0.2243, and RMSE = 0.0908)**, reflecting weak but non-zero correlation. Feature analysis revealed fruit color as the most influential visual trait. The study highlights both the potential and limitations of non-destructive, image-based sweetness estimation for agricultural quality assessment.

I. Introduction

Watermelon (Citrullus lanatus) originated in Africa over 5,000 years ago and has evolved from hard, pale, bitter types into the sweet dessert cultivars known today. The Sweet Gold watermelon (C. lanatus 'Sweet Gold') is a yellow-fleshed, high-sugar cultivar recognized for its bright β-carotene pigmentation, crisp texture, and consistently high Brix value of 14–16°. This variety performs well in tropical Philippine conditions, especially in Laguna during the dry season where yields can reach 7,000 kg per cycle. However, the rainy season reduces sweetness, causes skin defects, and lowers price value from ₱35/kg to ₱15–₱20/kg due to over-ripeness and fruit damage. Despite its value, farmers continue to rely on subjective visual signs of maturity rather than scientific assessment, making post-harvest quality evaluation inconsistent and prone to error.

To address this inconsistency, this study focuses on predicting sweetness level of post-harvest Sweet Gold watermelon based solely on visual appearance using machine learning. Although every harvested fruit is technically market-ready, external appearance is not always a reliable indicator of true sweetness due to environmental variability such as rainfall, soil composition, and sunlight exposure. Machine learning can identify subtle visual patterns that humans cannot detect, allowing for non-destructive, objective, and data-driven sweetness estimation. This approach has the potential to support precision agriculture, reduce labor and grading errors, improve yield value, and enhance market consistency by providing a more accurate, standardized basis for post-harvest fruit classification.

1.1 Research Objectives

The primary objective of this study is to develop and evaluate a hybrid machine learning model that demonstrates the correlation between the sweetness level (Brix value regression) of Sweet Gold watermelon (*Citrullus lanatus* 'Sweet Gold') and its external aesthetics such as color, texture, and shape, and size using image processing techniques. Specifically, the study aims to:

1. Capture 1000 high-resolution images of Sweet Gold watermelons using a controlled imaging setup.
2. Analyze and extract visual features from the images to investigate their correlation with sweetness levels measured using a manual Brix refractometer.
3. Train and compare multiple hybrid machine learning algorithms to predict sweetness levels based on image processing techniques.
   a. Convolutional Neural Networks (CNNs)
   b. Random Forest (RF)
   c. Support Vector Regression (SVR)
   d. Extreme Gradient Boosting (XGBoost Regressor)
4. Assess model performance using evaluation metrics.
   a. Coefficient of determination (R^2 score)
   b. Mean Absolute Error (MAE)
   c. Root Mean Squared Error (RMSE).
5. Determine the extent to which external aesthetic features can reliably predict the sweetness level of Sweet Gold watermelons, and present the results through a visual demonstration of the trained model.

1.2 Scope and Limitations

This study aims to develop a regression model that estimates the sweetness level of Sweet Gold watermelon (Cucumis lanatus 'Sweet Gold') based on external visual features using image processing and machine learning techniques. The data is collected from harvested Sweet Gold watermelon grown in a farm located in Brgy. Wakat, Calauan - Nagcarlan Rd, Laguna, capturing 900 images in 150 watermelons. The study involves extracting features such as color, shape, size, and texture, and applying machine learning algorithms including Convolutional Neural Networks (CNN), Random Forest (RF), Support Vector Regression (SVR), and XGBoost Regressor. The dataset will be divided into training (70%), validation (15%) and testing (15%) set to assess model performance using standard evaluation metrics like R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The study is limited to post-harvest fruit analysis along with a prototype, a demo interface, excluding development system, or evaluation of non-visual traits such as aroma, internal texture, nutritional value, and environmental factors.

II. Related Literatures

This study lies in its comprehensive review of current machine learning and computer vision applications for fruit quality assessment, particularly through image-based analysis. Literature widely supports deep learning, particularly CNNs, for accurately classifying visual traits like ripeness and detecting spoiled fruits. However, most existing models overlook internal qualities like sweetness, which are equally important for consumer satisfaction and market value (Sangeetha et al., 2024; Aherwadi et al., 2022; Dhanya et al., 2022). Some studies explored sweetness using non-image data like acoustic signals (Chawgien & Kiattisin, 2021) or environmental inputs (Helyes et al., 2024), but these lacked full automation or real-time visual analysis. To address these gaps, this research proposes a hybrid machine learning approach that uses image-based features to estimate sweetness levels in *Citrullus Lanatus* (Sweet Gold watermelon). This approach builds upon current models by emphasizing internal quality prediction without destructive testing in addition to visual ripeness.

This research provides a timely and useful answer for farmers, retailers, and quality manage researchers requiring quick, non-invasive tools to guarantee product quality through an image-based system that correlates aesthetic features with sweetness.

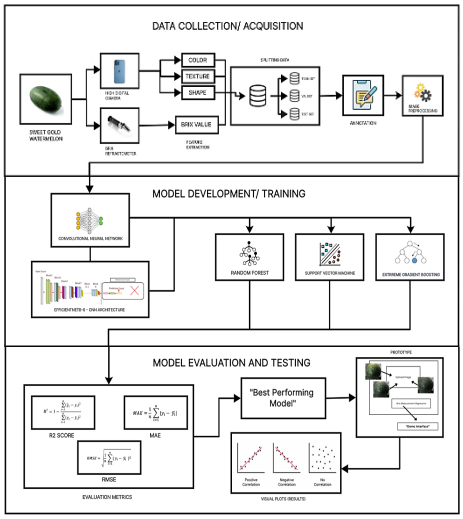


Figure 1: Conceptual Framework of the Study

III. Methodology

3.1 Data Acquisition Implementation

Data acquisition involved capturing high-resolution images of each Sweet Gold watermelon from six different angles to document variations in color, texture, and shape. In parallel, physical attributes including weight, height, and width were measured using calibrated instruments to generate supporting metadata. Sweetness levels, serving as the target variable, were determined by extracting juice from each sample and measuring °Brix values using a handheld manual refractometer. All visual, physical, and sweetness measurements were

systematically recorded to ensure a comprehensive and reliable dataset for analysis.

Table 4: Sweetness/Brix Measurement Basis (Santos, 2010)

Brix Scale	Description
8-10	Moderately sweet
11-13	Acceptably sweet
14-16	Very sweet

After data collection, all high-resolution images were annotated, standardized, and linked to their corresponding physical metadata (weight, height, and width) using a unique Fruit ID, ensuring consistency across samples. The dataset was split at the fruit level to prevent data leakage, assigning all images of a single fruit exclusively to the training (70%), validation (15%), or test (15%) sets, resulting in 105 training fruits (630 images), 22 validation fruits (132 images), and 23 test fruits (138 images). Preprocessing steps were applied to improve data quality, including background removal using the GrabCut algorithm to isolate the fruit, resizing images to 224 × 224 pixels, and normalizing pixel values to the [0,1] range. Data augmentation was applied only to the training set, where four additional samples were generated per image, expanding the training dataset fivefold to a total of 3,150 images and enhancing model generalization without affecting validation and test integrity.

3.2. Convolutional Neural Network (CNN)

A deep learning architecture widely used for computer vision tasks due to its ability to automatically extract hierarchical visual features from raw images. Traditionally, CNNs consist of a feature extraction stage using convolutional and pooling layers and a prediction stage using fully connected layers. In this study, CNNs are applied in a hybrid regression-based framework rather than classification, where the final layers are designed to output continuous °Brix values instead of discrete classes. A pre-trained EfficientNetB0 model is employed as the backbone for robust feature extraction, followed by dense layers that perform sweetness level regression.

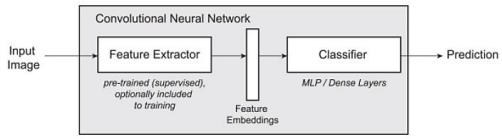


Figure 2: Convolutional Neural Network (Vashwani et al., 2017

3.3. Random Forest

A non-parametric ensemble learning method that combines multiple decision trees to model complex, non-linear relationships and deliver robust predictive performance. In prior studies, CNN–Random Forest hybrid models have demonstrated improved accuracy by replacing traditional fully connected layers with Random Forests for final prediction, particularly when working with high-dimensional CNN-extracted features and limited datasets. This approach helps reduce overfitting and enhances generalization. In this study, a similar hybrid framework is employed, where EfficientNetB0 serves as the feature extractor for capturing layered visual characteristics of Sweet Gold watermelon images, and Random Forest is used to regress continuous °Brix values based on the extracted features.

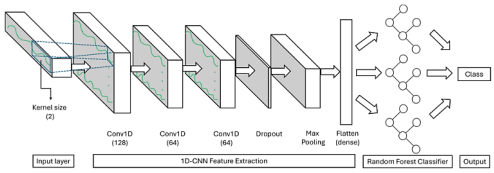


Figure 3: Random Forest, Hybrid CNN feature extraction (Sudiana et al., 2025)

3.4. Support Vector Regression (SVR)

Statistical learning theory–based method derived from Support Vector Machines (SVMs), originally developed for binary classification and later extended to handle continuous-valued prediction tasks. SVR has been widely applied in regression and time-series prediction due to its strong theoretical foundation and ability to model non-linear relationships. In previous studies, CNN–SVR hybrid models were implemented by using CNNs as automatic feature extractors and replacing fully connected layers with an SVR regressor to generate final predictions, often incorporating dimensional control, pooling operations, and sparse regularization to improve efficiency and generalization. In this study, a similar hybrid framework is adopted, where EfficientNetB0 serves as the backbone for visual feature extraction, and SVR is employed as the regression model to predict continuous °Brix values from the extracted features.

Formula 1: $$D_{j_1} = D_{j_1-1} - C_i + 1,\ j_1 = 2,4,6;\ \ i = 1,2,3 \quad (1)$$

Formula 2: $$D_{j_2} = \frac{D_{j_2-1}}{P_i},\ j_2 = 3,5,7;\ \ i = 1,2,3 \quad (2)$$

Formula 3: $$D_8 = n_k \times D_7 \times D_7 \quad (3)$$

Formula 4: $$\tilde{E}^n = E^n + \sum_{i,j} |\sigma_{ij}| \quad (4)$$

Formula 5: $$\frac{\partial \tilde{E}}{\partial W^l} = \frac{\partial E}{\partial W^l} + \frac{\partial \varphi}{\partial W^l} \quad (5)$$

3. Extreme Gradient Boosting (XGBoost)

A machine learning algorithm that enhances predictive performance by combining sequential decision trees with regularization to reduce overfitting. Built upon the Gradient Boosted Decision Tree (GBDT) framework, XGBoost trains trees iteratively, with each model learning from the errors of its predecessor, while incorporating optimization and regularization strategies for improved efficiency and robustness. In this study, XGBoost is integrated into a hybrid learning framework, where EfficientNetB0 is used to extract high-level visual features from Sweet Gold watermelon images, and XGBoost performs

continuous °Brix regression based on these extracted features.

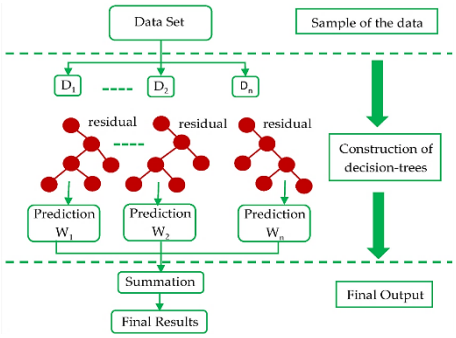


Figure 11: Structure of XGBoost Algorithm (Amjad et al., 2022 & Chen & Guestrin)

IV. Results and Discussion

The outcome compares the predictive performance of CNN-only and hybrid models using R^2, MAE, and RMSE to evaluate the effectiveness of external visual features from Sweet Gold watermelon images for sweetness prediction. The hybrid CNN + Random Forest (CNN + RF) model achieved the best overall performance, yielding the highest R^2 (−0.0295) and the lowest MAE (0.1623) and RMSE (0.0388). Although the R^2 value remained negative, CNN + RF demonstrated more stable and consistent predictions than the other models, likely due to the ensemble capability of Random Forest in modeling non-linear relationships within high-dimensional CNN-extracted features.

The CNN-only and CNN + XGBoost models showed moderate performance with R^2 values of −0.1648 and −0.1331, respectively, indicating limited effectiveness in capturing the relationship between visual appearance and sweetness. In contrast, the CNN + SVR model performed poorly (R^2 = −0.3280), suggesting difficulties in generalization when handling complex, high-dimensional feature spaces.

Models incorporating metadata generally exhibited degraded performance. In particular, CNN + Metadata + SVR recorded the lowest R^2 (−1.7725) and the highest error values, while CNN + Metadata + RF and CNN + Metadata + XGBoost also performed poorly. These results indicate that the additional metadata did not contribute meaningful predictive information beyond visual features.

Table 2: Hybrid Machine Learning Models Performance Results

Model	R ² score	MAE	RMSE
CNN-only	-0.1648	0.1743	0.0439
CNN + RF	-0.0295	0.1623	0.0388
CNN + SVR	-0.3280	0.1884	0.0500
CNN + XGB	-0.1331	0.1690	0.0427
CNN + Metadata	-0.2399	0.1786	0.0467
CNN + Metadata + RF	-1.4112	0.2243	0.0908
CNN + Metadata + SVR	-1.7725	0.2478	0.1044
CNN + Metadata + XGB	-1.4735	0.2270	0.0931

Despite the capabilities of the proposed model, exact °Brix prediction was not expected, as fruit sweetness is governed by internal biochemical processes that are not fully observable from surface-level visual features. This inherent limitation explains the occurrence of negative correlations during testing and underscores the difficulty of relying solely on visual information for precise sweetness estimation. Experimental observations indicate that the model primarily relies on color as the dominant visual cue, reinforcing the conclusion that color provides a stronger predictive signal than other aesthetic traits such as shape, size, or texture.

The results further demonstrate that visual features alone, even when combined with basic physical metadata (e.g., weight, height, and width), are insufficient for accurately predicting internal fruit quality. While CNN-based image analysis can capture general sweetness trends and relative differences, it cannot fully model the complex biological mechanisms influencing °Brix levels. Consequently, the model is more suitable for relative assessment rather than exact measurement.

Overall, the CNN + Random Forest system proves effective as a rapid, non-destructive screening tool for estimating relative sweetness levels. It can support agricultural monitoring, quality control, and post-harvest decision-making by identifying fruits that are likely sweeter or less sweet and tracking general sweetness trends over time. However, its outputs should be interpreted as indicative rather than precise, emphasizing that direct biochemical testing remains essential for accurate °Brix determination.

V. Conclusion and Recommendation

5.1 Conclusion

Based on the study objectives and results, it is concluded that controlled high-resolution image acquisition effectively captured consistent external aesthetic traits of Sweet Gold watermelons, and data augmentation improved model learning despite limited raw samples. Sweetness values were successfully measured using a manual Brix refractometer; however, analysis confirmed that visual features alone, even when combined with basic physical measurements such as weight, height, and width, are insufficient to fully explain internal sweetness due to complex biological processes. Multiple regression-based models were trained and compared using standardized preprocessing techniques, including background removal, resizing, normalization, and metadata alignment, to assess the contribution of visual and numerical features. Among all evaluated approaches, the CNN + Random Forest model demonstrated the most stable and reliable performance based on R^2, MAE, and RMSE metrics, leading to its selection as the core model for the developed prediction prototype. The results further

revealed that skin color is the most influential external feature related to sweetness, while other visual traits showed weaker correlations. Overall, the study demonstrates that CNN-based hybrid models are effective as non-destructive screening tools for estimating relative sweetness trends, with results visualized through clustering, correlation, and feature-importance analyses, although precise °Brix prediction still requires direct biochemical measurement.

5.2 Recommendations

Expand the dataset: Increase the number and diversity of samples to reduce overfitting and improve model generalization across different farms, growing conditions, and fruit appearances.

Include field-acquired images: Capture images directly from the field to enhance robustness under natural lighting and environmental conditions and enable early, pre-harvest sweetness assessment.

Integrate with a physical deployment device: Develop an Arduino or microcontroller-based prototype with a camera module to validate real-world usability in farms and markets beyond laboratory settings.

Collect richer and standardized metadata: Incorporate additional structured variables (e.g., farm origin, planting method, environmental factors) and standardize measurement protocols to reduce noise and improve regression reliability.

Extend to other fruit crops: Apply the hybrid CNN + Random Forest approach to different fruit types to evaluate cross-crop generalizability and increase agricultural applicability.

Incorporate additional non-invasive sensing: Combine visual analysis with technologies such as near-infrared or hyperspectral sensing to better capture internal biochemical properties and improve sweetness prediction accuracy.

References

[1] A Fruit Ripeness Detection Method using Adapted Deep Learning-based Approach - ProQuest. (2023). Proquest.com. https://www.proquest.com/openview/4474b00348fc6513dddadf424eb69bdc/1?pq-origsite=gscholar&cbl=5444811

[2] Achour, B., Belkadi, M., Filali, I., Laghrouche, M., & Lahdir, M. (2020). Image analysis for individual identification and feeding behaviour monitoring of dairy cows based on Convolutional Neural Networks (CNN). Biosystems Engineering, 198, 31–49. https://doi.org/10.1016/j.biosystemseng.2020.07.019

[3] Adainoo, B., Thomas, A. L., & Krishnaswamy, K. (2023). Correlations between color, textural properties and ripening of the North American pawpaw (Asimina triloba) fruit. Sustainable Food Technology. https://doi.org/10.1039/d2fb00008c

[4] Aguirre-Radilla, J., De, E., José Luis Hernández-Hernández, Carranza-Gómez, J., José Antonio Montero-Valverde, & Martínez-Arroyo, M. (2022). Texture and Color-Based Analysis to Determine the Quality of the Manila Mango Using Digital Image Processing Techniques. Communications in Computer and Information Science, 93–106. https://doi.org/10.1007/978-3-031-19961-5_7

[5] Aherwadi, N., Mittal, U., Singla, J., Jhanjhi, N. Z., Yassine, A., & Hossain, M. S. (2022). Prediction of Fruit Maturity, Quality, and Its Life Using Deep Learning Algorithms. Electronics, 11(24), 4100. https://doi.org/10.3390/electronics11244100

[6] Ajay, A. (2024, September 30). *What are the Advantages and Disadvantages of Random Forest?* Pickl.AI. https://www.pickl.ai/blog/advantages-and-disadvantages-random-forest

[7] Alenazi, M. M., Shafiq, M., Alsadon, A. A., Alhelal, I. M., Alhamdan, A. M., Solieman, Talaat. H. I., Ibrahim, A. A., Shady, M. R., & Saad, M. A. O. (2020). Non-destructive assessment of flesh firmness and dietary antioxidants of greenhouse-grown tomato (Solanum lycopersicum L.) at different fruit maturity stages. Saudi Journal of Biological Sciences, 27(10), 2839–2846. https://doi.org/10.1016/j.sjbs.2020.07.004

[8] Amjad, M., Ahmad, I., Ahmad, M., Wróblewski, P., Kamiński, P., & Amjad, U. (2022). Prediction of Pile Bearing Capacity Using XGBoost Algorithm: Modeling and Performance Evaluation. Applied Sciences, 12(4), 2126. https://doi.org/10.3390/app12042126

[9] Barbosa Júnior, M. R., Santos, R. G. dos, Sales, L. de A., Vargas, R. B. S., Deltsidis, A., & Oliveira, L. P. de. (2025). Image-based and ML-driven analysis for assessing blueberry fruit quality. Heliyon, 11(3), e42288. https://doi.org/10.1016/j.heliyon.2025.e42288

[10] BD. (2025, March 5). Watermelon Production Guide: How to Grow High-Yield and Disease-Resistant Watermelons. Business Diary. https://businessdiary.com.ph/33117/watermelon-production-guide-how-to-grow-high-yield-and-disease-resistant-watermelons/

[11] Bergstra, J., & Bengio, Y. (2012). Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research*, *13*, 281–305. https://www.jmlr.org/papers/volume13/bergstra12a/bergstra12a.pdf

[12] Boukili, M., & Nawfal Jebbor. (2024). Performance Optimization of Convolutional Neural Networks (CNN) in Multilabel Fruit Classification. 182, 1–6. https://doi.org/10.1109/iraset60544.2024.10548403

[13] Calauan agri-trading post to boost farmers' income. (2019). Pna.gov.ph. https://www.pna.gov.ph/articles/1057983

[14] Cesar. (2025). My Laguna. Blogspot.com. https://laguna-living.blogspot.com/2007/06/my-laguna.html

[15] Chakraborty, S. K., A., S., Dubey, K., Jat, D., Chandel, N. S., Potdar, R., Rao, N. R. N. V. G., & Kumar, D. (2023). Development of an optimally designed real-time automatic citrus fruit grading–sorting machine leveraging computer vision-based adaptive deep learning model. Engineering Applications of Artificial Intelligence, 120, 105826. https://doi.org/10.1016/j.engappai.2023.105826

[16] Chandra, S. S., Devarapalli Dharmika, Guntupalli Vijayadurgarao, & V, S. S. (2024). Fruit Classification based on Shape, Color and Texture using Image Processing Techniques.

ResearchGate, 10(02), 100–107. https://www.researchgate.net/publication/389597736_Fruit_Classification_based_on_Shape_Color_and_Texture_using_Image_Processing_Techniques

[17] Chawgien, K., & Kiattisin, S. (2021). Machine learning techniques for classifying the sweetness of watermelon using acoustic signal and image processing. Computers and Electronics in Agriculture, 181, 105938. https://doi.org/10.1016/j.compag.2020.105938

[18] *Citrullus lanatus (Watermelon, Watermelons) | North Carolina Extension Gardener Plant Toolbox*. (2025). Ncsu.edu. https://plants.ces.ncsu.edu/plants/citrullus-lanatus/

[19] Cui, H., Ding, Z., Zhu, Q., Wu, Y., & Gao, P. (2020). Population structure and genetic diversity of watermelon (Citrullus lanatus) based on SNP of chloroplast genome. 3 Biotech, 10(8). https://doi.org/10.1007/s13205-020-02372-5

[20] Deepika Bairwa, & Gaurav Sharma. (2017). Classification of Fruits Based on Shape, Color and Texture using Image Processing Techniques. International Journal of Engineering Research And, V6(12). https://doi.org/10.17577/ijertv6is120057

[21] Dhanya, V. G., Subeesh, A., Kushwaha, N. L., Vishwakarma, D. K., Nagesh Kumar, T., Ritika, G., & Singh, A. N. (2022). Deep learning-based computer vision approaches for smart agricultural applications. Artificial Intelligence in Agriculture, 6. https://doi.org/10.1016/j.aiia.2022.09.007

[22] Dhiman, P., Kaur, A., Balasaraswathi, V. R., Gulzar, Y., Alwan, A. A., & Hamid, Y. (2023). Image Acquisition, Preprocessing and Classification of Citrus Fruit Diseases: A Systematic Literature Review. Sustainability, 15(12), 9643. https://doi.org/10.3390/su15129643

[23] Ergün, E. (2025). High precision banana variety identification using vision transformer-based feature extraction and support vector machine. Scientific Reports, 15(1). https://doi.org/10.1038/s41598-025-95466-0

[24] SWEET GOLD WATERMELON F1. (2025, November 3). Flickr; SWEET GOLD WATERMELON F1 | VERY PROLIFIC HIGH YIELDING VARIE… | Flickr. https://www.flickr.com/photos/52749860@N02/4865380506/in/photostream/

[25] Tan, M., & Le, Q. V. (2019, May 28). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ArXiv.org. https://arxiv.org/abs/1905.11946

[26] Tian, H., Wang, T., Liu, Y., Qiao, X., & Li, Y. (2020). Computer vision technology in agricultural automation —A review. Information Processing in Agriculture, 7(1), 1–19. https://doi.org/10.1016/j.inpa.2019.09.006

[27] Trends of Fresh Watermelon production in Philippines. (2024). Tridge. https://www.tridge.com/intelligences/watermelon/PH/production

[28] Ultralytics. (n.d.). Preprocessing Annotated Data. Docs.ultralytics.com. https://docs.ultralytics.com/guides/preprocessing_annotated_data/

[29] wadii. (2025, May 3). Watermelon Cultivation: Complete Guide for Sowing, Growing, and Profit. PermaTerra Farm. https://permaterrafarm.com/en/watermelon-cultivation-guide/

[30] Wakat, Nagcarlan, Laguna Profile – PhilAtlas. (1990, May). Philatlas.com. https://www.philatlas.com/luzon/r04a/laguna/nagcarlan/wakat.html

[31] Wang, F., Zhao, Y., Zheng, J., Qi, K., Fan, Y., Yuan, X., & Ruan, H. (2021). Estimation model of soluble solids content in bagged and non-bagged apple fruits based on spectral data. Computers and Electronics in Agriculture, 191, 106492. https://doi.org/10.1016/j.compag.2021.106492

[32] Welcome To Zscaler Directory Authentication. (2025). Dergipark.org.tr. https://dergipark.org.tr/en/pub/bajece/article/973129?fbclid=IwY2xjawN0mXpleHRuA2FlbQIxMABicmlkETE0NUdEeFZicjd3YkFRSTRVAR5_spv02D0AwGzKhxbVNCDO1L344tXA0C4S1pHLoA8RavR3pLhHUTFnenCZEw_aem_pWBXc1mUF8Eo0ijhy_y-rQ

[33] Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). *How transferable are features in deep neural networks?* ArXiv.org. https://arxiv.org/abs/1411.1792

[34] You, W., Shen, C., Guo, X., Jiang, X., Shi, J., & Zhu, Z. (2017). A hybrid technique based on convolutional neural network and support vector regression for intelligent diagnosis of rotating machinery. 9(6), 168781401770414-168781401770414. https://doi.org/10.1177/1687814017704146

[35] Zhu, Y., Gu, Q., Zhao, Y., Wan, H., Wang, R., Zhang, X., & Cheng, Y. (2022). Quantitative Extraction and Evaluation of Tomato Fruit Phenotypes Based on Image Recognition. Frontiers in Plant Science, 13, 859290. https://doi.org/10.3389/fpls.2022.85929

APPENDIX H

CURRICULUM VITAE

JUSTINE MILES E. DEL ROSARIO

12487 Junction, Mayondon, Los Baños, Laguna

Contact No.: 09649180908

Email Address: justinemiles26.lspu@gmail.com



EDUCATIONAL ATTAINMENT

Tertiary

Laguna State Polytechnic University-Los Baños Campus

Bachelor of Science in Computer Science

Barangay. Malinta, Los Baños, Laguna (2022-2026)

Secondary

Collegio De Los Baños

Barangay, Los Baños, Laguna (2020 – 2022)

Los Baños Integrated School

Barangay Mayondon, Los Baños, Laguna (2016 – 2020)

Mayondon Elementary School (2010 – 2016)

PERSONAL PROFILE

Sex	: Male
Birthday	: September 26, 2004
Birthplace	: Paciano Rizal, Bay, Laguna
Height	: 5`7
Weight	: 57 kg
Name of Father	: Michael O. Del Rosario
Occupation	: Security Guard
Name of Mother	: Lea E. Del Rosario
Occupation	: House Wife
Status	: Single
Citizenship	: Filipino
Religion	: Roman Catholic

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

JUSTINE MILES E. DEL ROSARIO

LOUISE ANDREI SAPIN

11985 Mayondon, Los Baños, Laguna

Contact No.: 09633621228

Email Address: louise.sapin5@gmail.com



EDUCATIONAL ATTAINMENT

Tertiary

Laguna State Polytechnic University-Los Baños Campus

Bachelor of Science in Computer Science

Barangay. Malinta, Los Baños, Laguna (2022-2026)

Secondary

Los Baños Senior High School

Barangay San Antonio, Los Baños, Laguna (2020 – 2022)

Los Baños Integrated School

Barangay Mayondon, Los Baños, Laguna (2016 – 2020)

Mayondon Elementary School (2010 – 2016)

PERSONAL PROFILE

Sex	: Male
Birthday	: October 18, 2002
Birthplace	: Mayondon, Los Baños, Laguna
Height	: 5`6
Weight	: 51kg
Name of Father	: N/A
Occupation	: N/A
Name of Mother	: Aizalyn Q. Sapin
Occupation	: N/A
Status	: Single
Citizenship	: Filipino
Religion	: Jehova`s Witnesses

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

LOUISE ANDREI SAPIN

JOHN DAVE A. SILVA

Ruliog Comp. 1 San Isidro Calauan, Laguna

Contact No.: 09267234926

Email Address: johndavesilva00@gmail.com



EDUCATIONAL ATTAINMENT

Tertiary

Laguna State Polytechnic University-Los Baños Campus

Bachelor of Science in Computer Science

Brgy. Malinta, Los Baños, Laguna (2022-2026)

Secondary

CARD - MRI DEVELOPMENT INSTITUTE INC.

Brgy. Tranca Bay, Laguna (2020-2022)

Dayap National Integrated High School

Brgy, Kanluran, Calauan, Laguna (2016-2020)

San Isidro Elementary School (2009-2016)

PERSONAL PROFILE.

Sex	: Male
Birthday	: May 31, 2003
Birth Place	: San Pablo City, Laguna
Height	: 5‘7
Weight	: 75 kg
Name of Father	: Catalino Silva
Occupation	: Deceased
Name of Mother	: Lema Silva
Occupation	: Peanut Vendor
Status	: Single
Citizenship	: Filipino
Religion	: Roman Catholic

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

JOHN DAVE A. SILVA